

RACEMIC CONGLOMERATE: An equimolar mechanical mixture of crystals each one of which contains only one of the two enantiomers present in a racemate. The process of its formation on crystallisation of a racemate is called spontaneous resolution, since pure or nearly pure enantiomers can often be obtained from the conglomerate by sorting.

RACEMIFORM: An inflorescence with the form of a raceme, but not necessarily the structure of a true raceme.

RACEMISATION (RACEMIZATION): The formation of racemic mixtures from optically active compounds. It can be achieved by the action of heat or light or chemical reagents.

RACEMOSE INFLORESCENCE (INDEFINITE INFLORESCENCE): A type of flowering shoot in which the meristematic activity continues at the apex of the main stem and primary laterals and flowers are developed from the axillary meristems. As a result, the youngest flowers are at the top and the oldest are at the stalk.

RACHIS: 1. The main axis or shaft such as the main stem of a compound leaf or an inflorescence. 2. The shaft of a feather. 3. The backbone.

RAD: A unit of energy absorbed from ionising radiation, defined as the dose when one kilogram of matter absorbs 0.01 joule of radiation energy. It is now replaced by the gray (one rad is equal to 0.01 gray).

RADAR: A device for locating objects in space, that uses reflected radio waves to detect, find the range, altitude, direction or speed of both moving and fixed objects such as aircrafts, ships, spacecrafts, mountain ranges, radio and TV towers, guided missiles, motor vehicles and weather formations. The radar dish or antenna, transmits pulses of radio waves or microwaves which bounce off any object in their path. The object returns a tiny part of the wave's energy to a dish or antenna which is usually located at the same site as the transmitter.

RADAR ASTRONOMY: The use of radar techniques in which short bursts of radio waves are transmitted in the direction of an object. It is used to study and map astronomical objects in the solar system.

RADAR METEOR: A meteor detected by bouncing a radar signal off the train of ionised air left in its wake.

RADDLE: A long, flexible length of wood which is interwoven with others, between upright posts or stakes, used for making hurdles or fences.

RADIAL: 1. Of or arranged like rays. 2. Of, relating to or situated near the radius or the thumb side of the hand or forearm. 3. A longitudinal section that passes through the centre of a cylindrical organ. In a radial section down a woody stem the growth rings appear parallel and flecks of rays may be seen. 4. Of or relating to structures that radiate from a central point, such as the arms of a starfish. 5. Having pistons moving inward and outward from a central point or shaft. 6. In mathematics, of lines, extending from a common central point; arranged like the radii of a circle.

RADIAL ARTERY: A major artery that follows the course of the radius on the ventral aspect of the forearm to the wrist, where it is felt as a pulse and enters the hand. It supplies blood mostly to the thumb, index finger and muscles of the forearm.

RADIAL CIRCUIT: A circuit used in household wiring for lights or sockets where power is taken from a main switch and transmitted from point to point by a single length of cable linking each point to the next.

RADIAL CLEAVAGE: A type of holoblastic cleavage in which embryonic development in deuterostomes having planes of cell division that transform the zygote into a ball of cells are either parallel or perpendicular to the polar axis, thereby aligning each upper tier directly over those of the next lower tier resulting in radial symmetry around the pole to pole axis of the embryo.

RADIAL COMPONENT: The component of a vector quantity in the direction of the radius vector. In polar coordinates, the radial component of velocity is dr/dt and the radial component of acceleration

is given by: $\frac{d^2r}{dt^2} = r \left(\frac{d\theta}{dt} \right)^2$.

RADIAL DISTRIBUTION FUNCTION (RDF): The variation of atomic density as a function of the distance from one particular atom. In the case of a crystal, the radial distribution function is a series of regular spikes, corresponding to the atomic positions in the lattice. Calculating the radial distribution function is of fundamental importance in the theory of liquids.

RADIAL ELECTROSTATIC FIELD ANALYSER: An arrangement of two conducting sheets forming a capacitor and giving a radial electrostatic field which is used to deflect and focus ion beams of different energies. The capacitor may be cylindrical, spherical or toroidal.

RADIAL FIELD: A field in which the field lines diverge radially outward from a point source or converge radially inwards towards a point source. The field strength reduces as the distance from the centre increases and increases towards the centre. Examples are electric field surrounding an isolated point electric charge or the gravitational field surrounding an isolated point mass.

RADIAL NERVE: One of the main nerves of the arm, running from the shoulder to the hand. The radial nerve controls the muscles that straighten wrist. It also conveys sensation from the back of the forearm, the thumb, second and third fingers and the base of the thumb.

RADIAL SYMMETRY: 1. A line or symmetry about one of several lines or planes through the centre of an object or organism along which it can be cut to obtain two equal halves. 2. The arrangement of parts in an organ or organism which resembles a pie and exhibits no left or right sides such that cutting through the centre of the structure in any direction produces two halves.

RADIAN: A unit of measurement whose angle is equal to 57.3° , equivalent to the angle subtended at the centre of a circle by an arc equal in length to the radius. Angle can also be measured in degrees, with the symbol $^\circ$.

RADIATA: A taxonomic rank that has been used for members of the radially symmetrical animal phyla, including cnidarians, coelenterates, ctenophores and echinoderms. Members have all the parts arranged radially around the vertical axis of the body and the various organs repeated symmetrically in each ray or spheromere.

RADIATION: Energy travelling in the form of particles or electromagnetic waves through a medium or space. There are two types of radiation: ionising and non-ionising. **Ionising radiation** are energetic particles such as alpha particles, beta particles, neutrons and cosmic rays or electromagnetic waves that detach electrons from atoms or molecules, thus ionising them. Ionising radiation can cause damage to cellular structure and cause changes to the environment.

Non-ionising radiation is the radiation of energy such as radio waves, microwaves, ultra violet light (such as that radiated from the Sun) or visible light from a source. This type of energy causes molecular excitations and can cause changes to the RNA and DNA molecules and may result in certain disorders such as skin cancer.

Radiate is to send out energy such as heat or light, in the form of rays or waves; or having parts spreading from a central point, as in the flower heads of daisies and sunflowers.

RADIATION BELTS: A toroidal region within a planet's magnetosphere, where charged particles become trapped for long periods, spiralling backwards and forwards between 'mirror points' at opposite ends of magnetic field lines. These high-energy charged particles result from cosmic rays. Energetic particles in planetary radiation belts can present a hazard to spacecraft, causing damage to electronic components. Example of radiation belts is Van Allen Belts.

RADIATION BIOLOGY: The study of how living things are affected by ionising radiations such as x-rays, gamma rays or cosmic rays and particulate radiation such as alpha particles, beta particles, neutrons or heavy charged ions and by non ionisation radiation which includes ultraviolet radiation, microwaves and extra low frequency electromagnetic radiation.

RADIATION CHEMISTRY: The part of chemistry which deals with the chemical effects of ionising radiation, as distinguished from photochemistry associated with visible and ultraviolet electromagnetic radiation.

RADIATION COUNTER: Radiation measuring assembly comprising a radiation detector in which individual ionising events cause electrical pulses and the associated equipment for processing and counting the pulses. Often an expression is added indicating the type of radiation detector, for example, scintillation and semiconductor.

RADIATION DAMAGE: Damage caused by changes in the properties of liquids, gases and solids by any type of radiation to inanimate materials which can cause them to become radioactive, leading to radiation poisoning, radiolysis within the material, formation of reactive compounds, nuclear transmutation of the elements within the material, etc. In humans, this can lead to radiation sickness, burns and some forms of cancer. Radiation damage is caused by energetic particles from cosmic rays, energetic particle beams released from a particle accelerator, gamma radiation (X rays), particles or rays released by radioactive decay of elements from a nuclear reactor, nuclear weapons, etc.

RADIATION DETECTOR: A device in which incident radiation produces a measurable effect. If this effect is a rise in temperature it is called a **thermal detector**. If it is a rise in pressure it is called a **photoacoustic detector**. In the case where an electrical signal is produced it is called a **photoelectric detector**. Photoelectric detectors can be classified as photo-emissive detectors and semiconductor detectors. Where the radiation produces a chemical reaction, it is termed a **photochemical detector**. A detector yielding an output signal that is independent of the wavelength of the radiation over a specific region is called a **nonselective detector**. Where it is wavelength specific it is a **selective detector**. A detector having a quantum efficiency independent of the wavelength is a **nonselective quantum counter**.

RADIATION HAZARD: Hazard that exists in a region where there is a radiation field, other than what is considered to be normal background radiation.

RADIATION PRESSURE: The pressure exerted on objects by an electromagnetic radiation because it transmits energy and possesses momentum. This pressure is negligible on large objects but very considerable on small objects and can have some effects on the object such as altering its course. For example, the radiation pressure from the Sun can force a smaller object such as a comet away from it.

RADIATION SICKNESS (RADIATION POISONING): A damage to organ tissue caused by over-exposure to ionising radiation such as x-rays, gamma rays, neutrons and other nuclear radiations such as from weapons and fallout. Radiation sickness ionises atoms in the body and causes nausea, vomiting, diarrhoea, etc.

RADIATION SPECTRUM: The components of radiation arranged in order of their wavelengths, frequencies or quantum energies. For particle radiation they are arranged in order of their kinetic energies.

RADIATION TEMPERATURE: The surface temperature of an object in space, measured over a narrow region of the electromagnetic spectrum, such as the visible range. The radiation temperature is the same as the effective temperature but is usually measured over a narrow region of the electromagnetic spectrum, such as the visible range.

RADIATION THERAPY: The use of ionising radiation, in the form of rays from an outside source or from radium or other radioactive metal implants, in the treatment of malignant diseases such as cancer, lymphoma and leukaemia.

RADIATION TRAPPING (RADIATION IMPRISONMENT): A process by which the resonance radiation emitted in a lamp is absorbed and re-emitted many times in the plasma prior to striking the phosphor. Radiation trapping lengthens the effective lifetime of emission as viewed from outside the lamp. The control of this

trapping is, therefore, an important design consideration for low-pressure lamps.

RADIATION UNITS: Units of measurement for radioactivity and radiation doses and for quantifying the amount of radiation striking other objects. Radioactivity is measured in the SI unit of Becquerel (Bq) per second, which has replaced the former unit Curie (Ci). The becquerel (Bq) is defined as the activity of a radionuclide decaying at a rate, on average, of one spontaneous nuclear transition per second; $1 \text{ Bq} = 1 \text{ s}^{-1}$. The curie (Ci), is equal to $3.7 \times 10^{10} \text{ Bq}$. Radiation absorbed dose is measured in Gray, rad, rem and Sievert (Sv). The gray (Gy) is the SI unit of absorbed dose, which is the absorbed dose when the energy per unit mass imparted to matter by ionising radiation is 1 joule per kilogram. $1 \text{ gray (Gy)} = 100 \text{ rad (rad)}$. The former unit, the rad (rd), is equal to 10^{-2} Gy . In SI units, exposure to ionising radiation is expressed in coulombs per kilogram, the quantity of x- or gamma-radiation that produces ion pairs carrying 1 coulomb of charge of either sign in 1 kilogram of pure dry air. The roentgen (R), the former unit, is equal to $2.58 \times 10^{-4} \text{ C kg}^{-1}$. The sievert (Sv), the SI unit of dose equivalent, is the dose equivalent when the absorbed dose of ionising radiation multiplied by the stipulated dimensionless factors is 1 J kg^{-1} . The former unit, the roentgen (R), is equal to $2.58 \times 10^{-4} \text{ C kg}^{-1}$.

RADIATIONLESS DEACTIVATION: Loss of electronic excitation energy without photon emission or chemical change. Often, ultrafast radiationless decays are mediated by conical intersections.

RADIATIONLESS DECAY: The decay of an atom or molecule from an excited state to a lower energy state without photon emission or chemical change. An example of a radiationless decay is the Augér effect. The Augér effect is the emission of an electron rather than x-rays or gamma rays from an excited positive ion resulting in a doubly charged ion. It was named after Pierre Augér (1899-1993), French physicist.

RADIATIONLESS TRANSITION: A transition between two states of a system without photon emission or absorption.

RADIATIVE ABSORPTION: A process by which a particle in the ground state or an excited state may undergo transition to a higher energy level by absorption of a photon. For a given particle in the lower state of probability per second of such a transition in a field with a continuous spectrum, it is proportional to the spectral radiant energy density of the absorption line. The proportionality constant is termed the transition probability for absorption. The transition probability for stimulated emission is defined in a similar way for the reverse radiative de-excitation process that is induced by the same radiation field.

RADIATIVE CAPTURE: The capture of a particle by a nucleus followed by immediate emission of gamma radiation.

RADIATIVE COLLISION: A collision between two charged particles in which part of the kinetic energy is radiated in the form of photons or electromagnetic radiation.

RADIATIVE DE-EXCITATION: The change in the internal energy of a particle may be due to radiative processes, i.e. the emission or absorption of a photon. A particle in an excited state may undergo a transition to a lower energy level by emission of a photon. This is known as radiative de-excitation. If such a transition occurs spontaneously its probability per second for a given excited particle is termed the transition probability for spontaneous emission.

RADIATIVE ENERGY TRANSFER: Transfer of excitation energy by radiative deactivation of a donor molecular entity and reabsorption of the emitted radiation by an acceptor molecular entity. Radiative transfer results in a decrease of the donor fluorescence intensity in the region of spectral overlap. Such a distortion of the fluorescence spectrum is called inner-filter effect. Radiative energy transfer depends on the shape and size of the vessel.

RADIATIVE FORCING: The change in the value of the net radiative flux (i.e., the incoming flux minus the outgoing flux) at the top of the atmosphere in response to some perturbation, such as a change in the concentration of carbon dioxide or the output of the Sun. Radiative forcing can be quantified in units of the extra energy in watts per metre squared (W m^{-2}) entering the Earth near the top of the atmosphere (TOA). Usually radiative forcing is computed after allowing for stratospheric temperatures to readjust to radiative equilibrium, but with all tropospheric properties held fixed at their unperturbed values. If no climate change is taking place, the earth is in **radiative equilibrium**. This means that the amount of solar radiation it absorbs is exactly balanced by the amount of infrared radiation it emits, with the balance achieved by the adjustment of the surface and atmospheric temperatures. The influences that cause changes to the Earth's climate system altering Earth's radiative equilibrium, forcing temperatures to rise or fall, are called **climate forcings**. Positive radiative forcing means Earth receives more incoming energy from sunlight than it radiates to space. This net gain of energy will cause warming. Conversely, negative radiative forcing means that Earth loses more energy to space than it receives from the sun, which produces cooling.

RADIATIVE FORCING SCENARIO: A plausible representation of the future development of radiative forcing associated, for example, with changes in atmospheric composition or land-use change, or with external factors such as variations in solar activity. Radiative forcing scenarios can be used as input into simplified climate models to compute climate projections.

RADIATIVE LIFETIME: Lifetime of an excited molecular entity in the absence of radiationless transitions. The reciprocal of the first-order rate constant for the radiative step or the reciprocal of the sum of these rate constants if there is more than one such step. The equivalent term, natural lifetime, is discouraged.

RADIATIVE TRANSITION: A transition between two states of a molecular entity, the energy difference being emitted or absorbed as photons. In principle, radiative and radiationless transitions can be distinguished in molecules. The first occur by absorption or emission of light quanta and the latter is the result of the transformation of electronic excitation energy into vibrational/rotational energy.

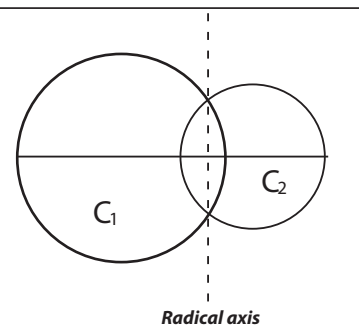
RADIATORS: 1. A device that cools an internal combustion engine and prevents it from overheating. It consists of tubes through which heated water from the engine is circulated. An important aspect of a cooling radiator is an installed fan, which takes the heat away from the radiator. 2. A room-heating device that emits heat by radiation, consisting of a heating medium such as a boiler, which generates hot water, steam or hot oil which is circulated under pressure to heat-emitting units that emit heat by radiation.

RADICAL: 1. Also called **free radicals**, it is a group of atoms, molecules or ions with unpaired electrons forming part of a compound which acts as a single unit and takes part in most chemical reactions without disintegration. Examples are sulfate, chloride, chlorate, cyanide, silicate, nitrate and carbonate. Free radicals are very reactive and unstable and are important in many chemical processes such as combustion, atmospheric chemistry and polymerisation. Free radicals also play an important role in human physiology. They occur in processes such as the destruction of invading organisms by white blood cells. Free radicals are produced in the body by natural biological processes or introduced from outside from sources such as in tobacco smoke, toxins, or pollutants and can damage cells, proteins, and DNA by altering their chemical structure. Free radicals might also play a role in various maladies, such as arthritis, heart disease and Alzheimer's disease. When natural enzyme controls fail, free radicals in the body attack lipids, proteins and nucleic acids. 2. In mathematics, **radical sign** is a symbol placed over a number or expression, called

the **radicand**, to indicate a root of the radicand. For example, in the expression $\sqrt[3]{15}$, the symbol $\sqrt[3]{}$ is the radical sign and 15 is the radicand. 3. A **radical expression** is an expression containing a square root. Examples of radical expressions are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and $\sqrt{a^2 + b^2}$. When used without a sign or index number, it designates the positive square root of the radicand such as 3. If both square roots are meant, the radical sign is preceded by $\pm$. To indicate higher roots of the radicand such as cube or fourth roots, an index number is used. The radical sign is generally taken to indicate the principal root of the radicand, although any radicand will have n different n th roots. 4. In mathematics, **Jacobson radical** of a left module is the intersection of all maximal (proper) submodules of the torsion module. Also, **Jacobson radical** of a ring is the intersection of all maximal left ideals or equivalently of all maximal right ideals; this is a two-sided ideal. In a commutative Banach algebra, it is the intersection of all maximal ideals. 5. Noetherian radical of a ring is the union of all two-sided nilpotent ideals, for which $J^n = 0$ for some natural number n . In a left Noetherian ring, this is the largest two-sided ideal with all of its elements nilpotent. In a left Artinian ring, the Noetherian and Jacobson radicals coincide. 6. In a commutative Banach algebra, it is the set of generalised nilpotent. 7. Denoting removal of the source of a disease, for example by surgery rather than treating the symptoms. 8. Relating to the roots; arising from or near the roots.

RADICAL AXIS: The locus of points (P) at which tangents drawn to two circles have the same length, which can be expressed as $R^2 = d_1^2 - r_1^2 = d_2^2 - r_2^2$ where r_1 and r_2 are the radii of the two circles, d_1 and d_2 are distances from P to the centres of the two circles and R is the radius of the unique orthogonal circle centred on P. In other words, when two circles intersect, it is the line joining the two points of intersection. If the equations of the two circles are respectively

$x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$, the radical axis has the equation: $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$.



RADICAL CENTRE: 1. The point of intersection of the respective radical axes of three intersecting circles. 2. The atom (or group of atoms) in a polyatomic radical on which an unpaired electron is largely localised. Attachment of a monovalent atom to a radical centre gives a molecule for which it is possible to write a Lewis formula in which the normal stable valencies are assigned to all atoms.

RADICAL FRACTION (RADIX FRACTION): The generalisation of decimal fractions to other bases of place-value notation.

RADICAL ION: A polyatomic ion that has a positive charge called **radical cations** or a negative charge called **radical anions**. An example is the benzene radical cation (C_6H_6^+ or C_6H_5^+).

RADICAL PAIR: The term is used to identify two radicals in close proximity in solution, within a solvent cage. They may be formed simultaneously by some unimolecular process, for example, peroxide decomposition or they may have come together by diffusion. While the radicals are together, correlation of the unpaired electron spins of the two species cannot be ignored: this correlation is responsible for the CIDNP phenomenon. A radical pair is called geminate, provided that each radical partner is a descendant of the same parental pair.

RADICAND: A number or quantity, usually put under the radical sign from which a root is to be extracted. For example, in the expression $\sqrt[6]{15}$, the symbol $\sqrt{}$ is the radical sign and 15 is the radicand.

RADICANT: A root arising from or above the prostrate stem.

RADICICOLOUS: Positioned directly upon the root crown or in roots.

RADICLE: The embryonic root of seed plants which is the first structure to emerge from the seed during the process of germination, which develops into the primary root. There are two types of radicles: **antitropous**, which points away from the seed coat scar (helium); and **syntropous**, which points towards the seed coat scar (helium).

RADII: Plural of radius. The **radius** is half a diameter or the distance from the centre of a circle to its circumference.

RADIO: A means of transmitting signals by modulation of electromagnetic waves with frequencies below those of visible light. In radio transmission, a microphone converts sound waves into oscillating electromagnetic waves that pass through the air and the vacuum of space and are then picked up by a receiving aerial and fed to a loudspeaker which converts them back into sound waves.

RADIO ASTRONOMY: A branch of astronomy that studies the radio-frequency radiation emitted by celestial objects such as stars, radio galaxies, quasars, pulsars and masers. This is achieved by using radio telescopes, used as a single unit or interlinked with other telescopes.

RADIO BEACON: A fixed ground-based radio transmitter that sends out radio frequency, operating in the band of 190 to 535 kHz. It transmits a continuous radio wave to help aircrafts and ships fix their position.

RADIO BUTTON: In computing, it is a dialogue box that presents two or more options displayed on screen allowing users to choose only one.

RADIO FREQUENCIES (RF): The range of electromagnetic waves with frequencies between about 3 KHz and 300 GHz used in radio transmission. The amplitude modulation (AM) broadcast band has frequencies ranging from 540 to 1,800 kHz and the Frequency modulation (FM) broadcast frequencies range from 88 MHz to 108 MHz.

RADIO FREQUENCY

IDENTIFICATION (RFID): A small tag that uses radio-frequency electromagnetic fields to identify objects carrying tags when they come close to a reader.

RADIO GALAXY: An active galaxy that strongly radiates energy in the form of radio waves or electromagnetic radiation with wavelengths longer than visible light and infrared radiation. The radio emission from the galaxies is due to a process by which atoms and molecules radiate energy known as **synchrotron process**. Although all galaxies emit some radio waves, radio galaxies emit between 1000 and 100 million times more radio energy than an ordinary galaxy such as Milky Way, the Earth's galaxy.

RADIO INTERFEROMETER: An array of antennas that operate together as if they were a single radio telescope of very large aperture.

RADIO SOURCE: Astronomical objects that emit radio waves. Radio sources include Jupiter, the Sun, pulsars, radio galaxies, quasars and spiral galaxies.

RADIO STAR (STELLAR RADIO SOURCES, RADIO SOURCE STARS): Stellar objects that produce copious emissions of various radio frequencies, whether constant or pulsed. Radio emissions from stars can be produced in many varied ways. Examples of radio stars are pulsars, a type of neutron star and quasars.

RADIO TELESCOPE: A system of instruments for detecting radio waves emitted by natural or celestial sources such as satellites and

space probes. Radio telescopes usually consist of a metal bowl that collects and focuses radio waves the way a concave mirror collects and focuses light waves. Radio telescopes are much larger than optical telescopes, because the wavelengths they are detecting are much longer than the wavelength of light. The largest single radio telescope is at Arecibo Observatory, Puerto Rico.

RADIO TRANSMISSION: The transmission of signals from a transmitting aerial through space at radio frequencies by means of radiated electromagnetic waves to a receiving aerial.

RADIO TRANSPARENT (RADIO-WAVE TRANSPARENT MATERIAL): Materials that are transparent to radiation, especially to x-rays and gamma rays. It is a type of in-homogeneous dielectric structural material, made up of one or several layers, which does not significantly alter the amplitude and phase of radio-frequency electromagnetic waves transmitted through it. Examples of radio transparent materials are ceramics, glass and plastics in the form of fibreglass laminates.

RADIO WAVES: Electromagnetic wavelengths in the electromagnetic spectrum with a long wavelength ranging from about 10^3 to 10^4 m and a low frequency from about 10^5 to 10^{11} Hz. The radio-wave part of the spectrum includes mobile radio communication, broadcasting, radar and other navigation systems, satellite communication, computer networks, microwaves used for both communications and for cooking, etc.

RADIO WINDOW: A range of frequencies in the range of 10 000 to 40 000 megahertz of electromagnetic radiation or radio waves that can be transmitted through the Earth's atmosphere without significant reflection or attenuation by constituents of the atmosphere.

RADIOACTIVE: Referring to a substance such as uranium or plutonium that emits energy in the form alpha, beta or gamma particles or rays. This occurs due to the decay of its unstable atoms. Radioactive emission is harmful to people or animals exposed to it and can cause death.

RADIOACTIVE AGE: The time, estimated from measurement of the isotopic composition, during which the content of a radioactive species within an object has remained unchanged except for nuclear decay.

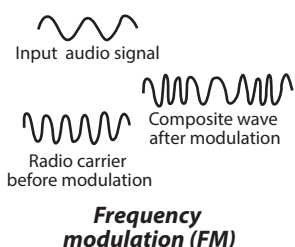
RADIOACTIVE COLLISION: A collision between charged particles in which a neutron is absorbed by a nucleus and kinetic energy is radiated as gamma ray.

RADIOACTIVE CONTAMINATION: A radioactive substance in a material or place where it is undesirable.

RADIOACTIVE COOLING: The cooling of the earth's surface and the air near the surface, occurring mainly at night and caused by heat loss engendered by terrestrial radiation.

RADIOACTIVE DATING: A method of estimating the age of an ancient object or specimen using radioactive decay based on the ratio of the amounts of mother and daughter isotopes present in a radioactive element. Examples of radioactive dating include carbon-14 dating, potassium-argon dating, rubidium-strontium dating and uranium-lead dating. The process is based on the fact that atoms of most common naturally occurring radioactive isotopes such as potassium-40, thorium-232 and uranium-238 take billions of years before undergoing alpha or beta decay. The steady decay of these isotopes and other, more stable atoms allows scientists to determine the age of substances in which these isotopes occur. The amount of isotope called the mother isotope that was present when the mineral was formed can be estimated. The amount of isotope that has decayed, called the daughter isotope is also determined. Knowing the rate at which the isotope decays, can determine how much time has passed since the formation of the object.

RADIOACTIVE DECAY: A process of continuous spontaneous disintegration in which the unstable nuclei of radioactive elements lose energy by emission of electromagnetic radiation or alpha particles and beta particles and gamma rays. Examples of such elements are radium and various isotopes of uranium and also those having atomic



numbers higher than that of uranium, known as the **transuranic elements**. This changes the element's atomic number and transforms an element commonly referred to as the **parent isotope** into another called the **daughter isotope**. The four common forms of decay are alpha decay, beta decay, inverse beta decay and isomeric transition. Radioactivity is measured in the SI unit of Becquerel (Bq) per second, which has replaced the former unit, Curie (Ci). The becquerel (Bq) is defined as the activity of a radionuclide decaying at a rate, on average, of one spontaneous nuclear transition per second; $1 \text{ Bq} = 1 \text{ s}^{-1}$. The curie (Ci), is equal to $3.7 \times 10^{10} \text{ Bq}$.

RADIOACTIVE EMISSIONS: Alpha and beta particles and gamma rays emitted during radioactive decay. Alpha particle is a positively charged nuclear particle consisting of two protons bound to two neutrons. Alpha particles are emitted spontaneously in some types of radioactive decay. They are also produced when helium-4 atoms are completely ionised. Its symbol is α . Beta particle is electrically charged particle emitted by certain radioactive elements, such as uranium and berkelium. Beta particles are produced when the nucleus of an unstable atom spontaneously decays or transforms, changing the atom into an atom of a more stable element. Its symbol is β . In radioactive emissions, gamma emission is usually found in association with alpha and beta emission. Gamma rays possess no charge or mass; thus emission of gamma rays by a nucleus does not result in a change in chemical properties of the nucleus but merely in the loss of a certain amount of radiant energy. The emission of gamma rays is a compensation by the atomic nucleus for the unstable state that follows alpha and beta processes in the nucleus.

RADIOACTIVE EQUILIBRIUM: The condition of equilibrium reached by a radioactive series in which the rate of decay of the parent nuclide is equal to the rate of decay of every intermediate daughter nuclide. An example is when uranium-238 decays to thorium-234.

RADIOACTIVE FALLOUT: The deposition on the ground of radioactive substances from nuclear explosions and other injections of radioactive material into the atmosphere.

RADIOACTIVE DECAY LAW: A law stating that the rate at which radioactive nuclei in a sample decay (the activity) per unit time is directly proportional to the number of radioactive nuclei present in the sample at any one time. In other words, the probability per unit time that a nucleus will decay is a constant, independent of time. If the number of nuclei in a sample is N and the number of radioactive decays per unit time Δt is ΔN then, $-\frac{\Delta N}{\Delta t} \propto N$ or $-\frac{\Delta N}{\Delta t} = \lambda N$, where

λ is the radioactive decay constant or disintegration constant.

RADIOACTIVE MATERIAL: A material that emits radiation due to radioactive decay. Examples are uranium, thorium and potassium and any of their decay products, such as radium and radon. Radioactive materials which occur naturally and where human activities increase the exposure of people to ionising radiation are known by the acronym '**NORM**' (**Naturally Occurring Radioactive Material**), which potentially includes all radioactive elements found in the environment. NORM results from activities such as burning coal, making and using fertilisers, oil and gas production.

RADIOACTIVE SERIES (DISINTEGRATION SERIES; RADIOACTIVE CHAIN; RADIOACTIVE DECAY SERIES): A series of radioactive decay in which each nuclide is transformed by the radioactive decay of the nuclide that precedes it, until a stable state is reached. This process happens because most radioactive elements do not decay directly into a stable state, but undergo a series of decays until a stable isotope is reached. The following are the three main radioactive series occurring in nature: (i) the **thorium series** also called **$4n$ series** ($4n$ represents the mass number of the isotopes), in which the unstable thorium-232 decays in stages to the stable isotope lead-208. (ii) The **actinium series** also known as **$4n + 3$ series** ($4n + 3$ represents the mass number of the isotopes), in which the unstable uranium-235 decays in stages to the stable isotope lead-207. (iii) The **radium-uranium series** also

known as **$4n + 2$ series** ($4n + 2$ represents the mass number of the isotopes), in which the unstable uranium-238 decays in stages to the stable isotope of lead, element 82. (iv) The **neptunium series** also known as **$4n + 1$ series** ($4n + 1$ represents the mass number of the isotopes), starts with curium-241, which is the artificial isotope of curium and decays to neptunium-237 and ends with bismuth-209.

RADIOACTIVE SOURCE: Any quantity of radioactive material which is intended for use as a source of ionising radiation.

RADIOACTIVE STANDARD: A radioactive-isotope of a known number and rate of radioactive atoms at some definite time used for the calibration of radiation measuring instruments.

RADIOACTIVE TRACER: A substance or compound containing radioactive isotope concentrations or some radioactive atoms that can be introduced into a substance such as a cell or a tissue in small amounts and tracked within the body to study disease and biochemical processes.

RADIOACTIVE TRACER TECHNIQUE: A technique for investigating recovery and loss of a microcomponent, in which a radioactively labelled element or compound chemically identical with the microcomponent is added to the sample before preconcentration and its behaviour is followed by sensitive, rapid and selective radioactivity measurements.

RADIOACTIVE WASTE (NUCLEAR WASTE): Any material such as a solid, liquid or gaseous waste, which is the waste product of mining and processing of radioactive ores, scientific, military and industrial applications of nuclear energy such as research reactors and nuclear power reactors that contain radionuclides. Many nuclear wastes remain radioactive for thousands of years and pose a danger to human and animal life and must, therefore be properly disposed off. The safe disposal methods include dumping concrete-encased containers filled with radioactive waste in the ocean and burying the waste underground in old salt mines.

RADIOACTIVITY: The spontaneous disintegration of unstable nuclei to produce a more stable nucleus with emission of radiation. The types of radiation emitted are alpha particles, beta particles and gamma rays. The rate of disintegration is not influenced by chemical reactions or any normal changes such as temperature and pressure in the environment. On the other hand, the bombardment of a previously stable nuclei with high-speed subatomic particles (protons, neutrons, etc.) also creates unstable nuclei which disintegrates with emission of radiation, a process known as induced or stimulated nuclei reaction, which used to be called artificial radioactivity. Radioactive isotopes are useful in archaeology for carbon-dating or measuring the age of carbon-bearing items using carbon-14, in medicine for diagnosis and treatment of diseases such as tumors, studying living organisms and sterilizing medical equipment, producing nuclear weapons, generating energy for heat and electric power, in agriculture for pest control, as tracers to monitor fluid flow and filtration as well as in all types of industrial processes.

RADIOBIOLOGY (RADIATION BIOLOGY): A branch of science that studies the radioactive effects of substances on living organisms and the use of radioactive tracers to study metabolic processes.

RADIOCARBON CYCLE: Production and recycling of the radioisotope carbon-14 (^{14}C). The radioisotope occurs when a neutron flux, caused by cosmic radiation bombarding the upper atmosphere, reacts efficiently with nitrogen present. Carbon-14 intake by living organisms eventually returns to the atmosphere when dead vegetation or animal flesh decomposes, except when it is locked in preserved organic artefacts and remains. Radioactive decay occurs, forming the basis of the radiocarbon dating method.

RADIOCARBON DATING (CARBON DATING): The method of dating organic remains such as bone or wood, based on the carbon 14 it contains by which the ratio of radio-carbon to regular carbon is measured and the date of the material calculated. Radiation from

space produces neutrons that enter the Earth's atmosphere and react with nitrogen to produce the carbon isotope C-14 (carbon 14). All living organisms accumulate this isotope through their metabolism until it is in balance with levels in the atmosphere, but when they die they absorb no more. Because the nucleus of C-14 decays at a known rate, scientists can determine the age of organic substances such as bones, plant matter, shells and charcoal by measuring the amount on C-14 that remains in them. A device called **accelerator mass spectrometer** directly counts C-14 atoms, rather than counting rates of disintegration. This method can date items that are up to 90,000 years old.

RADIOCHEMICAL ACTIVATION ANALYSIS: A kind of activation analysis in which, after the irradiation, chemical or physical separation is applied.

RADIOCHEMICAL PURIFICATION: Chemical separation applied to a radioactive preparation in order to improve the radiochemical purity.

RADIOCHEMICAL PURITY: For a material, the fraction of the stated isotope present in the stated chemical form.

RADIOCHEMICAL SEPARATION: Separation by a chemical means of the radioactive isotopes of a specific element from a mixture of radionuclides.

RADIOCHEMICAL YIELD: The yield of a radiochemical separation expressed as a fraction of the activity originally present.

RADIOCHEMISTRY: The branch of chemistry that is concerned with radioactive substances. It includes the study of the chemical effects of radioactive isotopes and the preparation and use of compounds containing radioactive isotopes.

RADIOCHROMATOGRAPH: A measuring assembly designed to measure the spatial or time distribution of the activity of a mixture of radioactive components after separation by a chromatographic method.

RADIOGENIC: An isotope produced by the process of radioactive decay. An example is the stable isotope lead-208 produced from unstable thorium-232 decay.

RADIOGRAPHY: The process of producing images of an opaque object on photographic frame by the penetration of radiation such as gamma rays, x-rays, neutrons or charged particles. Special plates with thick emulsions are used to increase sensitivity. It is used in examining the internal structure of objects such as bones for fractures or other defects. It is also used to examine the heart or chest, pipes and other structures for cracks. **Radiograph** is the image or visual representation produced from radiography.

RADIOGRAVIMETRIC ANALYSIS: A kind of quantitative analysis in which the activity of a precipitate is used as a measure of its mass.

RADIOIMMUNOASSAY (RIA): A sensitive method for measuring the level of antigens such as enzymes, hormones and steroids in blood, in which a radioactively labelled form of the antigen is mixed with a known amount of antibody for that antigen, with the two binding to each other. A specimen of the fluid or serum to be tested is mixed with the combination and the concentration of antigens in the specimen is determined by the decrease in the amount of radioactive antigen bound to the antibody, which is measured by a gamma counter. RIA is used to diagnose diseases such as diabetes, thyroid disorders and hypertension.

RADIOIODINATION: The process of incorporating the radionuclides of iodine, usually ^{125}I , ^{131}I or ^{123}I into or covalently linking a radioiodinated substance to a substance.

RADIOISOTOPE (RADIOACTIVE ISOTOPE): An isotope that is radioactive and having an unstable nucleus that decays, emitting alpha, beta or gamma rays producing a stable isotope of another element. For example, unstable thorium-232 decays to produce the stable isotope lead-208.

RADIOISOTOPE IMAGING: The formation of images by administering radioisotopes internally to obtain information on

biochemical processes in the body of living organisms. It is used in medicine to locate diseases such as tumours or cancers and also to examine body fluids.

RADIOISOTOPE INDUCED X-RAY EMISSION ANALYSIS: A kind of analysis based on the measurement of the energies and intensities of characteristic X-radiation emitted by a test portion during irradiation with a radioactive source.

RADIOISOTOPE SCANNING: The use of a radioactive substance, which is administered internally to enhance the image of an organ or tissue, which is recorded, displayed and examined for diseases or other defects.

RADIOLARIAN: Any protozoan of the class *Polycystinea* (superclass *Actinopoda*), found as zooplankton in the upper layers of all oceans. They are mostly spherically symmetrical and known for their tiny, complex, perforated skeletons, known as tests, which are usually composed of silica.

RADIOLOCATION: The location of distant objects by means of radar using radio waves and also for detecting underground items such as cables and public utilities like water pipes.

RADIOLOGY: The study and use of x-rays, gamma rays, radioactive isotopes and other forms of radiation in medicine, for diagnosis and the treatment of diseases such as cancer, tumour, bone fracture and internal abnormalities of the body.

RADIOLUMINESCENCE: Luminescence arising from excitation by high-energy particles or radiation such as beta particles. For example, self-glowing paints containing zinc sulfide are used to make watch and aircraft instrument dials glow in the dark and make them readable.

RADIOLYSIS: The cleavage of one or several bonds resulting from exposure to high-energy radiation. The term is also often used loosely to specify the method of irradiation ('pulse radiolysis'), used in any radiochemical reaction, not necessarily one involving bond cleavage.

RADIOMETER: An instrument that detects and measures the intensity of radiant energy, especially that of infrared radiation. Satellites often carry radiometers to measure radiation from clouds, snow, ice, bodies of water, Earth's surface and the Sun.

RADIOMETRIC ANALYSIS: A kind of quantitative analysis in which measurement of the activity is an essential step.

RADIOMETRIC DATING (RADIOACTIVE DATING): A method of dating geological features such as rocks by the amount of radioactive decay of naturally occurring isotopes compared to its decay products, using known decay rates.

RADIOMETRIC TITRATION: A titration in which a radioactive indicator is used to monitor the end-point of the titration.

RADIOMETRY: The measurement of quantities associated with radiant energy. The quantities may also describe the variation of the energy with respect to other variables such as wavelength, time, position, direction (solid angle), area normal to the light or projected area of emitting or receiving surfaces.

RADIONUCLIDE (RADIOACTIVE NUCLIDE): A naturally occurring or artificially produced atom with an unstable nucleus, capable of radioactive decay. These radionuclides try to achieve a more stable structure by the emission of one or more different kinds of particles, including alpha particles, beta particles and gamma rays, in the process known generally as radioactivity or radioactive decay.

RADIONUCLIDE SCANNING: A diagnostic technique that is based on the detection of radiation emitted by radionuclides (radioactive materials) introduced into the body. Different substances are taken up in different concentrations by different tissues, allowing specific organs to be studied. For example, iodine is taken up mainly by the thyroid gland, so by "tagging" a sample of iodine with a radioactive marker, the thyroid glands uptake of iodine can be monitored to investigate the functioning of the gland.

RADIONUCLIDIC PURITY: For a material, that fraction of the total activity which is present in the form of the stated radionuclide, including daughter products.

RADIOPAQUE: Any material such as a metal object that does not allow electromagnetic radiation especially x-rays to pass through it. It is used in radiographic examination to enhance the x-ray images of internal body structures. For example, radiopaque dye is injected intravenously into the body or barium meal is introduced into the body orally before a radiographic examination to enhance the images. On x-ray films, radiopaque materials appear light grey to white.

RADIOPHARMACEUTICAL: A radioactive drug or substance, such as radioactive iodine or cobalt used for diagnosis and treatment of diseases such as some cancers.

RADIORESISTANT: Able to withstand high doses of ionising radiation without serious injury, as demonstrated by certain bacteria such as *Micrococcus radiodurans*.

RADIOSCOPY: The use of x-rays or other electromagnetic radiation to study the internal structure of optically opaque objects.

RADIOSENSITIVE: Quickly injured by exposure to ionising radiation.

RADIOSONDE: A package of meteorological instruments launched into the atmosphere by balloon, rocket or parachute to altitudes of about 27,000 metres that measure temperature, pressure and humidity in the upper atmosphere and transmit the data by radio to meteorological stations.

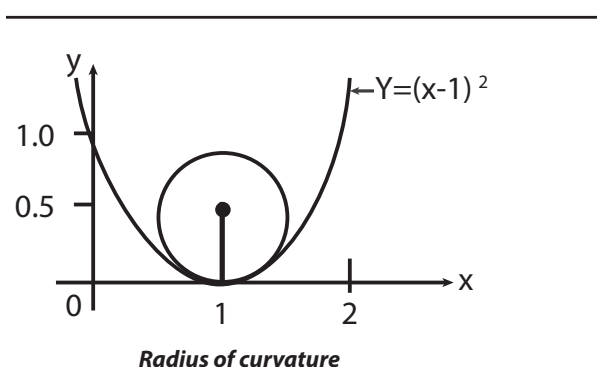
RADIOTHERAPY: The treatment of diseases such as cancerous growths by which the affected part is exposed to radiation from x-rays or other radioactive sources, killing the cells and stopping growth. The unaffected parts of the body are usually protected from the radiation.

RADIUM: A white highly radioactive metallic element in Group 2 (IIA) of the Periodic Table and one of the alkaline earth metals. It is formed by the radioactive disintegration of uranium and found in all uranium ores. Its symbol is Ra, atomic number 88, melts at about 700 °C (973 K) and has a relative density of 5. The most stable isotope is radium -226 which decays to radon. It is used as a radioactive source in atomic energy technology, in medicine for the treatment of cancer and also for luminous coatings.

RADIUS: 1. The shorter of the two long bones of the lower arm of humans and many other vertebrates. 2. The distance from the centre of a circle or sphere to its circumference. It is half the diameter of the circle. 3. Acronym for **Remote Authentication Dial-In User Service**, a client/server protocol that enables servers located remotely to communicate with the central server, thereby allowing dial-in users to authenticate themselves to gain access to the service.

RADIUS OF CONVERGENCE: The radius of the largest circle or interval in the real case around a given point such that a given power series converges absolutely at all points strictly inside the circle. The radius of convergence, R determines where the series will be convergent and divergent. There is always an interval $|x - a| < R$ in which a power series converges. For a power series, $\sum C_n(x - a)^n$, if $R > |x - a|$, the series converges; if $R < |x - a|$, the series diverges.

RADIUS OF CURVATURE: The radius of a circle that touches the inside of the curve at that point. The radius of curvature of a plane curve at a point is the radius of a circle that is equal to that of the given curve at that point. The rate of change of the tangent to the curve with respect to the arc length is the absolute value of the reciprocal of the curvature of a curve at a given point. For $y = f(x)$, it is given by $\frac{d}{ds} \left(\frac{df}{dx} \right)$.



RADIUS OF GYRATION: The distance between the point about which rotation of a body occurs and the point at which the body's mass is distributed, having the maximum effect. It is expressed as $k^2 = I/m$ where k is the radius of gyration and I the body's moment of inertia and m is the mass.

RADIUS OF NORMAL CURVATURE: Of a surface in a given direction, it is the absolute value of the reciprocal of the curvature of the normal section of the surface at the point in the given direction.

RADIUS VECTOR: 1. A directed line from the origin of a coordinate system to a point in space that is regarded as variable and sweeping out a curve. 2. A line connecting the centre of an astronomical object such as the Sun with the centre of another astronomical object in orbiting around it.

RADIX (BASE): 1. The number that is to be raised to a power. For example, in 16^2 , 16 is the radix or the base. 2. A number that is the base of a number system or of a system of logarithms. For example, in the decimal system, the radix is ten, because it uses the ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9; and in the binary system, it is 2.

RADIX FRACTION (RADICAL FRACTION): The generalisation of decimal fractions to other bases of place-value notation.

RADIX POINT (RADIX CHARACTER): The symbol or decimal part, usually a dot (.) used to separate the integer part of a number from its fractional part.

RADON: A heavy, colourless and gaseous radioactive element with a short half-life. Its symbol is Rn, atomic number 86, melting point -71 °C (202 K) and boiling point -61.8° C (211 K). It is formed by the disintegration of radium and uranium. It is used to treat cancer by radiotherapy.

RADON-NIKODYM DERIVATIVE: In mathematics, the function f , denoted $d\lambda/d\mu$, that possesses a Lebesgue integral and is unique up to measure μ , such that for each measurable set E $\lambda(E) = \int_E f d\mu$, where measures λ and μ satisfy the conditions of the Radon-Nikodym theorem.

RADON-NIKODYM THEOREM: The theorem that, given a sigma-finite measure μ , and a signed measure λ that is absolutely continuous with respect to μ , there is a function f that possesses a Lebesgue integral, such that, for each measurable set E , $\lambda(E) = \int_E f d\mu$ that is unique up to a set with μ -measure zero. This function is called the Radon-Nikodym derivative of λ with respect to μ and is denoted $d\lambda/d\mu$.

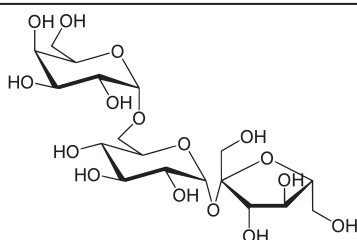
RADON'S THEOREM: The theorem that any $n + 2$ points in an n -dimensional vector space can be partitioned into two non-empty sets whose convex hulls are disjoint.

RADULA: A tongue-like organ of molluscs, used for feeding. It consists of rows of tiny teeth on a strip for boring or scraping food before it enters the oesophagus.

RAFFINATE: A liquid purified by solvent extraction to remove solutes from the original liquor.

RAFFINOSE (MELITOSE; MELITRIOSE; GOSSYPOSE; O- α -D-GALACTOPYRANOSYL(1,6)-O- α -D-GLUCOPYRANOSYL-(1,2)- β -D-FRUCTOFURANOSIDE):

A colourless or white, crystalline trisaccharide carbohydrate with the chemical formula $C_{18}H_{32}O_{16}$. It is composed of galactose, fructose and glucose and occurs in sugar beet, cottonseed meal and molasses.



Chemical Structure of Raffinose

RAGWORT: A weed of the genus *Senecio* in the composite family with yellow flower heads found in grassland. The variety *Senecio jacobea* is poisonous to livestock.

RAID (REDUNDANT ARRAY OF INDEPENDENT DISKS):

Formerly called redundant array of inexpensive disks, it is an array of disks connected to a controller that allows the same data to be stored in multiple places (redundantly) in a balanced way to improve data security and performance. RAID can be configured in different ways, depending on the application. RAID I, for example, is for disk mirroring, while RAID 5 spreads every character between disks.

RAIN: A liquid precipitation in which water condensed from vapour in the atmosphere falls to the Earth's surface from clouds as separate drops. When air masses acquire moisture on passing over warm bodies of water or over wet land surfaces, the moisture or water vapour is carried upward into the air mass by turbulence and convection. Raindrops vary in size, from about 0.5 mm to about 5 mm in diameter. When the drops are smaller than 0.5 mm, the precipitation is usually called **drizzle**. There are three main types of rainfall: **convective orographic** or **relief** and **frontal** depending on how the air rises. In **convective** rainfall, the sun heats up the ground, which heats the air above it. This then rises and cools and the water vapour condenses into water droplets, forming cumulonimbus or cumulus congestus clouds. The droplets fall as rain under the influence of gravity. It falls as showers with rapidly changing intensity over a certain area for a relatively short time. In **orographic** or **relief rainfall**, the presence of hills or mountains forces warm air to rise and cool. The water vapour condenses, forming clouds; the droplets then fall on the windward side of mountains as rain, sleet, hail or snow, depending on the atmospheric conditions. The other side of the mountain called the **leeward** side does not receive much rainfall and have sparse vegetation. **Frontal** rainfall occurs at occluded fronts, when warm and cold air masses meet. The warm air is pushed upwards over the cold air and, as it rises, it cools down. The water vapour condenses to form clouds, which fall as rain.

RAIN FOREST: A region characterised by high annual rainfall and evergreen tall trees with broad-leaves. It has a vegetation of four layers: emergent (with trees as tall as 60 m), canopy, understory, and forest floor. Rainforests are diverse and complex and home to more than half of the world's plant and animal species: There are two types of rainforests: tropical and temperate. Temperate rain forests are much cooler than their tropical ones with average temperatures of between 10° and 21°C and an annual rainfall of between 150-500 centimetres. Temperate rain forests are found mostly in coastal, mountainous areas in the mid-latitudes. They are not as biologically diverse as tropical rain forests. Tropical rain forests are the most biologically diverse terrestrial ecosystems in the world with temperatures of between 21° and 30°C, high humidity and annual rainfall of between 200 - 1000 centimetres. The Amazon rain forest is the world's largest tropical rain forest. It is home to around 40,000 plant species, nearly 1,300 bird species, 3,000 types of fish, 427 species of mammals, and 2.5 million different insects.

RAIN GAUGE (PLUVIOMETER): An instrument for measuring precipitations such as rain, snow, hail or sleet over a set period of time. It consists of a cylinder that catches rain covered by a funnel-like

lid. If 10 cm of rain collects in the cylinder, it means 10 cm of rain has fallen. **RAIN GAUGE (PLUVIOMETER):** An instrument for measuring precipitations such as rain, snow, hail or sleet over a set period of time. It consists of a cylinder that catches rain covered by a funnel-like lid. If 10 cm of rain collects in the cylinder, it means 10 cm of rain has fallen.

RAIN OUT: The mechanism by which small particles in the clouds are removed by the formation of raindrops; this is a different mechanism from wash out which is a process which occurs below cloud level. Both the terms rain out and wash out have not always been used in accordance with these definitions. For clarity they should be replaced by the terms, in-cloud scavenging and below-cloud scavenging, respectively.

RAIN SHADOW: An area of low rainfall on the lee side of a mountain barrier, which shelters it from prevailing rain-bearing winds. Rainfall on the corresponding windward side of the barrier is extremely high.

RAIN TREE (COCO TAMARIND; ACACIA PRETA; FRENCH TAMARIND; SAMAN; MONKEY POD; *Albizia saman*; *Samanea saman*):

A large tropical tree of about 60 metres tall belonging to the Mimosaceae (Leguminosae) family. It has a fissured bark with a symmetrical broad umbrella-shaped crown about 80 metres wide. Its leaves are bipinnately compound with small pinkish-green flowers. Traditionally, the roots are used against stomach cancer and the seeds can be chewed to cure sore throat. One of the most important uses of the rain tree is as a shade tree, especially in parks, pastures and roadsides.

RAINBOW: An optical effect of the sky appearing as spectrum of light when the Sun's rays are refracted and reflected by drops of rain. By convention, the rainbow colours are described as red orange, yellow, green, blue, indigo and violet. Two bows are usually displayed, the inner ring and the brighter of the two, known as the **primary bow**, produced by light that is reflected once inside the drop; and the outer ring, known as the **secondary bow**, in which the colours are reversed and produced by light that is reflected twice inside the drop.

RAISE: In mathematics, to raise to a power is to multiply a number (or quantity) by itself a certain number of times. For example, 3 raised to the power 3, written as 3^3 equals 27.

RAM: 1. An abbreviation in computing for **Random-access memory**, which is a form of computer data storage that allows stored data to be accessed in any order. 2. Abbreviation in physics for **relative atomic mass**, defined as the ratio of the average mass per atom of an element to one twelfth of the mass of a carbon-12 atom. Its symbol is Ar. 3. In agriculture, it is the male of a sheep.

RAMAL (RAMEAL): Relating to the branches.

RAMAN EFFECT (RAMAN SCATTERING): The change in wavelength and phase exhibited by monochromatic light as it passes through a transparent medium due to a change in rotational or vibrational energy of the scattering molecules. It is used in Raman spectroscopy to obtain information about the structure of molecules.

RAMAN SPECTROSCOPE: A type of spectrometer based on the Raman effect. Unlike most methods of analysing a substance, it does not require destructive testing. By firing a laser at the sample, atoms are excited, which then emit a very weak light with a pattern characteristic of the material.

RAMAN, CHANDRASEKHAR VENKATA (1888–1970): Indian physicist who was awarded the 1930 Nobel Prize in Physics for his discovery of the Raman effect, that when a medium is exposed to a beam for electromagnetic radiation, light scattered at right angles to the beam has a range of frequencies characteristic to the medium. This is the basis of the Raman spectroscope. Raman was professor of physics (1917–33) at Calcutta University. He was director of the Indian Institute of Science, Bangalore and in 1946 founded and directed the Raman Institute.

RAMAPITHECUS: A genus of extinct primates that lived about 12 to 14 million years ago. Fossil remains have been found in India, Pakistan and East Africa.

RAMENTUM: Any of the small brown chaff-like that cover the young fronds of a fern. The ramenta are shed as the frond unfurls but some persist on the rachis.

RAMIFICATION: The arrangement of branches or branching parts. **Ramiform** means having the form of a branch.

RAMJET: A jet engine in which air is drawn in and compressed by the forward motion of the aircraft for combustion. Forward thrust is obtained as a reaction to the rearward rush of high velocity hot exhaust gases. Ramjet engines can operate at speeds above 320 km/h but the practical operating speed is supersonic and higher and the best minimum operating speed is twice the speed of sound (Mach 2).

RAMOSE (RAMOUS): Having many branches.

RAMSEY THEOREM: In mathematics, the theorem, which states that for any two positive integers k and l there is a least integer $R(k, l)$ such that given any simple graph with $R(k, l)$ vertices, either the graph contains a clique of k vertices with all vertices adjacent or an independent set with l vertices with no vertices adjacent. The integers $R(k, l)$ are called Ramsay numbers.

RAMSAY, SIR WILLIAM (1852-1916): British chemist, who became a professor at University College, Bristol (1880-87) and University College, London (1887-1912). In the early 1890s he worked on the inert gases of the atmosphere and discovered helium. In 1894 he discovered argon together with Lord Rayleigh. In 1898, he discovered neon, krypton and xenon together with Morris Travers (1872-1972). He also discovered radon. He was awarded the Noble Prize for Chemistry in 1904, for his work on gases.

RAMSDEN EYEPiece: An eyepiece used for optical instruments consisting of two identical plano-convex lenses with their plane sides facing outward. They have the same power and focal length and are separated by a distance of two thirds of the focal length of either lens.

RAMUS COMMUNICANS (RAMI COMMUNICANTE): A nerve which connects two other nerves. It is most often used to describe the nerve structure that runs from the spinal nerve to a ganglion of the sympathetic nervous system and usually refers to the grey ramus communicans or white ramus communicans.

RANCH: A large-scale commercial farm where cattle, sheep, horses or other livestock are raised on large tracts of open land, especially in North and South America and Australia.

RANCID: The strong unpleasant taste or smell associated with decomposing oils or fats.

RANDALL-SUNDRUM SCENARIO (5-DIMENSIONAL WARPED GEOMETRY THEORY): A variant of the brane world picture in which gravity is trapped by warping of the higher dimensions, proposed in 1999 by Lisa Randall and Raman Sundrum. There are two popular models called RS1, which has a finite size for the extra dimension with two branes, one at each end; and RS2, similar to the RS1, but one brane has been placed infinitely far away, so that there is only one brane left in the mode.

RANDOM: 1. In statistics, describing a set in which all the members have the same probability of occurrence. It is a description for any process in which results may not be certain. 2. Involving or relating to variables that have definite probability but undetermined value.

RANDOM ACCESS MEMORY (RAM): A type of primary computer memory in the form of integrated circuits, consisting of a collection of cells, each capable of holding one bit of data used for temporary data storage, which can be written and read and accessed in any order. The time for accessing any location is fast and constant.

RANDOM ACCESS MEMORY DIGITAL-TO-ANALOGUE CONVERTER (RAMDAC): The portion of a computer video card that converts the digital data to an analogue signal to be displayed on a computer monitor.

RANDOM COIL (STATISTICAL COIL): The complete set of spatial arrangements of a chain molecule with a large number of segments that randomly change mutual orientation with time, under conditions

in which it is free from external constraints that would affect its conformation. If the solution of the chain molecules is not in theta state, the segments change mutual orientation only approximately randomly.

RANDOM COINCIDENCE: A coincidence of events occurring in not physically connected nuclei.

RANDOM COPOLYMER: A copolymer consisting of macromolecules in which the probability of finding a given monomeric unit at any given site in the chain is independent of the nature of the adjacent units.

RANDOM DRAW: Taking an object from a set of objects in which each object has an equally likely chance of being chosen. Examples are drawing a card from a deck, and drawing a domino from a bag of dominoes.

RANDOM ERROR: Result of a measurement minus the mean that would result from an infinite number of measurements of the same measurand carried out under repeatability conditions. In other words, it is the difference between predicted values and observed values treated as a random variable.

RANDOM EVENTS: Events which do not have deterministic regularity but possess some degree of statistical regularity. It can also be referred to as favourable outcome of a random experiment and is a subset of a sample space.

RANDOM EXPERIMENT: An experiment in which the outcome is not predictable in advance. A random experiment can be characterised as follows: (i) it is possible to describe the set of all possible results (called the sample space of the random experiment); (ii) it is not possible to predict the result with certainty; and (iii) it is possible to associate each possible outcome to a probability of appearing.

RANDOM NUMBERS: A sequence of numbers that have no detectable pattern in their progression and no member can be predicted from the preceding elements. Random numbers find application in computer games.

RANDOM NUMBER GENERATOR: A part of the software of most computers and many calculators, used to produce a sequence of random numbers (they lack any pattern and the outcome is unpredictable). It does not produce real random numbers but pseudo-random numbers for checking and comparisons and used, for example, in computer games and generating data encryption keys. There are many methods for random number generation, for example, congruential methods by taking remainders modulo a large prime; physical methods by periodically sampling some physical quantity such as voltage; and from a probability distribution.

RANDOM NUMBER TABLE: A table made up of a sequence of randomly chosen digits from 0 to 9, each digit has an equal probability of appearing in a particular position and choices for different positions are independent.

RANDOM SAMPLE: The sample so selected that any portion of the population has an equal (or known) chance of being chosen. Haphazard or arbitrary choice of units is generally insufficient to guarantee randomness.

RANDOM UNCERTAINTY: An uncertainty produced by unknown and unpredictable variations in the experimental situation, such as temperature fluctuations and estimations when reading instruments. This affects the precision of results and can be reduced by taking repeated trials but not eliminated.

RANDOM VARIABLE (CHANCE VARIABLE; STOCHASTIC VARIABLE): A variable whose value is determined by the result of a random experiment and may take any range of values, which are either continuous or discrete that cannot be predicted with certainty but only described probabilistically. It may also be defined as a measurable function defined on a probability space having range in the interval $[0, 1]$.

RANDOM VECTOR: An n -tuple of random variables, whose values depend on the outcome of experiments. In other words, it is an ordered set of n random variables that often represents the outcomes

of a repeated experiment. For example, the outcome of the experiment of throwing a dice six times can be described by the vector $\{x_1, x_2, x_3, x_4, x_5, x_6\}$, where each x_i is a uniform distribution on the set $\{1, 2, 3, 4, 5, 6\}$.

RANDOM WALK: 1. In statistics, a route taken by a point or quantity, consisting of successive and unconnected steps in which each step is randomly determined without any influence from any previous step. It is applicable to various processes, such as diffusion and the related problem of Brownian motion and also in problems involving the structures of polymers and disordered solids. It can also be used to describe the path traced by a molecule as it travels in a liquid or a gas, the search path of a foraging animal and the price of a fluctuating stock. The progress of the object is determined by the net distance DN , travelled in N steps. The average distance travelled away from the starting position, root-mean-square value, $Drms$, is given by $\sqrt{N}$. 2. A mobility model used in mobile adhoc networks to model the movement of mobile nodes in an area of interest.

RANDOMIZE: To select or order data or individuals in a deliberate way, usually to enhance the reliability of any statistical results obtained.

RANDOMLY AMPLIFIED POLYMORPHIC DNA (RAPD): A polymerase chain reaction, used to amplify segments of an organism's DNA in which the segments of DNA that are amplified are random. Short primers (8-12 nucleotides) effectively select all regions of DNA that happen to lie within inverted copies of the primer sequence. RAPD has been used to characterise and trace the phylogeny of diverse plant and animal species.

RANDOMNESS: The lack of accuracy in a pattern or order in a physical system or a set of numbers.

RANEY NICKEL: A spongy form of nickel, made by treating aluminium-nickel alloy with concentrated sodium hydroxide (NaOH) which dissolves most of the aluminium out of the alloy. It is used as a catalyst in hydrogenation.

RANGE: 1. In mathematics, it is the set of all values that can be taken by a function or a variable. That is, it is the set of all the images of the domain. Also, a set of points such as P, Q, R, S lying on a straight line. 2. In statistics, it is the difference between the smallest and the largest value in a distribution. 3. An extensive area of open land on which farm animals can graze. 4. An area in which a species of organism normally lives or grows. 5. In geography, a **mountain range** is a number of mountains or hills forming a connected row or group, which extends over several kilometres. The Andes Mountains is the longest mountain range in the world, covering 7242 km, stretching along the western coast of South America. The Himalayas is the highest mountain range in the world and contains Mount Everest, the tallest mountain in the world with a height of 8850 metres, located on the border of Nepal and Tibet, China.

RANGE OF DATA: The difference between the highest and the lowest data points in a given data set.

RANGE OF FUNCTION: The complete set of all possible outputs (values of the dependent variable, usually y -values) of a function, given a specified domain for the function.

RANGE OF SIGNIFICANCE (DOMAIN OF DEFINITION): The set of subjects for which a given predicate is intelligible.

RANGEFINDER: An optical instrument used to determine the distance of an object from the observer to a target point. It is used to focus a camera or to sight a gun accurately. A rangefinder has two windows spaced apart. The distance between the windows is called the **base length**. In a rangefinder camera, a rotating mirror or prism alters the image seen through the secondary window. The rangefinder operates as an angle-measuring device for solving the triangle comprising the rangefinder base length and the line from each window to the target point.

RANGELAND: A collection of native grasses and shrubs that cover an arid or semi-arid area, occupying between 40–50 percent of the land area of the Earth.. Rangeland can include ecosystems such as forests, woodlands, savannahs, tundra, marshes and wetlands. Much

of these rangelands are unsuitable for land uses such as cultivating crops due to soil quality and low rainfall levels. Rangelands are largely used for livestock grazing or reserved for a conservation program.

RANGING POLE: A pole usually held vertically, used to mark a specific position when surveying a plot of land.

RANITIDINE (HYDROCHLORIDE; ZANTAC; ZINETAC): The name for an anti-ulcer drug with the chemical formula $C_{13}H_{22}N_4O_3S$. It is a histamine H_2 -receptor antagonist that inhibits stomach acid production. It is used to treat peptic ulcers and relieve heart burns.

RANK: 1. Also known as **category**, it is a position or status of a taxon in a classification hierarchy. Examples of ranks are the class order, family, genus and species. 2. In biology, leaves are said to be in a rank when they are arranged into two vertical rows. 3. In mathematics, it is the greatest number of linearly independent rows or columns in a matrix; the number of rows or columns in the highest order non-zero minor of the matrix. Also, it is the rank of the matrix associated with a given quadratic form. 4. The number of elements in the basis of a free module. 5. In statistics, to array a set of objects as a sequence, especially in terms of the natural arithmetic ordering of some measure of their elements. For example, one may rank students by their test scores, by their dates of birth or by their heights.

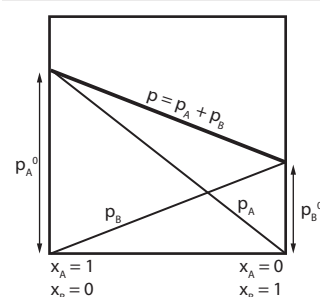
RANKINE CYCLE: A cycle of operations in a heat engine consisting of heat addition at constant pressure, isentropic expansion, heat rejection at constant pressure and isentropic compression. It is used as an ideal standard for the performance of heat-engine and heat-pump installations such as a steam power plant operating with a condensable vapour as the working fluid.

RANKINE SCALE: A temperature scale expressing absolute temperature in Fahrenheit degrees. It was devised by the Scottish engineer William Rankine (1820–1872). For Fahrenheit readings above zero, 460° is added. Thus, 72°F equals 460° plus 72° or 532° absolute (532°R). If the Fahrenheit reading is below zero, it is subtracted from 460° . Thus, -40°F equals 460° minus 40° or 420° absolute (420°R).

RANKSITE: A mineral with the chemical formula $2\text{Na}_2\text{CO}_3 \cdot 9\text{Na}_2\text{SO}_4 \cdot \text{KCl}$ consisting of mixed sodium carbonate, sodium sulfate and potassium chloride.

RAOULT'S LAW: A law established by François-Marie Raoult in 1882,

which states that the partial vapour pressure of a solution is proportional to its chemical component and the mole fraction. A solution that obeys Raoult's law is said to be an **ideal solution**. Generally, the law holds only for mixtures of liquids such as benzene and toluene that are very similar in chemical structure and described as perfect solutions. Some mixtures of liquids conform to



the law over a whole range of concentrations. Deviations in the law occur in most cases making one component only slightly soluble in the other and its volatility is abnormally increased. For example, the volatility of alcohol in dilute aqueous solution is several times as great as predicted by Raoult's law. For a mixture of two solutions A and B with partial vapour pressures P_A and P_B respectively and with respective mole fractions X_A and X_B , the total pressure above the mixture according to Raoult's law is $P = P_A X_A + P_B X_B$. Since $X_A = 1 - X_B$, the equation can also be written as: $P = P_A X_A + (1 - X_A)P_B = (P_A - P_B) X_A + P_B$ etc. Diagrammatically, the equation is as represented in the illustration.

RAPHE: 1. A line or ridge visible on a seed coat which marks the fusion between an ovule wall and the funicle, as found in castor oil fruit. 2. A groove on certain diatom valves. 3. A groove, ridge or line which marks the fusion between two similar parts of an organ, such as the scrotum or separating structures.

RAPHIDE: A needle-shaped crystal, often of calcium oxalate, found in clusters in plant cells. The presence and form of raphide clusters are sometimes useful as diagnostic characters.

RAPID ACCESS COMPUTING ENVIRONMENT (RACE): A software service platform by the United States Department of Defense for developing and testing of software applications.

RAPID CLIMATE CHANGE: Sudden unexpected changes in the climate due to non-linearity of the climate system, sometimes called **abrupt events**. Some such abrupt events may be imaginable, such as a dramatic reorganisation of the thermohaline circulation, rapid deglaciation, or massive melting of permafrost leading to fast changes in the carbon cycle. Others may be truly unexpected, as a consequence of a strong, rapidly changing, forcing of a non-linear system.

RAPID-EYE-MOVEMENT SLEEP (REM SLEEP): One of the stages of sleep that recurs several times in most humans, characterised by rapid eye movements beneath closed lids and is associated with dreaming. It is also characterised by irregular breathing and heart rate; and involuntary muscle jerks. REM occurs at intervals of between 1-2 hours and may last for about 5 minutes or over an hour.

RAPIFORMIS: Shaped like a turnip.

RAPTOR: 1. A bird of prey, such as an eagle or hawk. 2. A small or medium-sized dinosaur that ate other animals.

RARE GASES (NOBLE GASES; INERT GASES): Any of the seven chemical elements that make up Group 18 (VIIIA) of the Periodic Table. They are extremely unreactive, gaseous, colourless, odourless, nonflammable, monoatomic and stable elements occurring in the atmosphere in small quantities. They contain eight electrons in their stable electronic configuration. They include helium, neon, argon, krypton, xenon and radon.

RARE-EARTH ELEMENTS: Series of elements in the Periodic Table, with atomic numbers of between 57 and 71. They include lanthanum, cerium, praseodymium, neodymium, promethium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium and lutetium. Sometimes yttrium (atomic number 39) and scandium (atomic number 21) are included in the group. Cerium (58), with the symbol Ce, is the most abundant and Lutetium (71), with the symbol Lu, the rarest.

RAREFACTION: A reduction in the pressure or density of a fluid or gas.

RAS PROTEIN: A small G protein that controls cell growth and other activities. RAS protein acts as a molecular switch in signalling transduction pathways and turns itself off by rapidly hydrolysing nucleotide guanosine triphosphate (GTP) to guanosine diphosphate (GDP) and returning to its inactive state. If RAS is left in its active form as GTP-bound and not converted to GDP, it can lead to excessive cell growth, which can cause cancer. Activation of RAS is controlled by a guanine-nucleotide exchange factor (GEF).

RASCHIG PROCESS: An industrial process for making chlorobenzene and phenol by a gas-phase reaction between benzene vapour, hydrogen chloride and oxygen (air) at 230 °C (503 K):
 $2\text{C}_6\text{H}_6 + 2\text{HCl} + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + 2\text{C}_6\text{H}_5\text{Cl}$. The catalyst is copper(II) chloride. The chlorobenzene is mainly used for making phenol by the reaction $\text{C}_6\text{H}_5\text{Cl} + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{C}_6\text{H}_5\text{OH}$. This reaction proceeds at 430 °C (703 K) with a silicon catalyst.

RASMOL: A computer program originally developed by Roger Sayle in the early 90s commonly used for visualising and exploration of biological macromolecule molecules. It is particularly good at viewing and rotating protein molecules, although it also works perfectly with smaller molecules.

RASTER: The pattern of horizontal lines called scan lines of a television; or a computer monitor that provides a visual display.

RASTER GRAPHICS: One of the two main formats in which graphical images are stored and transmitted; the other being vector graphics. In raster graphics, the images are stored as a bit map consisting of grids of small, independently controlled dots or pixels arranged in rows and columns in a computer's memory, which can be viewed through

any display medium such as a monitor. Raster graphics include TIFF (Tagged Image File Format), JPEG (Joint Photographic Experts Group) and PNG (Portable Network Graphics), which are typically used for painting applications, which allow the user to create artwork on a computer screen much as if they were painting on paper or canvas.

RASTER IMAGE PROCESSOR (RIP): A program in a laser printer or other high-resolution printer that converts images described in the form of vector graphics statements into raster graphics images or bitmaps. A separate program is required for each type of printer and for each page description language such as Post Script.

RAT: 1. Common name for any large member of the mammalian order of rodents (order Rodentia, family Muridae) of the of the genera *Rattus*, *Praomys*, *Mastomys*, *Arvicanthis*, *Lophuromys*, etc., with dull-coloured, coarse fur, long tail, large ears and pointed snout. It is highly destructive and a carrier of many diseases such as bubonic plague. The common house rat or black rat is *Rattus rattus* and the brown rat is *Rattus norvegicus*. 2. Acronym for **Remote Access Trojan**, a malware program usually downloaded by a user that is invisible to the individual when downloaded. This gives the intruder administrative control over the target computer.

RATCHET: A mechanism used especially in lifting devices and some hand tools consisting of a metal wheel operating with a catch that transmits motion or permits a shaft to rotate in only one direction. Its uses include in socket wrench or spanner handles for loosening and tightening bolts and nuts and in mechanical jacks.

RATE: 1. A measurement, quantity, or frequency, typically one measured against another quantity or measure. 2. A fixed price or payment made for something over a period of time, expressed as a percentage of another amount, or as a basis of calculation. 3. The occurrence of events over a specific interval in time. 4. The change in a quantity with respect to time.

RATE CONSTANT (SPECIFIC RATE CONSTANT): The mathematical constant of proportionality for a chemical reaction at a particular temperature that relates the concentrations of the reactants in a chemical reaction.

RATE EQUATION (RATE LAW): An equation that relates the rate of a chemical reaction to the concentrations or pressures of reactants and constant parameters of the individual reactants. Rate equation can be expressed as: $k[A]^m[B]^n$, where k is the rate constant, A and B , the concentration of the reactant and m and n the order of reaction.

RATE LAW EXPRESSION: The mathematical expression relating the rate of reaction to the rate constant and the concentration of the reactants.

RATE OF CHANGE: The rate at which one quantity changes with respect to another variable, for example, speed. In other words, it is the ratio of the difference between values of a variable quantity at different times to the difference between the times. The rate of change in distance with respect to time may be expressed per unit time as $\frac{f(t) - f(t_0)}{t - t_0}$. If the quantity is continuously variable and this

ratio tends to a limit as both endpoints of the time interval tend to the same point, the instantaneous rate of change is the derivative of the function with respect to time, at the point that represents that quantity. Any derivative dy/dx is the rate of change of y with respect to x .

RATE OF CONVERGENCE (ORDER OF CONVERGENCE): The speed at which a convergent sequence approaches its limit. It is often measured by the number of terms or evaluations involved in obtaining a given accuracy. Convergence of a sequence subject to the condition, for $p > 1$ that $\frac{x_{n+1} - x}{x - x^p} = O(1)$ as n increases is called p^{th} -order convergence. An example is quadratic convergence when $p = 2$.

RATE OF COOLING: The rate of fall in temperature, which equals the slope of a temperature-time graph. It is not synonymous with the rate of heat loss.

RATE OF MIGRATION: The distance of migration divided by time, sometimes called velocity of migration. Its symbol v is also used for velocity.

RATE OF NUCLEATION: The number of nuclei formed in unit time per unit volume of the liquid phase.

RATE OF REACTION: The speed at which a chemical reaction takes place in a reactant. For example, in the oxidation of iron under atmospheric conditions, the rate of reaction may take several years, whilst the combustion of benzene takes a very short time (seconds). It is usually expressed in terms of the concentration of a reactant consumed or product formed in a unit time.

RATE TABLE: A display of rate information. In a rate table, the fractions formed by the two numbers in each column are equivalent fractions.

RATE UNIT: A compound unit such as kilometres per hour, dollars per kilogramme, and words per minute for a rate.

RATE-DETERMINING STEP (RDS; RATE LIMITING STEP; RATE-CONTROLLING STEP): The slowest step in a mechanism, which is the step that determines the overall rate of reaction. For example, in pouring a liquid into a bottle through a funnel, the rate of flow is determined by the width of the neck and not by the speed at which water is poured into the bottle. RDS, therefore is the width of the neck of the funnel.

RATEMETER: An instrument used to measure the count rate of a radioactive source. It gives the number of particles emitted from the source and captured by a detector in a unit time.

RATIO: A measure of the relative size of two quantities a to b expressed as a proportion, which can be written as $a:b$ or as the fraction $\frac{a}{b}$. For example, if the population of boys in a class is 30 and that of girls is 20, the ratio of boys to girls is 30 to 20, written as 3:2 or $\frac{3}{2}$.

RATIO OF SIMILITUDE (SIMILITUDE): In mathematics, it is the ratio of proportionality between similar figures. In other words, it is the ratio of the sides of one figure to the sides of the other figure.

RATIO SCALE: In statistics, it is a scale of measurement of data that has a fixed zero value and allows the comparison of differences. For example, although time itself cannot be measured on a ratio scale, differences in time can be, since it makes sense to talk of one pair of events being twice as far apart as another.

RATIO TEST: 1. Also known as **Cauchy's ratio test**, it is a test for a complex series $\sum_{n=1}^{\infty} a_n$ to determine whether the series is absolutely convergent by testing the limit of the ratio of the absolute values of successive terms. If the term $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ is less than unity, the series

converges absolutely; if it is greater than unity, it diverges. 2. Also known as **d'Alembert's ratio test** or **generalised ratio test**, it is a test of convergence for an infinite series by testing whether there is a $k < 1$ such that the ratio of the absolute values of its predecessors are less than k , for all terms after some terms; if this ratio is always greater than unity, then the series diverges.

RATIO THEOREM (SECTION FORMULA): The theorem that if a point P divides a directed line segment, AB, in the ratio $m : n$, then the position vector, p , of P, can be expressed in terms of A and B as $p = \frac{ma + nb}{m + n}$.

RATION: A fixed amount of food given to an animal or person per day.

RATION FORMULATION: The process of putting together different amounts of feed ingredients to be included in a ration to provide the amount of nutrients required by an animal.

RATIONAL: 1. In mathematics, expressing as a ratio of two integers involving or consisting of rational numbers. It can be written as a fraction with both the numerator and the denominator being integers. 2. A rational expression, equation or function is one containing no variable either in an irreducible radical form or raised to a fractional power, it can be expressed as a ratio of polynomials.

RATIONAL ELEMENT: A Dedekind cut corresponding to a rational number in Dedekind's construction of the real numbers.

RATIONAL EXPONENTS: For any non zero number b and any integer m and n , with $n > 1$, $b^{\frac{m}{n}} = (\sqrt[n]{b})^m$ except $\sqrt[n]{b}$ is not a real number.

RATIONAL FORM: A square matrix in which the only non-zero entries are in submatrices arranged down the main diagonal, that are companion matrices of non-scalar monic polynomials such that each polynomial divides the next. It can also be defined as a quotient of two polynomials with integral coefficients.

RATIONAL FUNCTION: A function of the form $\frac{f(x)}{g(x)}$, $g(x) \neq 0$, where $f(x)$ and $g(x)$ are polynomial functions.

RATIONAL NUMBERS: Any number that is the result of dividing one integer by another. For example, the numbers, 5 (5/1) and -1.4 (-7/5) are rational numbers because they are quotients (results of division) of two integers. Rational numbers are a subset of the real numbers, which also include the set of irrational numbers. Rational numbers include all positive and negative integers as well as all fractions and decimals. In the decimal form they can have a terminating point such as 2 or 2.5 or repeating decimals (e.g. 2.333...). **Irrational numbers** are numbers such as pi (π), the square root of two ($\sqrt{2}$) and the mathematical constant e that are not the quotient of any two integers.

RATIONAL POLYNOMIAL: Any ratio of two polynomials.

RATIONAL ROOT THEOREM: It states that, if the polynomial equation $a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0 = 0$, where $a_0, a_1, \dots, a_n$ are integers, has any rational roots, then each rational root can be expressed as a fraction in which the numerator is a factor of a_0 and the denominator is a factor of a_n . This theorem sometimes makes it easier to find the roots of complicated polynomial equations but provides no help if there are no rational roots to begin with.

RATIONALISE (RATIONALIZE): 1. The process of eliminating radicals from an irrational fraction, algebraic expression or equation without changing its value or roots so that the denominator contains only rational numbers. For example, the fraction $\frac{1}{\sqrt{a} - \sqrt{b}}$ may have

its denominator rationalised by multiplying the top and bottom by $\sqrt{a} + \sqrt{b}$ so that the denominator contains only rational numbers as

$$\frac{1}{\sqrt{a} - \sqrt{b}} \times \frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} + \sqrt{b}}{a - b}$$

2. To structure and run according to rational or scientific principles in order to achieve desired results. 3. To attempt to explain or justify (behaviour or an attitude) with logical reasons, even if these are not appropriate.

RATIONALISED UNIT (RATIONALIZED UNIT): A unit for which the defining equation conforms to the geometry of the system. Equations involving circular symmetry contain the factor 2π , those involving spherical symmetry 4π .

RATITAE (PALAEOGNATHAE; RATITES): Any of flightless species of birds including the ostrich (*Struthio camelus*), rhea (*Rhea*), kiwi (*Apteryx*), cassowary (*Casuarus*) and emu (*Dromaius*). They are characterised by long legs, heavy bones, small wings, a flat sternum or breastbone and curly feathers. The sternum does not have the ridge-shaped part known as **keel** or **carina** to which flight muscles can be anchored, thus preventing flight.

RATOON: The regrowth of a crop after it has been harvested once. Examples are the sugar cane and banana plants which are cut leaving

the roots to mature and produce other crops and can be harvested many times.

RATTLE: 1. The sound-producing organ on a rattlesnake's tail, consisting of shed skin, used to deter predators. 2. A device that makes a rattling sound (such as shaking a box filled with pebbles) that is put on an animal so its location can be heard traced.

RAUNKIAERIAN SYSTEM: A system of plant classification in which plants are grouped according to the position of their perennating buds, in relation to the soil surface.

RAUWOLFIA (RAUVOLFIA): A genus of evergreen flowering shrub of between 3-6 m high; or a small tree, seldom more than 10 m high, with whorled branches and branchlets, rich in white bitter-tasting latex. Some of the species include *Rauvolfia serpentina*, *Rauvolfia tetraphylla*, *Rauvolfia canescens* L., *Rauvolfia vomitoria*, etc. *Rauvolfia serpentina*, commonly known as Indian snakeroot or serpentine root is cultivated on commercial scale in India, Sri Lanka, America and Thailand. The bark is greyish-brown and the branchlets are covered with white lenticels. The leaves occur in whorls of 3-5, mostly 4, variable in shape but mostly elliptic to slightly oblanceolate, acuminate with a narrowly triangular acute acumen, narrowly cuneate at base, with 8-16 pairs of lateral nerves. The various alkaloids found in *Rauwolfia* are ajmaline, ajmalicine, reserpine, rescinnamine, serpentine and serpentinine. Reserpine and deserpidine belong to the general class of medicines called antihypertensives and is used to treat high blood pressure or hypertension. Ajmalicine is helpful in fighting against circulatory diseases. Traditionally, the freshly pounded roots mixed with lemon juice and clay is reputed to be an excellent medicine against psoriasis.

RAW DATA: Data that has not been processed or ordered or arranged in any way. Example is text that is not stored in any proprietary format.

RAW FILE: A collection of unprocessed data or a file that has not been altered, compressed or manipulated in any way by the computer. Raw files are often used as data files by software programs that load and process the data. An example of raw file is "Camera RAW," which is generated by a digital camera. Instead of processing the image captured by the camera, the data is left unprocessed and uncompressed until it is opened with a computer program.

RAY: 1. In physics, it is a narrow beam of radiating light from the Sun or an artificial light source. 2. In mathematics, it is a straight line from a point extending infinitely in one direction. Also, it defines any one-dimensional subspace of a finite-dimensional vector space over the given field. 3. In botany, it refers to the following: (i) the radial line of parenchyma cells in the secondary xylem and phloem, derived from the ray initials of the vascular cambium. The ray system of gymnosperms often also contains tracheids, distinguished from the ray parenchyma by their bordered pits and lack of protoplast; (ii) the portion of a lingulate flower that has a flat, strap-shaped corolla or the lingulate itself, as seen in composite flowers. (iii) a branch of an umbel; (iv) a distinct strand of tissue running radially through the conducting tissues in the stem of a plant. 4. In astronomy, a bright streak that radiates from a crater on the Moon. 5. In zoology, it is an arm of a starfish or any other animal with body parts radiating from the centre; or a bony spine supporting the membrane of a fish's fin. Also, it refers to a type of cartilaginous mostly marine shark-like fish, also known as **skate**, with body dorso-ventrally compressed.

RAY DIAGRAM: A diagram used to locate the paths of selected rays through an optical system.

RAY FUNGUS: A group of prokaryotic organisms called **actinomycetes**, which are both fungi and bacteria found in soil. Actinomycetes produce many antibiotic compounds including streptomycin, actinomycin and chloromycin. Actinomycetes cause a disease called **actinomycosis**. It is an inflammatory bacterial disease of cattle and other animals, sometimes caught by humans. It resembles a fungus and causes swelling of the abdomen, chest and especially the jaw.

RAY-FINNED: A group of bony fish such as trout, tuna, salmon, goldfish, cod, seahorse and bass, belonging to the class Actinopterygii, with fins supported by long rays of bone. They make up about 50% of aquatic vertebrate species and about 96% of all known fish species. They are found in fresh water and marine habitats of all conditions.

RAYLEIGH CRITERION: The criterion for spontaneous pressure oscillations to accompany combustion, namely, that combustion progresses more rapidly or efficiently during the compression phase of the pressure oscillation than during the rarefaction phase.

RAYLEIGH DISTRIBUTION: The distribution of the distance between a point and its nearest neighbour in a spatial Poisson process. Also, the distribution of the distance from the origin in n -dimensional space to the point $(x_1, x_2, \dots, x_n)$, where $x_1, x_2, \dots, x_n$ are independent normal variables, each with expectation 0 and variance σ^2 .

RAYLEIGH SCATTERING: Named after the British physicist Lord Rayleigh (1842-1910), it is the scattering of electromagnetic radiation by particles much smaller than the wavelength of the light and in which there is no change in frequency of the scattered radiations from the original beam of radiation. It is mostly observed in gases, but can also occur when light permeates solids and liquids. An example is the blue colour of the sky which occurs due to the scattering of sunlight by molecules in the Earth's atmosphere.

RAYLEIGH QUOTIENT: The ratio $r_A(x) = \frac{\langle x, Ax \rangle}{\langle x, x \rangle}$ of which, for a positive definite matrix A , the supremum produces the maximum eigenvalue and the infimum produces the minimum eigenvalue.

RAYLEIGH-JEANS FORMULA (RAYLEIGH-JEANS LAW): A law giving the intensity of radiation emitted by a blackbody within a narrow band of wavelengths. It states that this intensity is proportional to the temperature divided by the fourth power of the wavelength. It is a good approximation to the experimentally verified Planck radiation formula only at long wavelengths.

RAYLEIGH-RITZ METHOD: A method of obtaining approximate solutions to functional equations or variational problems by replacing the functions by appropriate finite combinations of basic elements and finding the minimum solution within this class.

RAYLEIGH-TAYLOR INSTABILITY: A type of fluid instability that occurs any time a dense, heavy fluid is being accelerated by light fluid. Rayleigh-Taylor instabilities are common in supernova remnants, such as the Crab Nebula, in which hot gas from the explosion is ramming into the surrounding interstellar medium, giving rise to the familiar clumpy appearance of material in these objects.

RAYLEIGH, JOHN WILLIAM STRUTT (1842-1910): British physicist, educated at the University of Cambridge and professor of experimental physics and director of the Cavendish Laboratory at Cambridge from 1879 to 1884. In 1887-1905, he was professor of natural philosophy at the Royal Institution in London. In 1908 he became the chancellor of University of Cambridge. Rayleigh researched into electromagnetic radiation, acoustics, electricity and optics. He collaborated with Sir William Ramsay to discover argon in 1894, for which he was awarded the Nobel Prize for Physics in 1904.

RAYNAUD'S DISEASE: A chronic condition in which the blood supply to the skin is reduced due to the narrowing of the smaller arteries causing some areas of the body such as the fingers and toes to have pallor, pain, numbness and sometimes, gangrene. This is caused by constant exposure to cold, stress or vibration.

RAYON: A silky artificial fibre produced from naturally occurring polymers such as cellulose obtained from surplus cotton fibres or from pulp and used in textiles.

RCA CABLE (COMPOSITE CABLE): An audio/video cable with yellow, red and white connectors on each end that allows the colour video information from one device such as VCR, DVD player, gaming system or other audio/video devices to be linked to a monitor or

television. The yellow connector is for video and the red & white connectors are for audio: red for the right speaker, white for the left speaker.

RE-ENTRANT ANGLE: In an irregular polygon, it is an interior angle that is greater than 180° .

RE-ENTRANT POLYGON: A concave three dimensional figure with many sides and an internal angle greater than 180° .

REABSORPTION (TUBULAR REABSORPTION): The return to the blood of most of the glomerular filtrate such as water, sodium, amino acids and glucose that were removed from the glomerulus in the Bowman's capsule during filtration. The glomerular filtrate including water, sodium, amino acids and glucose in liquid form moves from the Bowman's capsule through the renal tubule and are reabsorbed by the network of blood vessels surrounding the tubule.

REACHABLE SET (REACHABLE POINTS): 1. In mathematics, the set of state values that can be attained at a given time in a problem in control theory. 2. The set of vertices s that can be connected to a given vertex t in a digraph if there exists a sequence of adjacent vertices or a path which starts with s and ends with t .

REACTANCE: The opposition to the flow of alternating current by either a inductor (L) and capacitor (C) in a circuit, measured in ohms. There are two types of reactance: **inductive reactance (X_L)**, resulting from the changing magnetic field around a current carrying conductor; and **capacitive (X_C)**, which results from the changing electric field between two conducting plates separated by an insulator such as a capacitor. $X_L = 2\pi fL$ and $X_C = \frac{1}{2\pi fC}$.

REACTANT: A substance that directly reacts with another in a chemical reaction, for example in the equation $A + B \rightarrow C$, the reactants are A and B.

REACTING BOND RULES: 1. For an internal motion of a molecular entity corresponding to progress over a transition state (energy maximum), any change that makes the motion more difficult will lead to a new molecular geometry at the energy maximum, in which the motion has proceeded further. Changes that make the motion less difficult will have the opposite effect. This rule corresponds to the Hammond principle. 2. For an internal motion of a molecular entity that corresponds to a vibration, any change that tends to modify the equilibrium point of the vibration in a particular direction will actually shift the equilibrium in that direction. 3. Effects on reacting bonds (bonds made or broken in the reaction) are the most significant. The bonds nearest the site of structural change are those most strongly affected.

REACTION: 1. A transformation in which one or more chemical elements or compounds decompose, combine or react with the chemical components of other substances to form new compounds with different chemical and physical properties. 2. A response to some treatment, situation or stimulus. 3. The force that a body subjected to the action of a force from another body exerts in the opposite direction. 4. A **nuclear reaction** is a reaction, as in fission, fusion or radioactive decay, that alters the energy, composition or structure of an atomic nucleus. 5. Any effect produced by the action of a drug, especially an adverse effect or any effect produced by a substance (allergen) to which a person is allergic.

REACTION BARRIER: The energy barrier to chemical reaction. In vibrationally adiabatic transition-state theory its height is the zero-point energy of the activated complex minus that of the reactants.

REACTION CENTRE (PHOTOSYNTHETIC REACTION CENTRE): The site of the light reactions of photosynthesis in the chloroplast that receives the energy trapped by chlorophyll and accessory pigments and initiates the electron transfer process for higher energy level within the pigment.

REACTION CHROMATOGRAPHY: A technique in which the identities of the sample components are intentionally changed between sample introduction and detection. The reaction can take

place upstream of the column when the chemical identity of the individual components passing through the column differs from that of the original sample or between the column and the detector when the original sample components are separated in the column but their identity is changed prior to entering the detection device.

REACTION DYNAMICS (MOLECULAR DYNAMICS): A branch of chemical kinetics that is concerned with the intermolecular and intramolecular motions that occur in the elementary act of chemical change and with the details of the relationships between the quantum states of the reactant molecules and those of the product molecules.

REACTION FORCE: The equal and opposite force produced by an object in reaction to a force (action force) applied on it by another object. For example, if object A applies a force called **action force** to push object B, object B counters by producing a force called **reaction force** of the same magnitude but opposite in direction by the one produced by object A.

REACTION MECHANISM: The sequence of elementary reactions that lead to chemical transformation or a series of elementary reactions that occur in the course of a complex reaction.

REACTION NORM: The pattern of expression of a given phenotype of a single genotype across a range of environmental variable. Species of the same type raised in the same environment will produce varying norm of reaction to environmental variables such as the application of fertiliser.

REACTION ORDER (ORDER OF REACTION): The classification of chemical reactions based on the power to which the concentration of components of the reaction is raised in the rate law. The overall order is the sum of the powers of the concentration. For example, given a chemical reaction $2A + B \rightarrow C$ with a rate equation $r = k[A]^2[B]$ the reaction order with respect to A would be 2 and with respect to B would be 1. The total reaction order would be $2 + 1 = 3$. The order of reaction can be zero or even fractional. Reaction orders can be determined only by experiment.

REACTION PROPULSION: Propulsion by means of reaction to a jet of gas or fluid projected rearward, in which a jet nozzle converts potential energy of the working fluid into kinetic energy. Hot high-pressure gas escapes through the nozzle, expanding in volume as it drops in pressure and temperature, thus gaining rearward velocity and momentum. This process is governed by the laws of conservation of mass, energy and momentum and by the pressure-volume-temperature relationships of the gas-state equation. Reaction propulsion is used in turbojet, ramjet and rocket engines.

REACTION QUOTIENT (Q_r): The function of the activities of any set of conditions involved in a chemical reaction that is not in equilibrium.

REACTION STAGE: A set of one or more (possibly experimentally inseparable) reaction steps leading to and/or from a detectable or presumed reaction intermediate.

REACTION STEP: An elementary reaction, constituting one of the stages of a stepwise reaction in which a reaction intermediate (or, for the first step, the reactants) is converted into the next reaction intermediate (or, for the last step, the products) in the sequence of intermediates between reactants and products.

REACTION STOICHIOMETRY: It describes the quantitative relationships in substances as they undergo chemical reactions.

REACTION TIME: 1. Also known as **latent period**, it is the period of time that elapses between the detection of a sensory stimulus and the appropriate behavioural response by the affected organ. This delay is caused by the time taken for the impulse to travel across the synapses of adjacent neurons and is shorter when there is only one possible response but slows down when there are other response options. RT time is shorter in young people and declines with age, affected by factors such as fatigue and lack of concentration. 2. The period of time that elapses between the start of the reaction and the attainment of a given extent of a reaction.

REACTION WOOD: Structurally abnormal wood formed in response to various stresses. In gymnosperms it is termed **compression wood** and tends to form on the lower side of branches. Compression wood is dense in structure due to the heavy lignification of tracheid walls. Tension wood forms in angiosperms on the upper side of the branches and the fibres tend to be gelatinous rather than lignified.

REACTIVE (REACTIVITY): As applied to a chemical species, the term expresses a kinetic property. A species is said to be more reactive or to have a higher reactivity in some given context than some other (reference) species if it has a larger rate constant for a specified elementary reaction. The term has meaning only by reference to some explicitly stated or implicitly assumed set of conditions. It is not to be used for reactions or reaction patterns of compounds in general. The term is also more loosely used as a phenomenological description not restricted to elementary reactions. When applied in this sense the property under consideration may reflect not only rate, but also equilibrium, constants.

REACTIVE BLENDING: Mixing process that is accompanied by the chemical reaction(s) of components of a polymer mixture. Examples of reactive blending are: (i) blending accompanied by the formation of a polymer complex; (ii) the formation of block or graft copolymers by a combination of radicals formed by the mechano-chemical scission of polymers during blending. Reactive blending may also be carried out as reactive extrusion or reaction injection moulding.

REACTIVE COMPLEX: A species of very short life that occurs as an intermediate in a chemical reaction. An activated complex (or transition state) is a special case.

REACTIVE DYE: A method of dyeing introduced in 1956, which forms stable chemical links with textile materials to produce coloured fabrics with excellent overall fastness. Pigments are finely ground particles that can be bonded to fibres by means of synthetic resin binders. Pigments can be used to dye cotton, rayon and nylon fibres with good fastness to light, acids and alkalis. They are particularly useful for coating glass fibre, which, because of its inert nature, can be difficult to dye. Molten glass can also be coloured with metallic salts before it is spun into fibres.

REACTIVE OXYGEN SPECIES (ROS): Any of various highly reactive molecules that contain oxygen. Examples of ROS are superoxide anion (O_2^-) and peroxides. ROS is produced normally as a by-product of aerobic respiration inside mitochondria and play important roles in cell signalling. ROS levels can increase significantly in conditions such as exposure to heat, UV and some other environmental conditions, causing damage to living cells because of their highly reactive nature.

REACTIVE POLYMER: Polymer having reactive functional groups that can undergo chemical transformation under the conditions required for a given reaction or application.

REACTIVE SCATTERING: If in a molecular-beam experiment a chemical reaction occurs there is said to be reactive scattering.

REACTIVITY INDEX: Any numerical index derived from quantum mechanical model calculations that permits the prediction of relative reactivities of different molecular sites. Many indices are in use, based on a variety of theories and relating to various types of reaction. The more successful applications have been to the substitution reactions of conjugated systems where relative reactivities are determined largely by changes of π -electron energy.

REACTIVITY SERIES (ACTIVITY SERIES; ELECTROCHEMICAL SERIES): A series of chemical elements arranged in descending order of their chemical reactivity or electrode potentials, with the most active at the top and the less reactive or more noble metals at the bottom. The series shows the order in which metals replace one another from their salts. For example, electropositive metals will replace hydrogen in acids; zinc will displace hydrogen from hydrogen chloride. Basically, such an activity series need not be limited to the metals but may be carried on through the electronegative or nonmetallic elements as well.

(refer to page 25 for table).

REACTOR: 1. In electricity, it is a device such as inductor or capacitor that introduces reactance into an electrical circuit. Inductive reactance x is a function of the product of frequency f and inductance L ; thus, $x = 2\pi fL$. For this reason, a reactor is also called an **inductor**. Since a voltage drop across a reactor increases with frequency of applied currents, a reactor is sometimes called a **choke**. 2. In nuclear physics, it is also called **nuclear reactor** or **atomic reactor**, it is a device in which heat is generated from a controlled nuclear chain reaction, with the resulting heat typically used for power generation and the neutrons and fission products used for military, experimental and medical purposes. 3. In chemical engineering, it is a device or process vessel in which chemical reactions take place during a chemical conversion type of process.

READ: 1. In computing, it refers to accessing programme instructions or data from a computer storage device such as a disk drive into its main memory. 2. To decode or translate sections of messenger RNA into amino acid sequences and to assemble them into a polypeptide chain.

READ ERROR: In computing, it refers to a failure that occurs when the computer or device is attempting to read.

READ HEAD: Electromagnetic device that converts digital information stored on a magnetic tape, disk or drum into electrical signals usable by devices such as a tape recorder, video recorder or computer.

READ-ONLY MEMORY (ROM): A memory chip from which data can only be read by the CPU. The CPU may not store data to this memory. The advantage of ROM over RAM is that ROM does not require power to retain its program. This advantage applies to all types of ROM chips; ROM, PROM, EPROM and EEPROM.

READ-WRITE HEAD: Electromagnetic part of a disk drive that transforms the recorded magnetic field into electrical current or electrical current into magnetic field functions as both a read head and write head.

READABILITY: The smallest fraction of a division to which the index scale can be read with ease either by estimation or by the use of a vernier caliper. It should normally be expressed in divisions of the pointer scale.

READING FRAME: A sequence of bases in the messenger RNA (mRNA) that encodes for an amino acid. Each coding unit of the genetic codes consists of a group of triplets and the reading frame is determined by the point of initiation of reading.

README: A text file containing information for the user about the software program, utility, game or other software program. README files commonly contain instructions and additional help about the software program.

READTHROUGH: A very important process of information decoding, which is the continuation of transcription of DNA beyond a stop signal, due to failure of RNA (ribonucleic acid) polymerase to recognise the termination signal. In translation of mRNA, it is a mutation which converts a normal stop codon into one encoding an amino acid. It results in the synthesis of an extended polypeptide chain until the next stop codon is reached.

READY RECKONER: A set of mathematical tables, especially one used for ready calculation of rates of interest, discount, etc. to different sums.

REAGENT: A substance or compound substance taking part in a chemical reaction, especially one used to detect, measure or prepare another substance.

REAL: In mathematics, involving or consisting of real numbers without any imaginary part. The term is also used to denote real numbers.

REAL ANALYSIS: The branch of mathematics that deals with real functions. In particular, it deals with the analytic properties of real functions and sequences, including convergence and limits of sequences of real numbers, the calculus of the real numbers and continuity, smoothness and related properties of real-valued functions.

REAL ANALYTIC: The real analytic of a real function is one possessing derivatives of all orders and agreeing with its Taylor series locally.

REAL AXIS: The x-axis or horizontal axis of an Euclidean or Argand diagram.

REAL FUNCTION (REAL-VALUED FUNCTION): A function f that takes only real values, such as the numerical trace of a self-adjoint matrix. In other words, it is any function that takes only real values or assigns a real number to each member of its domain.

REAL GAS (IMPERFECT GAS): A gas that does not behave or exhibit the properties of an ideal gas due to interactions between gas molecules. An ideal gas is known only in theory and is supposed to obey the gas laws exactly. An ideal gas would consist of molecules that occupy negligible space and have negligible forces between them. All collisions made between molecules and the walls of the container or between molecules and other molecules would be perfectly elastic, because the molecules would have no means of storing energy except as translational kinetic energy.

REAL IMAGE: An optical representation of an object produced by refraction or reflection or when rays actually meet. An example of real image is the image produced on a human retina. Real image can be transferred onto a surface such as the film inside a camera.

REAL LINE (REAL NUMBER LINE): The real numbers regarded as the points on a line; it is usually denoted by R or $\mathbb{R}$. In other words, it is the set R of all real numbers, viewed as a geometric space (i.e. the Euclidean space of dimension one).

REAL NUMBER (REAL): Any number that is irrational or rational and include all positive and negative numbers including zero. It is usually denoted R . Unlike the rationals, the real numbers are not denumerable. The reals are dense and form a compact connected set (a continuum) and sometimes referred to as the **continuum**. A real and an imaginary number sum up to a complex number.

REAL PART: In complex number $z = x + iy$ the real number x is called the real part and it is part of a complex number that does not have an imaginary number.

REAL VARIABLE: A variable whose value ranges over the real numbers.

REAL-IS-POSITIVE CONVENTION: A convention used in optical formulae relating to lenses and mirrors in which the distances to the real points, images and foci are taken as positive and distances to virtual points are taken as negative.

REAL-SYMMETRIC: The real-symmetric of a square matrix, $n \times n$ is equal to its transpose and satisfies $A^T = A$, where A^T is the transpose, so that $a_{ij} = a_{ji}$.

REAL-TIME CLOCK, RTC (COMPLEMENTARY METAL-OXIDE SEMICONDUCTOR, CMOS; NON-VOLATILE RAM, NVRAM; CMOS RAM): An on-board semiconductor chip powered by a CMOS battery inside computers that stores information such as the system time and date and the system hardware settings for your computer.

REAL-TIME SYSTEM: Software and hardware systems in computing that process information and responds to operational events within a time frame as they happen. Real-time system uses include in automatic braking systems (ABS), air traffic control, auto pilot systems, over temperature control and also in medicine in devices such as blood flow monitor.

REAL-VALUED FUNCTION (REAL FUNCTION): A function f that takes only real values, such as the numerical trace of a self-adjoint matrix. In other words, it is any function that takes only real values or assigns a real number to each member of its domain.

REALGAR: An important ore of arsenic with the chemical formula As_2S_2 , specific gravity 3.56 and melting point 320 °C (593 K). It is orange-red in colour. It is found in ores of lead, silver, associated with orpiment and stibnite. It is used as a pigment and in pyrotechnics.

REALLY SIMPLE SYNDICATION (RDF SITE SUMMARY; RICH SITE SUMMARY; RSS): A method of providing website content such as news stories or software updates in a standard XML format.

Websites such as The Wall Street Journal and CNET's News.com provide news stories to various RSS directories that distribute them over the Internet. RSS content can be accessed with an RSS-enabled Web browser or other programs designed for retrieving RSS feeds.

REANNEALING (ANNEALING): In biology, it is the process in which two single strands of DNA hybridise to form a DNA double helix, after they have been separated by heat.

REARRANGEMENT: 1. In chemistry, it is a type of chemical reaction in which atoms or group of atoms migrate from one point to another to form new molecules. Examples include Hoffman rearrangement and Pinacol rearrangement. 2. In mathematics, rearrangement of infinite series is a series whose terms are a permutation of the terms of a given series. For example, if $\sum a_n$

is an infinite series and $f : \mathbb{N} \rightarrow \mathbb{N}$ is a bijection, then the series $\sum_{n=1}^{\infty} a_{f(n)}$ is a rearrangement of $\sum a_n$.

REARRANGEMENT ION: An electrically charged species, involving a molecular or parent ion, in which atoms or groups of atoms have transferred from one portion of a molecule or molecular moiety to another during the ionisation fragmentation processes.

REARRANGEMENT STAGE: The elementary reaction or reaction stage (of a molecular rearrangement) in which there is both making and breaking of bonds between atoms common to a reactant and a reaction product or a reaction intermediate. If the rearrangement stage consists of a single elementary reaction, this is a 'rearrangement step'.

REAUMUR TEMPERATURE SCALE: A temperature scale in which water freezes at 0°R and boils at 80°R. The scale is named after René Antoine Ferchault de Réaumur, French scientist who first proposed it in 1730. Réaumur's thermometer contained diluted alcohol and was constructed on the principle of taking the freezing point of water as 0° and graduating the tube into degrees, each of which was one-thousandth of the volume contained by the bulb and tube up to the zero mark.

REBOOT: Also known as **restart**, it is an instance of shutting down and restarting a computer system. It can be done without the need to interrupt the power supply, called a **warm boot**. Another type is **cold boot**, done by turning the computer off and on again. Rebooting reloads the operating system files and is often used as a quick solution to many minor problems such as slow processing or unresponsive computer. It may also be done after the installation of a new program, or a software update.

REBOUND REACTION: If in a chemical process, occurring in a molecular beam, the reaction products are scattered backwards with respect to the moving centre of mass of the system, the reaction is called a rebound reaction.

RECALCITRANT SEED: A seed that loses its viability if stored for any length of time, even under conditions that are normally conducive to seed longevity, i.e. low moisture content and low temperatures. Many tropical plants, for example, coconut, rubber and tea, have recalcitrant seeds. Such seeds can often only be kept for a year or less.

RECALESCENCE: The sudden increase in temperature and glowing of a metal which has been heated to white heat and allowed to cool. This is caused by the liberation of latent heat as the metal undergoes a change in its crystalline structure. The temperature at which this occurs is called the **recalcescence point**.

RECALL SURVEY: A survey in which data are gathered by asking people what they remember about a particular topic. For example, a recall survey might ask people to list what soft drinks they consumed in the previous week.

RECAPITULATION THEORY (HAECKEL'S LAW): The theory that some stages in an organism's embryonic development and differentiation represent evolutionary development of their remote ancestors. For example, the beginnings of gill clefts appear in both humans and fish, but while they are elaborated and eventually function in the fish, in humans, except for the modified gill cleft that becomes the Eustachian tube, they disappear as the embryo develops.

RECURRING DECIMAL (REPEATING DECIMAL): A decimal in which a pattern of one or more digits is repeated indefinitely, for example, 0.353535 ...

RECEIVED EXCHANGE DATA (RXD): In data transfer, the term refers to the line that transmits received data from one source to another, for example, data transmitted from a modem to a computer.

RECEIVER RADIO: Radio communication system that receives its input in the form of modulated radio waves from an antenna, filters the signal by removing interferences, amplifies it, demodulates and decodes it. The final signal is usually an audio signal which may be heard.

RECEPTACLE: 1. Also known as **thalamus** or **torus** in botany, it is the enlarged end of a stem which may be rounded, flattened or cup-shaped which bears the floral parts. 2. The point on a leaf or thallus where reproductive organs are borne. Examples are the swollen tips of the thallus of certain algae in the Fucales, which bear the conceptacles.

RECEPTIVE FIELD: A region of space in which stimulus influences the activity of a particular neuron. For example, a given ganglion cell in the retina of the eye transmits signals to the brain exclusively from the photoreceptors to which it connects.

RECEPTOR: 1. A specialised cell or group of cells that detect stimuli in an organism and respond by transmitting impulses via the sensory nerves or receptors. Receptors can detect stimuli in the body or externally and are found in large numbers in the eyes, ears, nose and skin and respond to stimuli such as cold, heat and pain. 2. Specially adapted protein molecule in the plasma membrane that can bind with specific substances such as a hormone, a neurotransmitter, antigens, a toxin or a drug. There are many receptors which control many functions in the human body and can effect a change in a cell.

RECEPTOR MEDIATED ENDOCYTOSIS (RECEPTOR-MEDIATED ENDOCYTOSIS): An endocytotic mechanism in which specific molecules are ingested into the cell by the inward budding of membranous vesicles containing proteins with receptor sites specific to the molecules being taken in. It enables a cell to acquire bulk quantities of specific substances.

RECEPTOR POTENTIAL: An initial response of a receptor cell to a stimulus, consisting of a change in voltage across the receptor membrane proportional to the stimulus strength. The intensity of the receptor potential determines the frequency of action potentials traveling to the nervous system.

RECEPTOR TYROSINE KINASES: A diverse group of transmembrane proteins that act as receptors for cytokines, growth factors, hormones and other signaling molecules. Receptor tyrosine kinases are expressed in many cell types and play important roles in a wide variety of cellular processes, including growth, differentiation and angiogenesis.

RECESSIVENESS: 1. A condition in which an allele fails to produce a characteristic effect in an organism carrying it when a corresponding dominant allele is present. The effect is masked when the matching allele is not identical, producing an effect only when the matching allele is identical. 2. A trait that is expressed only when the determining allele is present in the homozygous condition.

REACHABLE SET (ATTAINABLE SET): In a directed graph, the set of points that can be reached from a given vertex V .

RECHARGE ZONE: An area in which water travels downward to become part of an aquifer. In a recharge zone surface water or precipitation percolate through permeable rock, sediment or fractured materials that lie over an aquifer.

RECHARGEABLE BATTERY (SECONDARY CELL): An electrochemical cell in which the chemical reaction producing the electromotive force (e.m.f.) is reversible and the cell can therefore be recharged many times by passing a current through it. An example is lead-acid cell. Recharge is to restore electrical energy to a battery by connecting it to a power supply.

RECIPROCAL: 1. Also known as **inverse**, the reciprocal of a number is 1 divided by that number. For example, the reciprocal of 2 is $1/2$. 2. Of or relating to a multiplicative inverse. For example,

the multiplicative inverse for a number x , denoted by $1/x$ or x^{-1} , is a number which when multiplied by x yields the multiplicative identity; the multiplicative inverse of a fraction a/b is b/a . 3. Relating to nerve and muscle tissues in which the activity of one group of muscles results in the inhibition of another.

RECIPROCAL ALTRUISM: Altruistic behaviour between unrelated individuals in manner that temporarily reduces its fitness while increasing another organisms fitness. It is believed that it produces some benefit to the altruistic individual in the future when the current beneficiary reciprocates.

RECIPROCAL CROSS: A type of experimental hybrid breeding which reverses the roles of males and females. It is used to detect sex linkage and maternal inheritance. For example, in the first mating, a male of genotype A crosses with a female of genotype B; in the second mating a male of genotype B mates with female of genotype A.

RECIPROCAL LATTICE: A lattice for a crystal constructed by drawing perpendiculars to each plane through a common point such that the distance of those points to the common point is equal to the reciprocal of the real interplanar distance. Reciprocal lattice is a basic concept in the theory of x-ray diffraction and energy band theory.

RECIPROCAL LINK (LINK EXCHANGE): A mutual link between two websites. For example, if website A links to website B, then website B can add a reciprocal link back to website A. The result of a reciprocal link is two websites that link to each other. Part of how search engines like Google rank web pages in their search results is by how many different pages link to a page and how popular those pages are.

RECIPROCAL POLAR CURVE: A pair of curves such that the polar of each point of one is tangential to the other.

RECIPROCAL POLAR FIGURE: The figure related to a given configuration of points and lines in the plane in such a way that each point of either figure is the pole of a line in the other. Equivalently, each line of either figure is the polar of a point in the other with respect to some given conic, such as a pair of reciprocal polar curves.

RECIPROCATING ENGINE: An engine in which a piston oscillates in a cylinder, being driven by the pressure of the working fluid.

RECIPROCATION: In mathematics (geometry), it is the transformation of a configuration of points and lines into its reciprocal polar figure.

RECLAIM: To make land such as marshy land, desert and even water bodies such as the sea usable for agricultural or commercial purposes.

RECLINATE: Bent abruptly downward towards the base.

RECLINING: Bending downward; leaning or lying upon something and being supported by it.

RECOGNITION SEQUENCE: A specific sequence of nucleotides, typically composed of 4, 6 or 8 nucleotides, that is recognised by a restriction enzyme and cleaves a DNA molecule.

RECOGNITION SITE (CLEAVAGE SITE; RESTRICTION RECOGNITION SITE): The site in a polynucleotide chain as to where the restriction enzyme cleaves nucleotides by hydrolysing the phosphodiester bond between them. The restriction enzyme recognises the site as to where it cuts via a specific sequence of nucleotides along the polynucleotide chain. For example, when the restriction enzyme, Eco RI, recognises the palindromic sequence GAATTC, it cuts between G and A on the top and bottom strands.

RECOGNITION SPECIES CONCEPT: The idea that specific mating adaptations become fixed in a population and form the basis of species identification and that a species is a set of organisms that recognise one another as potential mates; they have a shared mate recognition system.

RECOIL: The motion acquired by a particle through a collision with or the emission of, another particle or electromagnetic radiation.

RECOIL LABELLING: Labelling by a chemical reaction initiated by recoil.

RECOMBINANT DNA (rDNA): A synthetic DNA that contains genes from unrelated organisms that does not occur through natural processes within the cell, but through genetic engineering. It is formed by transplanting genes from one species into the cells of a host organism of a different species to produce new genetic combinations.

RECOMBINATION: A process that regroups the genetic material during meiosis, ensuring genetic variation in the offspring.

RECOMBINATION ERA: The period, about 300,000 years after the big bang, when the universe was cool enough to allow protons to capture one electron and form a neutral atom of hydrogen for the first time. Before this period, the temperature was high enough that all of the hydrogen was ionised, freeing the electrons from the protons resulting in the scattering of the electrons in all directions making the universe 'opaque'. After "recombination", photons were able to travel through the universe relatively unimpeded and the universe became 'transparent'.

RECOMBINATION PROCESS: The process in which a neutral atom or molecule is formed by the combination of a positive ion and a negative ion or electron.

RECOMMENDED DAILY AMOUNT (RDA): The amount of nutrients such as vitamin, fat, carbohydrates and mineral consumed each day by a person or animal to meet the metabolic needs to stay healthy. RDA is different for children, adults, males and females and there are differences in the amount in various countries.

RECOMMENDED STANDARD-232 (RS-232 INTERFACE): A standard type of computer interface approved by the Electronic Industries Alliance (EIA) used to connect computers to serial devices. It connects to modems, mice, screens and serial printers through an EIA-232 port. The device that connects to the interface is called **Data Communications Equipment (DCE)** and the device to which it connects such as a computer is called **Data Terminal Equipment (DTE)**.

RECON: The smallest genetic unit of DNA consisting of a single pair of nucleotides capable of mutation.

RECONNAISSANCE SATELLITE (RECON SAT; SPY SATELLITE): An Earth observation satellite or communications satellite that is used for military or intelligence operations.

RECONSTRUCTIVE TRANSITION: A transition which involves a major reorganisation of the crystal structure and a change of local topology, during which primary bonds are broken and reformed so that there is no immediate relationship between the crystal structures of the parent and product phases. Examples include the transition of Mg_2SiO_4 (olivine) to Mg_2SiO_4 (spinel); the transition of diamond to graphite.

RECORD: In a database, it is a collection of related data items in a table. It is made up of rows of a group of fields.

RECOVERY: 1. In toxicology, the process leading to partial or complete restoration of a cell, tissue organ or organism following its damage from exposure to a harmful substance or agent. 2. A term used in analytical and preparative chemistry to denote the fraction of the total quantity of a substance recoverable following a chemical procedure.

RECREATIONAL DRUG: A psychoactive drug such as marijuana, cocaine and amphetamines, often illegal, used for recreation. Recreational drugs may also include alcohol and tobacco.

RECRUITMENT: 1. The activation of extra motor neuron to achieve maximum response when a stimulus of unaltered intensity is prolonged. 2. The amount of migratory or juvenile fish reaching exploitable age each year due to growth.

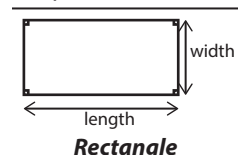
RECRYSTALLISATION (RECRYSTALLIZATION): 1. A purification method involving dissolving and crystallising to obtain an increasingly pure product. 2. A change in the structure of a crystal without a chemical reaction. 3. A process which takes place in metals and alloys

following distortion and fragmentation of constituent crystals by severe mechanical deformation, in which some fragments grow larger and strain-free grains are formed. This process takes place slowly at room temperature, but can be speeded by annealing. 4. The formation of new mineral grains in crystalline form in a rock under the influence of metamorphic processes.

RECTAL GLAND: A salt-secreting gland found in the cloacae of cartilaginous fishes such as sharks and the spiny dogfish that transmits excess sodium in the blood into the rectal gland for secretion.

RECTAL PALPATION: A physical examination technique of the posterior abdomen by feeling to detect pregnancy in animals.

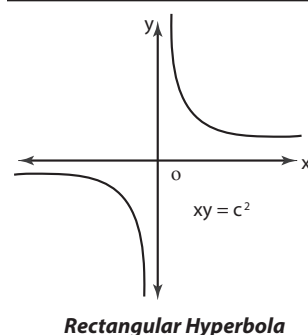
RECTANGLE (OBLONG): A four-sided plane figure with four right angles (90°). Its area (A) is the product of the length (L) and breadth (B): $A = L \times B$. Rectangular means shaped like a rectangle; also mutually perpendicular.



RECTANGLE METHOD: 1. A method for finding the area of a polygon in which one or more rectangles are drawn around all or parts of the polygon through its vertices. The sides of the drawn rectangle(s), together with the sides of the original figure, define regions that are either rectangles or triangular halves of rectangles. Add and/or subtract the areas of these rectangular and triangular regions to get the area of the original polygon. 2. Also called the **midpoint** or **mid-ordinate rule**, it computes an approximation to a definite integral, made by finding the area of a collection of rectangles whose heights are determined by the values of the function.

RECTANGULAR COORDINATES (CARTESIAN COORDINATE SYSTEM; CARTESIAN COORDINATES; RECTANGULAR COORDINATE SYSTEM): A set of coordinates that gives the position of an object with respect to two or three mutually perpendicular axes (x , y and z) that cross at some specified origin. In the two mutually perpendicular axes (x and y) system, the plane consists of two number lines, one horizontal (x -axis) and the other vertical (y -axis) intersecting at 90° of the origin. Their units are of equal measure. The two axes divide the plane into four quadrants numbered in the anticlockwise direction. The three mutually perpendicular axes (x , y and z) system consists of three axes x , y and z that intersect at the origin. (For diagrams see **cartesian coordinates**). In astronomy, heliocentric or geocentric rectangular coordinates are occasionally used to give positions of bodies in the Solar System.

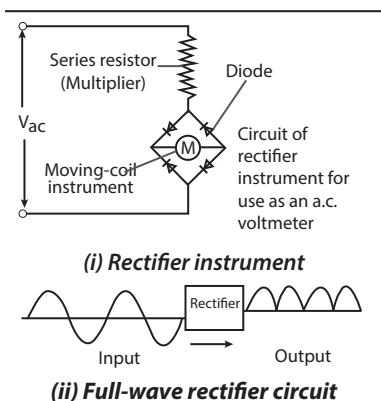
RECTANGULAR HYPERBOLA: A hyperbola whose asymptotes are perpendicular. If $xy = C^2$, then the asymptotes are the coordinate axes.



RECTANGULAR NUMBER: Any number that is not prime and can be expressed as a rectangular arrangement of dots with the number of columns being one more than the number of rows. In other words, rectangular numbers can be expressed as $a \times b$, where a and b are greater than 1 and hence equal to the number of unit-spaced points in a rectangular array since the total number of points in such an array equals the product of the numbers of points in the sides. An example of a rectangular number is 45 (5×9) that can be arranged in a rectangular form.

RECTIFIABLE CURVE: In mathematics, a curve that has a finite arc length such that, if $a_1, \dots, a_n$ are a set of points in order along the curve, the sum of the lengths of the chords from each a_i to a_{i+1} tends to a limit as n tends to infinity. In this context, to rectify is to determine the length of a curve.

RECTIFICATION: 1. The regulation of membrane conductance by the voltage across the membrane. Strong rectification implies that ionic channels carrying current are open at some membrane potentials and shut at others. Conductance is low at very negative membrane potentials, but increases as the potential becomes less negative. 2. The process of purifying a liquid by distillation. 3. The determination of a straight line whose length is equal to a portion of a curve. 4. A process that converts alternating current to direct current. The common device used for rectification is called a rectifier usually composed of semiconductor diode, which permits current to pass in only one direction, effectively blocking its flow in the other direction.

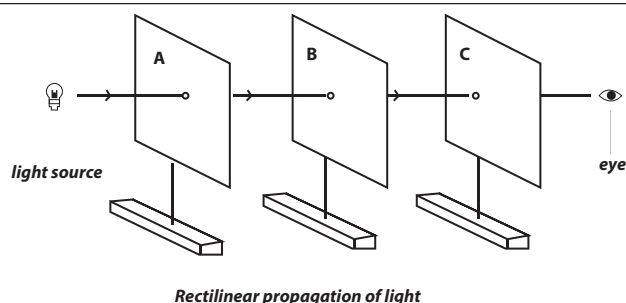


RECTIFIED CONCENTRATED GRAPE MUST: A type of natural grape sugar produced by distillation, which removes all the non-sugary components from surplus wine. It is used to add new wine to raise the alcohol level called **chaptalisation**.

RECTIFIED SPIRIT (RECTIFIED ALCOHOL): Solution of ethanol (ethyl alcohol) and water containing about 95% ethanol obtained by repeated distillation. It is used in medicinal applications, in alcohol production, etc.

RECTIFIER: 1. An electrical device that converts alternating current (AC) to direct current (DC). The common device used for rectification is the semiconductor diode, which permits current to pass in only one direction, effectively blocking its flow in the other direction. Uses of rectifiers include in power supplies, where direct current is needed to operate components in devices such as televisions, radios and devices. 2. An apparatus used for purifying or refining liquid by distillation.

RECTILINEAR: 1. Moving in or formed from a straight line, such as a rectilinear path. 2. A rectilinear figure is a closed 2-dimensional shape having line segments for sides. The interior angle at each vertex of a rectilinear figure is either 90° or 270° . Examples are polygons such as squares, triangles and pentagons. Some 3-dimensional solid shapes such as cubes, cuboids and prisms are also examples of rectilinear figures. 3. **Rectilinear propagation of light** is an experimental evidence that confirms that light travels in a straight line. In the diagram below, light travels from A to C.



RECTRICES: The large, stiff tail feathers of a bird that is used to navigate, when the bird is in flight.

RECTUM: The part of the large intestine between the colon and the anal canal which stores faeces temporarily before it is eliminated from the body by defecation.

RECUMBENT: Lying down or reclining on the surface from which it arises. It is used of an organ or other structure. For plants, it refers to

those that lie on the ground or grow close to the ground, for example, recumbent shrubs. For animals, it refers to a position of rest by lying down or leaning against an object. When applied to humans, **Dorsal recumbent** or **dorsal decubitus**, is lying in the supine position or lying on the back with the face upward. **Ventral recumbent** or **ventral decubitus** position is lying in the prone position or lying with the face down and the back upwards. The **lateral recumbent** position, also known as **semiprone position** or **Sims' position** is lying down on the side with the under arm behind the back and the upper thigh flexed. It promotes comfort by relieving pressure on the sacrum and heels. In medicine, it is usually done to facilitate vaginal and rectal examination.

RECUPERATIVE POTENTIAL: The ability of an organism's part to recover after being injured or from an illness.

RECUR: Of a digit or sequence of digits, to repeat an infinite number of times at the end of the decimal expression of a fraction. For example, in $10 \div 3 = 3.333\dots$, which can also be written as 3.3, 3 recurs and is called a **recurring decimal**.

RECURRENCE RELATION (DIFFERENCE EQUATION): An equation that defines a sequence based on a rule that gives the next term as a function of the previous term(s). The simplest form of a recurrence relation is the case where the next term depends only on the immediately previous term. If we denote the n^{th} term in the sequence by x_n , such a recurrence relation is of the form $x_{n+1} = f(x_n)$

for some function f . One such example is $x_{n+1} = \frac{2-x_n}{2}$. A recurrence

relation can also be higher order, where the term x_{n+1} could depend not only on the previous term x_n but also on earlier terms such as x_{n-1}

and x_{n-2} . A second order recurrence relation depends just on x_n and

x_{n-1} and is of the form $x_{n+1} = f(x_n, x_{n-1})$ for some function f with two inputs. For example, the recurrence relation $x_{n+1} = x_n + x_{n-1}$ can generate the Fibonacci numbers.

RECURRING DECIMAL (CIRCULATING DECIMAL; REPEATING DECIMAL): A rational number that is a decimal or radical fraction containing a pattern of digits that repeats indefinitely after the decimal point; for example, in 0.3333, 3 recurs and in 2.1818181..., the digits 81 recur. The repeating figure or figures of a recurring decimal fraction is called **repetend(s)**. Recurring decimals can also be written by drawing a horizontal line called a **vinculum** above the repetend, placing dots above the outermost numerals of the repetend or enclosing the repetend in parentheses. For example, 0.3333 can be written as $0.\overline{3}$, $0.\dot{3}$ or $0.(3)$; and 2.18, can be written as $2.\overline{18}$, $2.\dot{18}$ or $2.(18)$. **Recur** (a digit or sequence of digits) is to repeat an infinite number of times at the end of the decimal expression of a fraction.

RECURSION: The process of applying the output of a function or algorithm to its input values to generate an infinite sequence of values. The Fibonacci numbers and many fractal figures, for example, are defined recursively. Recursion is applied in computer programming in a procedure or routine that has the ability to call itself; or as a recursive function, algorithm or sequence of instructions that loops back to the beginning of itself until it detects that some condition has been satisfied.

RECURSION FORMULA (RECURSION CLAUSE; INDUCTIVE STEP): The formula for determining the next term of a sequence from one or more of the preceding terms. For example, the recursive formula for the sequence 5, 19, 75, ... is $4a_n - 1$.

RECURSIVE DEFINITION (INDUCTIVE DEFINITION): The definition of a sequence by specifying its first term by the base clause and an algorithm or the recursion clause to derive any term from its predecessor.

RECURSIVE FUNCTION: 1. A function defined in terms of the repeated application of a number of simpler functions to their own values, by specifying a base clause and a recursion formula. For

example, factorial can be defined as $x! = x(x-1)!$ with the base clause of $1!$ equal to 1. 2. Any of the class of functions generated from the four operations of addition, multiplication, selection of one element from an ordered n -tuple and determines whether $a < b$ (which are given to be recursive) by the two rules: (i) if F and $G_1, \dots, G_n$ are recursive, then so is $F(G_1, \dots, G_n)$; and (ii) if H is a recursive function such that for each a there is an x with $H(a, x) = 0$, then the least such x is recursively obtainable.

RECURSIVE PARTIAL FUNCTION: A function of which the graph (the set of points $\{x, f(x)\}$) is recursively enumerable.

RECURSIVE PREDICATE: A predicate that possesses a representing function that is recursive.

RECURSIVELY ENUMERABLE: Of a predicate, P , such that there is a recursive predicate, Q that satisfies $(\exists x)Q(a, x) \leftrightarrow P(a)$. A recursively enumerable language is a recursively enumerable subset in the set of all possible words over the alphabet of the language. A set is recursively enumerable if and only if it can be accepted by a Turing machine.

RECURSIVELY GENERATED: Of a sequence, able to be given a recursive definition; generated by an algorithm.

RECURVED: To be curved or bent backwards, as found in the teeth of snakes, which aid in swallowing large prey.

RECYCLE BIN: A location where deleted files are temporarily stored on Microsoft Windows 95, 98, ME, NT, 2000, XP, Vista and all later versions of Microsoft Windows. It is similar to the Apple Macintosh Trash. The Recycling Bin allows users to recover files that have been deleted in Windows.

RECYCLING: Recovering and processing of used or waste materials such as glass, paper, metal, plastic and textiles into new products. The recycling process reclaims the original material and uses it in new products. The use of recycled materials to make new products costs less and requires less energy than using new materials. Recycling can also reduce pollution, either by reducing the demand for high-pollution alternatives or by minimising the amount of pollution produced during the manufacturing process. Recycling decreases the amount of land needed for trash dumps by reducing the volume of discarded waste.

RED ALGAE: Members of the phylum Rhodophyta, growing mostly in tropical and subtropical coastal areas. It is one of the oldest groups of eukaryotic algae with characteristics such as reddish phycobilin pigments and eukaryotic cells without flagella and centrioles.

RED BLOOD CELLS (ERYTHROCYTES; RED BLOOD CORPUSCLES): The most common type of blood cells that carry oxygen from the lungs to the body tissues. They contain the red protein haemoglobin, which gives the cells and blood the red colour. Red blood cells develop in the bone marrow and stored in the spleen. They lack nucleus at maturity and live between 100 and 120 days. In a human adult, there are about 5.2 million red cells per cu mm.

RED BONE MARROW: Tissues that develop red blood cells, white blood cells and platelets. It is present throughout the skeleton during the developmental stages of the foetus. After birth, as the child ages, the red bone marrow is gradually replaced by yellow bone marrow.

RED CLAY: Fine-grained, reddish-brown sediment, covering about 50% of the mid-ocean floors, formed from materials such as meteoric and volcanic dust, pumice, shark teeth, manganese nodules and other materials that originated on land and carried far out to sea by rivers, melting ice and the wind.

RED DATA BOOK (IUCN RED LIST; RED DATA LIST): A catalogue which is the world's inventory of the global conservation status of plant and animal species published by the International Union for Conservation of Nature (IUCN), listing species which are in danger of being extinct and to inform people of the need for conservation.

RED DESERT SOIL: A type of coarse zonal soil rich in salts and lime (a pedocal) but poor in humus. They are formed in hot deserts and can be cultivated under irrigation.

RED DWARF: Any small low-mass (0.08 to 0.8 times that of the solar mass) star of spectral classes M and L, constituting over 70% of all stars. Because of their low masses, red dwarf stars have surface temperatures of about 1727 °C (2000 K) to 2527 °C (3800 K), making them relatively cool. They are faint (2×10^{-4} to 0.05 solar luminosity) and with a diameter and mass of one tenth that of the Sun. Because of the small size and low temperature they are faint and invisible to the naked eye and can only be seen with the aid of instruments. The closest red dwarf to the Earth is Proxima Centauri, which is about 80 light-years away (one light year is 9 460 730 472 580.8 km). Red dwarfs have very long life times, estimated at 100 billion years.

RED EFFECTS (FAR-RED EFFECTS): Those physiological responses exhibited by many plants as a result of being illuminated by either red or far-red light. Usually a period of red light will reverse the effect of a period of far-red light and vice versa. Thus the response elicited in the plant depends on which treatment was given most recently

RED FESCUE: A species of grass, *Festuca rubra*, a nutritious forage grass for animals and also used for lawns and turfgrass. It is found worldwide in many habitats and climatic conditions; although it needs some amount of sunlight to flourish.

RED GIANT: A stage in stellar evolution in which a star has used up its hydrogen by converting it to helium, making its outer layer to expand to a diameter of 10 to 1,500 times that of the Sun, cool and bright (about 2,000 times brighter than normal). Its surface temperature ranges from 1727 °C (2000 K) to 2727 °C (3000 K), which is considered relatively cool.

RED LEAD (LEAD TETRAOXIDE; LEAD(IV) OXIDE): The bright red to orange-red powdery oxide of lead with the chemical formula Pb_3O_4 , density 8.3 g cm⁻³ and melting point 500 °C (773 K). It is an oxidising agent and its uses include in the manufacture of storage batteries, red pigments and as a paint to protect iron and steel from rusting and also to manufacture lead glass.

RED LIGHT: Electromagnetic radiation of approximately 630 nm wavelength. Light in this part of the spectrum is the most effective in initiating various light-dependent reactions in plants. For example, short-day plants are prevented from flowering if given a short burst of red light during the dark period. Similarly, red light is best for promoting germination in light-sensitive seeds and for reversing etiolation

RED QUEEN HYPOTHESIS (RED QUEEN EFFECT): An evolutionary hypothesis that explains two similar ideas: microevolutionary and macroevolutionary. The microevolutionary idea is that, sexual reproduction may allow a species to evolve quickly just to hold onto the ecological niche that it already occupies in the ecosystem. The macroevolutionary version explains how the coevolution of competing species can lead to dynamic equilibrium, for which the probability of extinction remains fairly constant within the group over a long time. Hence, evolution is seen neither as progressive with a species' chances of survival improving over time nor as escalatory with increasing vulnerability to extinction over time. Instead, as one species evolves, improvements occur that make it more competitive, while its competitors experience selection pressure that forces them to evolve in order to keep pace with it. One that lags behind will become extinct.

RED SPIDERLING TARVINE (Boerhavia diffusa; PIG WEED; RED HOGWEED; SPREADING HOGWEED): A perennial plant in the Nyctaginaceae, the Four O'Clock family, that grows as common weed in wet areas, especially in the rainy seasons with sunshine. It occurs throughout India, Africa, the Pacific and southern United States and has become naturalised in many places. It has a fusiform root with a stem, which is prostrate, decumbent or ascending. The leaves are broadly ovate and obtuse at the base and arranged oppositely or sub-oppositely. The small red or white flowers are in pendunculate, glomerulate clusters arranged in slender, long stalked, axillary or terminal corymbs. The fruits are rounded above, obovoid, slightly cuneate below and glandular throughout. Medicinally, the various parts such as the root, leaves and seeds have been used to treat various ailments such as leucorrhoea, anaemia, inflammations, heart diseases,

asthma, diabetes, dyspepsia, tumours, spleen enlargement and abdominal pains. It also has antibacterial and antioxidants properties.

RED TIDE (ALGAL BLOOM): A sudden increase in the population of marine phytoplankton, notably dinoflagellate species. If the species are red, the sea is coloured red due to the high concentration of the organisms' photosynthetic accessory pigment. This is toxic to fish and fish consumers including other fishes and humans.

RED TROPICAL SOIL: A type of zonal soil formed in rain forests and in savannah with alternating dry and wet seasons. It is heavily leached but a balance is maintained between the nutrients produced by decomposition and those taken up by the plants. The red colour is due to the presence of iron, magnesium and aluminium oxides. Some red tropical soils may form a hard crust of chemicals on the surface.

RED WOOD (COAST REDWOOD, COASTAL REDWOOD; CALIFORNIA REDWOOD; *Sequoia sempervirens*): An evergreen, monoecious plant in the cypress family, Cupressaceae, native to the central and northern California coast in the United States. Its straight trunk is much enlarged and buttressed at the base and often with rounded swellings or burls, slightly tapering. It grows quickly and can reach heights of about 116 metres, when fully grown, with a trunk circumference of about 7 metres and regarded as the tallest tree in the world. It has symmetrical conical crown. It has a tough and fibrous red-brown bark that is about 30 centimetres thick. The leaves are spirally arranged and vary: young trees having long and flat leaves, whereas old trees have scale-like leaves. Inconspicuous male and female flowers are borne separately on different branches of the same tree. The cones are terminal and are 13 to 29 mm long, which open and shed seeds when ripe. The plant is resistant to fire and insect damage due to the presence of tannin in the bark. For this reason its timber is used for railroad ties and in construction. Its bark is used as hog fuel, insulation or garden mulch. The burls are used in the production of table tops, veneers and turned goods. The wood quality is excellent and the tree is much threatened by logging.

RED-GREEN-BLUE (RGB): An additive colour in which red, green and blue light are added together in various ways to reproduce a broad array of colours. It is used mainly for the sensing, representation and display of images in electronic systems, such as televisions and computers. All the colours displayed by the screen can be made up from three component colours: red, green and blue. RGB input devices include colour TV and video cameras, image scanners and digital cameras and the output devices are TV sets of various technologies (CRT, LCD, plasma, etc.), computer and mobile phone displays, video projectors, etc.

REDOX COUPLE: An electron donor such as a reducing agent and its corresponding oxidised form capable of reaccepting the electrons. The oxidised and reduced species in a redox half-reaction that forms redox couple. An example is Fe^{2+}/Fe .

REDOX ION EXCHANGERS: Conventional ion exchangers in which reversible redox couples have been introduced as counter-ions either by sorption or complex formation. They closely resemble redox polymers in their behaviour.

REDOX POTENTIAL (REDUCTION POTENTIAL; OXIDATION/REDUCTION POTENTIAL): The relative tendency of chemical species to release or accept an electron, measured in volts (V). The Standard redox potential, with the symbol ϵ_0 or E_0 is the redox potential of a solution containing the oxidant and reductant of the couple at standard concentrations. It is denoted by ϵ .

REDOX REACTION (REDUCTION-OXIDATION REACTION): Any of a large class of chemical reactions, including many familiar processes such as combustion, in which oxidation and reduction reactions occur simultaneously, resulting in changes in oxidation number of a molecule, atom or ion by gaining or losing an electron. In other words, this process involves one molecule, atom or ion becoming more positive and the other one becoming more negative to indicate that a redox reaction has occurred. They include many processes such as photosynthesis, respiration, combustion and corrosion or rusting.

REDOX SENSOR: An instrument capable of detecting the kind of non-equilibrium concentrations of reduced iron and manganese, along

with the oxides of nitrogen and sulfur, which are strongly indicative of life processes on Earth.

REDSHIFT: A systematic shift in the spectral lines of certain distant galaxies toward longer wavelengths or toward the red end of the spectrum. It is caused by its motion away from the Earth. There are three types: (i) **Doppler Effect** is a shift in the spectral lines of certain distant galaxies towards the red end of the visible spectrum, the longer wavelengths. Edwin Hubble, who first studied the redshift effects, concluded therefore, that all galaxies are receding from each other. (ii) **Gravitational or Einstein redshift** is the displacement of the spectral lines towards the red caused by a high gravitational field. (iii) **Cosmological redshift**, is redshift produced by the expansion of the universe and the reason most galaxies in the universe have red-shifts.

REDUCE: In mathematics, to simplify or convert an expression by substituting one term into an equivalent one. For example, to reduce a fraction is to remove a common factor that both the numerator and the denominator have.

REDUCED COMPLEXITY METHOD: Any algorithm that performs with a complexity below that of the usual or naive method. An example is afforded by multiplication using the fast Fourier transform.

REDUCED ECHELON FORM: An echelon form with the conditions that in a **row-reduced echelon** or **row echelon form** matrix, (i) all the zero rows are grouped together at the bottom of the matrix below the non-zero rows; (ii) the first non-zero entry in each row, called the leading entry is 1 and occurs in a column to the right of leading 1 in the row above; and (iii) the leading entry of each non-zero row after the first occurs to the right of the leading entry of the previous row. In a **column-reduced echelon form**, the reverse of the properties of the row echelon form apply.

REDUCED EQUATION OF STATE: A law, which relates the reduced parameters of a thermodynamic equilibrium system of a substance, based on the parameters of pressure, volume and temperature. It defines the ratio of their values to the critical state that if any two or more substances have the same reduced pressure then their pressures are the same fraction or multiple of their respective critical pressures. At equal reduced temperatures, their reduced volumes should be equal.

REDUCED FRACTION: A fraction that is simplified into its lowest terms, as both the numerator and denominator cannot be further divided by any common integer, for example, $3/5$.

REDUCED INSTRUCTION SET COMPUTING (RISC): Abbreviated RISC, it is a major type of microprocessor architecture usually having few instructions in the instruction set, with smaller and simpler instructions. In contrast, **Complex Instruction Set Computer (CISC)**, has an architecture that contains a large set of computer instructions and varies in size and complexity. Examples of RISC are the PowerPC processors and Macintosh computers RISC. Intel Pentium processors are mainly CISC-based, with some RISC facilities built into them.

REDUCED PRESSURE DISTILLATION (VACUUM DISTILLATION): A method of distilling in which the pressure above the liquid mixture to be distilled is reduced to less than its vapour pressure causing evaporation of the most volatile liquid to enable it boil at a lower temperature.

REDUCED SPECIES: A term used to characterise the degree of reduction (or oxidation) in atoms, molecules and ions. An atom in a molecule or an ion which has a low oxidation state. An element or atom in a compound can be reduced by the reaction of an element or compound with hydrogen, while it can be oxidised by reaction with oxygen. A reduced species can be formed also through the gain of electrons (either at the negative electrode in a cell or through transfer from another atom, ion or group of atoms in a chemical reaction). For example, the carbon atom in CH_4 and other hydrocarbons is in a reduced state, while the carbon in CO_2 is in an oxidised state. Similarly the sulfur atom in H_2S is in a reduced state while that in sulfuric acid (H_2SO_4) is in an oxidised state.

REDUCED-TILLAGE SYSTEM: A type of preparing land for planting which leaves 15-30 percent residue cover after planting. Most reduced-till systems are used in conjunction with cover crops or organic mulches to protect the soil surface at all times, helping to increase organic matter over time, or to help control weeds.

REDUCED WORD: A word on a non-empty set that contains no consecutive pair of letters of the form xx^{-1} or $x^{-1}x$.

REDUCIBLE: 1. In mathematics, a reducible surface or curve is one that is deformable within a given region to a point. 2. Of a polynomial, able to be factorised into polynomials of lower degree. It is irreducible otherwise. The polynomial is fully reducible if all irreducible factors are linear. 3. Of an ideal, being the intersection of two ideals that are both distinct from it; otherwise the ideal is irreducible. Every prime ideal is irreducible. 4. Of a linear transformation on a vector space, fixing two complementary subspaces; in the case of a Hilbert space, these are orthogonal. 5. Of a set of linear transformations on a vector space, fixing a nontrivial subspace.

REDUCING: Decreasing the size of the laboratory sample or individual particles or both. Division of the size of the laboratory sample may be accomplished manually by coning and quartering or by riffing or mechanically by rotary dividers. Reduction of particle size may be accomplished by milling or grinding. Simultaneous division and reduction may also be achieved with mills having stream diverters.

REDUCING AGENT (REDUCTANT, REDUCER): A chemical reagent such as hydrogen, carbon monoxide, carbon and metals in a redox (reduction-oxidation) reaction that causes reduction of a substance by donating electrons to it atom or group of atoms. It undergoes oxidation in the process.

REDUCING ATMOSPHERE: An atmosphere of a planet or moon which has a high hydrogen content, either in the form of free hydrogen or hydrogen-containing compounds, such as methane or ammonia.

REDUCING SUGAR: Any of the sugars such as monosaccharides, for example, glucose and fructose or disaccharides (maltose and lactose) that possess free ketone ($-\text{CO}-$) or aldehyde ($-\text{CHO}$) groups and can donate electrons to another molecule and therefore act as a reducing agent of many metallic salts such as copper, silver or bismuth. Reducing sugars can be detected by Benedict's solution test.

REDUCTASE: An enzyme that catalyses a reduction reaction. Examples are 5-alpha reductase, dihydrofolate reductase, hmg-coa reductase, methaemoglobin reductase and ribonucleotide reductase.

REDUCT: A structure derived from another by removing all occurrences of some set of symbols; A is thus a reduct of B if B is an extension of A.

REDUCTIO AD ABSURDUM: In logic, a method of proving the falsity of a premise by showing that its logical consequence is absurd or contradictory. The Euclidean proof of the infinitude of primes may be viewed as of this form, since it argues that, were any list $p_1, p_2, \dots, p_n$ to exhaust the primes, then the number $(p_1 p_2 \dots p_n) + 1$ would be neither prime nor composite.

REDUCTION: 1. A chemical reaction that brings about an increase in electrons, a loss of oxygen or a gain in hydrogen or the change in the metallic state of an ore or metal by removing impurities. 2. Simplification of an expression or equation without changing its value such as cancelling of common factors in the denominator and numerator of a fraction or the converting of a fraction into decimal form. 3. A type of cell division meiosis that halves the number of chromosomes in the two resultant cells.

REDUCTION DIVISION (MEIOSIS): The first cell division in sexually reproducing organisms that produces reproductive cells. The number of chromosomes in reproductive cells is reduced from diploid to haploid, resulting in each daughter cell receiving one member of each pair of chromosomes. This leads to the production of gametes in animals and spores in plants.

REDUCTION FORMULA: A formula that expresses a desired value in terms of more easily or previously calculated values. For example, any trigonometric function whose argument is $90^\circ \pm \theta$; $180^\circ \pm \theta$ and $360^\circ \pm \theta$ can be written simply in terms of θ . Integral reduction formula

often occur with the use of integration by parts as in integrating $\cos^x x$.

REDUCTION HALF-EQUATION: The separate equation which describes an oxidation or reduction half-reaction and the substance that is reduced.

For example, in the equation $\text{CuSO}_4 + \text{Zn} \rightarrow \text{ZnSO}_4 + \text{Cu}$; the oxidation half-reaction is given by $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$; the reduction half-equation is given by $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$.

The standard way of representing them is the oxidised form $+ \text{ne}^- \rightleftharpoons$ reduced form.

REDUCTION DIVISION (FIRST MEIOTIC DIVISION; FIRST MEIOSIS): The first cell division in meiosis, the process by which germ cells are formed. In reduction division, the chromosome number is reduced from diploid (46 chromosomes) to haploid (23 chromosomes).

REDUNDANCY: In computing, it refers to the duplication or mirroring of a device or data for backup purposes. The goal of redundancy is to prevent or recover from the failure of a specific component or system.

REDUNDANT: 1. In mathematics, implied by the remaining members of the system and so unnecessary in the sense that nothing derivable from the redundant element would fail to be derivable without it. 2. Spurious or vacuous; for example, when both sides of an equation containing roots are squared extraneous roots are introduced.

REDUNDANT ARRAY OF INDEPENDENT DISKS (RAID): Originally called **redundant array of inexpensive disks**, it is an assortment of hard drives connected and set up in ways to help protect or speed up the performance of a computer's disk storage. It is a method of storing the same data on different disks (places); it helps to increase fault tolerance of the system. In the OS, however, it appears as a single disk. There are about nine types of RAID in use that include RAID0, RAID 1 and RAID 2. RAID is commonly used on servers and high performance computers.

REDWATER (BABESIOSIS): A parasitic disease of the red blood cells caused by a parasite of the genus *Babesia* and transmitted among cattle by the common tick. The signs include fever, salivation, abortion of pregnant animals, diarrhoea followed by constipation and red urine. The acute form of the disease is often fatal.

REED: An aquatic plant of the family Graminae growing near shores of lakes. There are many varieties and uses include as thatch for roofs, instruments, bagpipes and reed organs.

REED SWITCH: An electrical switch invented by W. B. Ellwood in 1936 for the Bell Telephone Laboratories. It contains two or three thin iron strips or reeds inside a hermetically sealed tube. It is operated by an applied magnetic field, which may be an electrical coil, which induces magnetism in the reeds which then attract if a magnetic field is present or repel each other if there is no magnetic field. This makes or breaks contacts to supply or stop the flow of electric current to circuits. One practical application of a reed switch is as a proximity switch for sending signal for the activation of an alarm circuit.

REEF: A bar of rock, sand, coral or similar material, at or near the surface of water. **Coral reefs** are large underwater structures composed of the deposits of skeletons of corals found at or near the surface of warm, clear, shallow tropical waters. Coral reefs are economically and environmentally important ecosystems providing shelter and nutrients to diverse marine organisms such as fish, plants, and others. There are three types of coral reefs: (i) A **barrier reef** is one that surrounds an island and creates a lagoon between the reef and the beach; (ii) **atolls** are circular reefs with a central lagoon, formed on top of submerged volcano cones; (iii) **fringing reefs** are found close to newer volcanic islands.

REFECTION: Consumption by an animal of its own faeces. This may be beneficial to the recovery of the animal if it is suffering from any vitamin deficiency disease as the faeces is enriched with vitamins synthesised by intestinal bacteria.

REFERENCE: 1. In logic, the relation between a word, phrase, individual constant or other expression and the entity it refers to or picks out. 2. The entity referred to by an expression, the referent.

REFERENCE ANGLE: An acute angle in the first quadrant with vertex at the origin that is used to find the trigonometric functions of angle in other quadrants.

REFERENCE ATOM: The atom to which a primitive change is nucleophilic, electrophilic, nucleofugal or homolytic.

REFERENCE CONFIGURATION: An arbitrary fixed configuration of a body with respect to which its motion is described.

REFERENCE DOSE, Rfd (REFERENCE DOSE, Rfc): An estimate of daily exposure to the human population (including sensitive subgroups) that is likely to be without appreciable risk of deleterious effects during a lifetime.

REFERENCE ELECTRODE: 1. An external electrode system which comprises an inner element, usually mercury–mercury(I) chloride (calomel), silver–silver chloride or thallium amalgam–thallium(I) chloride, a chamber containing the appropriate filling solution and a device for forming a liquid junction, for example, capillary, ceramic plug, fritted disc or ground glass sleeve. 2. An electrode that maintains a virtually invariant potential under the conditions prevailing in an electrochemical measurement and that serves to permit the observation, measurement or control of the potential of the indicator (or test) or working electrode.

REFERENCE FRAME: 1. A system for locating numbers within a given context, usually with reference to an origin or zero point. Examples are number lines, clocks, calendars, temperature scales, and maps. 2. Also called **frame of reference**, it is any set of planes or curves, such as the three coordinate axes, used to measure the position, orientation, and other properties of a point in space.

REFERENCE MATERIAL: A substance or mixture of substances, the composition of which is known within specified limits and one or more of the properties of which is sufficiently well established to be used for the calibration of an apparatus, the assessment of a measuring method or for assigning values to materials.

REFERENCE METHOD: A method having small, estimated inaccuracies relative to the end use requirement. The accuracy of a reference method must be demonstrated through direct comparison with a definitive method or with a primary reference material.

REFERENCE PROCEDURE: Agreed method for determining one or more air quality characteristics where it is not practical to produce a reference material. The result obtained is defined as the measure of the air quality characteristic. For example, a gas such as NO₂ may be somewhat unstable when stored in a tank at low mixing ratios, so that the use of a more stable NO standard mixture in N₂ may be oxidised to NO₂ by O₃ in a specified manner referred to as the NO₂ reference procedure.

REFERENCE STATE: The state in which the element is stable at a chosen standard-state pressure and for a given temperature.

REFERENCE VALUE PH STANDARD: An aqueous solution of 0.05 mol potassium hydrogen phthalate per kg water which is the reference value for the definition of pH(RVS) values in the temperature range 0 - 95°C.

REFERENT: In logic, the object to which a formal or linguistic expression refers.

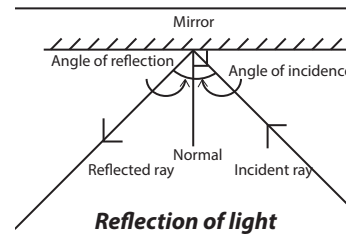
REFINEMENT: 1. In mathematics (topology), of a cover, another cover such that each member of the second lies inside a member of the first. 2. Of a partition of an interval on the line a new partition constructed by further division of members of the original partition. 3. Of a normal series, a normal series containing every member of the given normal series.

REFINING: The process of purifying substances by removing the impurities from it. Refining processes include **electrolysis** for refining of various metals such as copper; **amalgamation process** for metal refining, **fractional distillation** of petroleum, etc. Industrial or commercial refining of substances such as crude oil and sugar is done at a place called refinery.

REFLECTANCE (RADIANT REFLECTANCE; REFLECTIVITY): The ratio of the total amount of radiant flux reflected by a surface to that falling on it.

REFLECTING TELESCOPE: A telescope in which light from the object is initially reflected by a single or combination of concave mirrors.

REFLECTION: 1. A process in which radiation or waves such as light or sound in one or more medium meets the surface of a second and is thrown back to the first. The reflection of sound waves from a surface is called an **echo**. An object such as plastic, metal or glass that reflects light is called a **reflector**. 2. Geometrical transformation of a point or a set of points from one side of a point, line or plane to a symmetrical position on the other side.



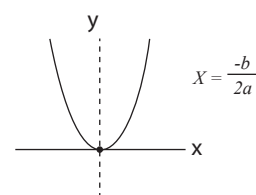
REFLECTION ELECTRON ENERGY LOSS SPECTROSCOPY (REELS): Any technique in which a specimen is bombarded with a focused low energy ($E_0 < 10$ eV) electron beam and the energy distribution of the reflected electrons is measured. This energy distribution contains features corresponding to discrete losses of energy of the reflected electrons due to excitation of vibrational and plasmon states and provides information on the type and geometric structure of compounds at the surface of the specimen.

REFLECTION HIGH ENERGY ELECTRON DIFFRACTION (RHEED): Any technique which measures the angular intensity distribution of electrons 'reflected' from a crystalline surface under bombardment with high energy electrons near grazing incidence. The diffraction pattern provides very surface sensitive information (information depth 1 nm) on the atomic arrangement of the top layers of a solid.

REFLECTION PRINCIPLE OF SCHWARZ: In complex functions, the theorem concerning analytic continuation that if a function is analytic on a domain that contains one or more admissible segments of a straight line or circle in its boundary, while it is continuous thereon and these segments are mapped to similar segments, then the function can be continued analytically across the segments by reflection or inversion. The reflection is most easily viewed for the case of a function that is real on a real segment. There is a corresponding principle for harmonic functions.

REFLECTION SYMMETRY (AXIS OF SYMMETRY; LINE OF SYMMETRY): A line through a plane or a 2-dimensional (2-D) shape so that each side of this line is a mirror image of each other. In other words, it is a line that divides a figure into two symmetrical parts in such a way that the figure on one side is the mirror image of the figure on the other side. For example, a quadratic curve and the letter **W** each has one axis of symmetry; the equilateral triangle has three lines of symmetry; and a circle has an infinite line of symmetry. The six axes of symmetry of a regular hexagon, for example, are all the bisectors of its angles and sides. The equation

$$y = ax^2 + bx + c$$



A parabola with its line of symmetry and its equation

is that of a parabola. Its line of symmetry passes through the vertex with its equation $x = -\frac{b}{2a}$.

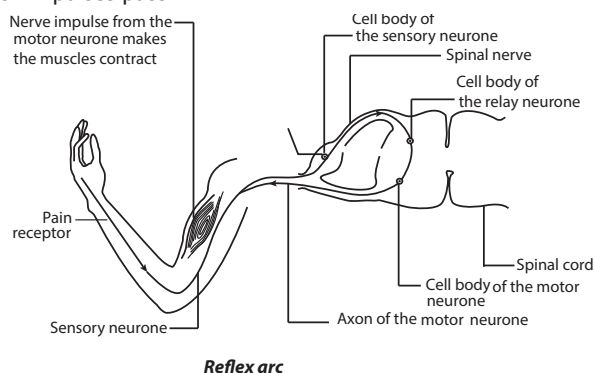
REFLECTIVE ANISOTROPY SPECTROSCOPY (RAS): A spectroscopic technique for studying surfaces by reflection of light or ultraviolet radiation. A beam of plane-polarised monochromatic radiation is directed at right angles onto a single-crystal surface and the reflected radiation along the surface is measured in two mutually perpendicular directions.

REFLEX ACTION: An automatic and involuntary response of the nervous system to a particular stimulus made by an organism. It is extremely rapid and the latency or the time taken to detect the

stimulus and the reflex action is about a fiftieth of a second ($1/50$ s) in humans. An example of reflex action is the removal of a body part such as the hand from a hot sensation such as fire.

REFLEX ANGLE: An angle between 180° and 360° .

REFLEX ARC: A nerve pathway consisting of receptors, sensory neurone, relay neurone and reflector muscles or glands through which impulses pass.



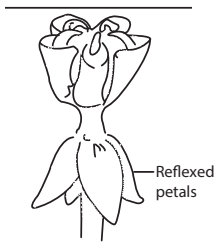
Reflex arc

REFLEX CAMERA: A camera for viewing and focusing with mirror and prisms to reflect light passing through the lens into the viewing screen, so that the object scene can be evaluated. At the moment the picture is taken, a spring action removes the mirror away from the path between the lens and film permitting light to reach the film.

REFLEX PROPERTY OF EQUALITY: An axiom that states an obvious fact $a = a$, for all a . That means that any number is equal to itself.

REFLEXED: 1. Bent backward or downward, as in milkweed, shown in the diagram. 2. An involuntary action or response, such as a sneeze, blink or hiccup. 3. An image produced by reflexion. 4. In psychology, it refers to an unlearned or instinctive response to stimuli.

REFLEXIVE: A mathematical relationship among the elements of a set such that they are self-related or a relation R defined on a given set is said to be reflexive if every member of the set has a relation to itself. Equality is an example of a reflexive relation. Equality is an equation in which the quantities on each side of an equal sign are the same. These are commonly denoted by letters or other symbols, as in the equations $x^2 + x - 4 = 8$, $y = \sin x + x$ and $3y = \log x$.



REFLUX: A distillation process for boiling in which vapour is condensed, cooled by a reflux condenser and returned as li

REFORESTATION: Planting of forests on lands that have previously contained forests but that have been converted to some other use or left fallow after cutting the trees. Reforestation could be a natural occurrence or done by man. Reforestation can help reduce the amount of CO_2 in the air, and also protect important species of animals.

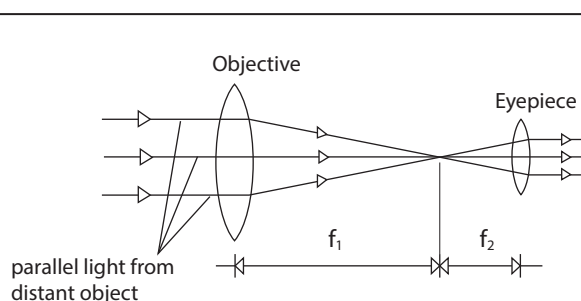
Afforestation refers to land that has been without forest for a much longer time, which is now being planted with trees.

REFOLIATE: Producing leaves, usually after rain, wind or diseases.

REFORMING: A thermal or catalytic process of re-arranging molecules of hydrocarbons in a chemical process to convert low octane number petroleum into products of higher octane number.

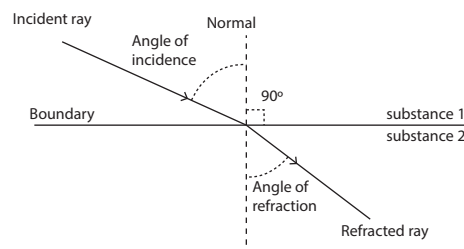
REFRACTED: 1. Bent backward from the base, as in leaves. 2. To defect from a straight path by refraction, as in lights.

REFRACTING TELESCOPE: A telescope in which a lens receives light and forms an image of the object, which is viewed through a magnifying lens.



Refracting telescope

REFRACTION: The change in direction or bending of a wave of light, heat or sound due to a change in its speed, when it passes from one medium to another.



Refraction of light

REFRACTION EFFECTS: The luminescence flux emitted from the interior of a rectangular sample reaching a photodetector placed at some distance from the sample is decreased by a factor of approximately n^2 (where n is the refractive index of the medium) compared with a medium whose refractive index is 1.0. Such effects are termed refraction effects.

REFRACTIVE INDEX: A measure of the refraction of a beam of light entering a denser medium from a transparent one. Refractive index is the ratio between the sine of the angle of incidence of the ray of light and the sine of the angle of refraction. For example, if i is the angle of incidence and r is the angle of refraction, then the refractive index, $\mu = \frac{\sin i}{\sin r}$. Refractive index is used as a measure of the purity of sugar, salt, diamond, oils, etc.

REFRACTOMETER: Any device or instrument used for the measurement of the refractive index of a substance or a medium. Examples are the Rayleigh and Pulfrich Refractometers.

REFRACTORY: A material that is able to withstand high temperatures without changing its chemical or physical structure and mostly used in lining furnaces, kilns, incinerators and reactors, etc. Refractories are mostly made from materials such as silica, magnesia, fireclay, alumina and lime.

REFRACTORY PERIOD: A brief period of time (lasting between 0.5 to 10 milliseconds) following the passage of an action potential, when the axon is no longer excitable. Refractory period has two periods: absolute refractory and relative refractory. **Absolute refractory period (ARP)** is a point of time after a nerve (cell) axon has produced an action potential that another action potential cannot be produced, no matter how strong the stimulus is. This occurs because the voltage-gated sodium channels which must open to produce an action potential have been inactivated. **Relative refractory period (RRP)** is a point of time after a nerve (cell) axon has produced an action potential during which another action potential cannot be produced with the threshold stimulus, but a larger than the threshold stimulus can produce an action potential.

REFRESH (RELOAD): In computer science, it is a command that is used to reopen a currently displayed window or Web page.

REFRESH RATE (SCAN RATE): A CRT monitor measurement in Hz that indicates how many times per second a monitor screen image is renewed. It has a vertical and horizontal value. The higher the refresh rate, the less image flicker will be noticed on the screen. Typical refresh rates for CRT monitors include 60, 75 and 85 Hz. Some monitors support refresh rates of over 100 Hz.

REFRIGERANT: A substance used in a heat transfer system such as an air conditioner or a refrigerator, which operates as a reversible phase change from a gas to a liquid. Examples are ammonia, sulfur dioxide, Freon and chlorofluorocarbons (CFCs). Some refrigerants such as ammonia, carbon dioxide and non-halogenated hydrocarbons are not destructive to the ozone layer; however some types such as R-12 and R-22 have been banned in some countries due to their destructive effects on the ozone layer.

REFRIGERATION: The heat exchange technique of removing heat from a substance in an enclosed area and maintaining it at a temperature below that of its surroundings. It is employed to preserve food by chilling and deep-freezing. Refrigerator and air conditioner are the devices often used to achieve this heat transfer and maintain temperature. The basic unit consists of a motor/pump called a compressor, which compresses and circulates a gas known as a refrigerant, which heats up under pressure. The heat is dissipated by heat exchanging coils outside and inside the unit and the refrigerant gas is condensed into liquid at high pressure. The high pressure liquid flows through an expansion valve, which reduces the pressure of the liquid, which boils and evaporises and drops to a very low temperature making the enclosed space such as a room or refrigerator cool. The cold gas is returned to the compressor for another cycle. A thermostat is often employed to control the temperature of the enclosed space. For an efficient working system, there must be no leakage in the system and the enclosed space must not be opened to the atmosphere.

R

REGELATION: A phenomenon in which water is refrozen after being melted under pressure, for example, beneath a glacier or under pressure from a skater.

REGENERATION: 1. Re-growth and restoration of normal functions of an organism's organ or tissue after loss or damage. This is common in some amphibians, fish, plants and even some mammals. Complex organisms such as man have less chances of regeneration. 2. Reversal of catalytic deactivation with restoration of the original catalytic activity.

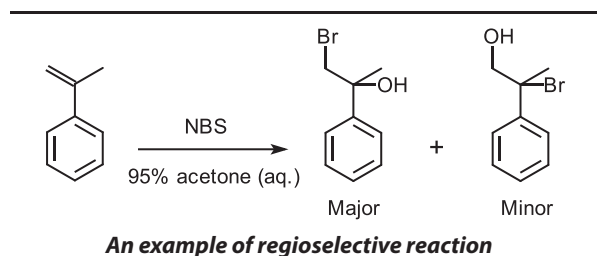
REGION: 1. In mathematics, a connected subset of two-dimensional space, such as the set of points (x, y) such that $x > 0$, $y \geq 0$ and $x^2 + y^2 < 1$. 2. Any set that is the union of an open connected set (domain) with some, none or all of its boundary points.

REGION OF DIVISION: The zone of cell division in the tip of a plant root composed of growing and dividing meristematic cells.

REGION OF ELONGATION: The zone in the tip of a plant root where one daughter cell elongates, pushing the root tip into the soil, increasing the length of the root.

REGION OF MATURATION (DIFFERENTIATION): The zone in a root where the cells of the primary tissues differentiate into specialised cells of the primary root such as the epidermis, the endodermis, the cortex, the xylem, the phloem and the pericycle.

REGIOSELECTIVITY (REGIOSELECTIVE): In chemical reactions, the preference of complex molecules to bond or break over all other possible positions, due to electronic distribution patterns. The reaction below shows regioselectivity:



REGISTER: In computer processor, it is a storage location of the central processing unit in a computer that can be accessed rapidly. Registers are 16, 32 or 64 bits wide.

REGISTERED JACK-11, RJ-11 (PHONE CONNECTOR; PHONE JACK): A four or six wire connection primarily used for telephones and computer modem connectors.

REGISTERED JACK-45 (RJ-45): An 8-pin connection used for Ethernet network adapters. This connector is most commonly connected to the end of Cat5 cable, which is connected between a computer network card and a network device such as a network router.

REGISTRY: A database used by Microsoft Windows to store configuration information about the software installed on a computer. This information includes settings such as the desktop background, program settings and file extension associations.

REGMA: A dry fruit that is characteristic of geranium family, consisting of three or more cells that break up into one seeded parts, to release seed.

REGNAULT'S METHOD: A technique for measuring the volume co-efficient of a gas, named after the French chemist Henri Victor Regnault (1810–78) in which a glass bulb of known volume is weighed and evacuated with a vacuum pump and hot air is used to dry any moisture in it. Gas at known pressure is admitted into the glass bulb and reweighed. The process is carried out at constant known temperature and the result corrected to standard temperature and pressure.

REGOLITH (MANTLE ROCK): The surface layer of loose, heterogeneous material composed of dust, soil, volcanic ash, broken rock and related materials that covers most rocks. Regolith is present on Earth and in various forms and compositions on the Moon and other planets.

REGRESS: In mathematics, a supposed definition or explanation each stage of which requires to be similarly defined or explained, often a recursion formula without a base clause. For example, if a natural number is defined simply as whatever is the successor of a natural number, then in order to show to be one, we have to first show this of $n - 1$ and to show that we first need to consider $n - 2$ and so on; this never terminates since there is no base clause to specify the first element, as in Peano's axioms.

REGRESSION: In statistics, a measure that determines the strength of the association between one dependent variable and a series of changing variables (known as independent variables) which have parametric coefficients that may enable future values of the dependent variables to be predicted.

REGRESSION ANALYSIS: A method used to model and calculate the relationships among several variables. It is used to predict how the outcome of any variable known as dependent variable changes when any of the prediction variables known as independent variables changes. Once the relationships have been established between the variables, a linear equation of the form $Y = a_0 + a_1X$ is used to predict the value of the outcome variable for different values. Its uses include forecasting and predicting events.

REGRESSION LINE: A line of the form $y = bx + c$, is the regression line of y on x . This gives the expected value of a random variable y conditional on the given value of a random variable x .

REGULA FALSI (FALSE POSITION; RULE OF FALSE POSITION):

A method of solving equations by using a trial and error values for the variables and then adjusting the values accordingly. For example, to find a solution of $x^2 + y^2 = 50$, test values like $x = 5, y = 6$ might be attempted, which gives $x^2 + y^2 = 25 + 36 = 61$, which is more than 50; therefore smaller values for the variables would have to be guessed until the equation is solved. There are two types of false position methods: (i) **simple false position** for solving problems involving direct proportion that can be written algebraically in the form $ax = b$, for the value of x to be found if the values of a and b are known. (ii) **Double false position** is used for solving more difficult problems that can be written algebraically in the form: $f(x) = b$ for the value of x to be determined, if it is known that $f(x_1) = b_1$ and $f(x_2) = b_2$.

REGULAR: 1. In botany, also known as **actinomorphic**, it refers to a radially symmetrical flower; or a flower in which all members of each set of the flowering parts such as the stamens, pistils, sepals and petals are similar in shape and size. Two mirror halves can be obtained by cutting a longitudinal section through many vertical planes. Examples include petunia, buttercup, wild rose, daisies and lilies. 2. A regular geometric figure has all of its sides and angles being equal. A prism has regular polygons as bases. A polyhedron has identical regular polygons as faces that make identical angles with each other. A pyramid has a regular polygon as a base and the altitude passing through the centre of the base.

REGULAR APPROXIMATING SEQUENCE: In mathematics, it is a strictly increasing sequence of real-valued functions that are all bounded and measurable, which converge to a given function almost everywhere.

REGULAR COKE: A petroleum coke with good graphitisability, characterised by a combination of properties which differ significantly from those of metallurgical coke but do not reach the quality level of premium coke. These properties are: optical anisotropy, medium reversible thermal expansion and low ash content. Regular coke is mainly used for the production of synthetic carbon and graphite materials.

REGULAR FUNCTION (ANALYTIC FUNCTION; HOLOMORPHIC FUNCTION): A complex function that is single-valued and differentiable at every point of its domain and in consequence possessing derivatives of all orders and agreeing with its Taylor series locally. The function may not be differentiable everywhere, it is only analytic at points where the function is differentiable. For example, $\exp z$ is holomorphic on the complex plane and $\log z$ is holomorphic on $\mathbb{C} \setminus (-\infty, 0)$.

REGULAR MACROMOLECULE: A macromolecule the structure of which essentially comprises the repetition of a single constitutional unit with all units connected identically with respect to directional sense.

REGULAR OLIGOMER MOLECULE: An oligomer molecule the structure of which essentially comprises the repetition of a single constitutional unit with all units connected identically with respect to directional sense.

REGULAR POINT: In mathematics, of a holomorphic function, f , analytic on an open disk, is a point on the boundary of the open disk that can be surrounded by another disk on which there is an analytic function, g , which agrees with f at all points on the latter disk. A boundary point that is not a regular point is a singular point.

REGULAR POLYGON: A closed two dimensional many sided plane figure with all its sides and internal angles equal. The sum of its interior angles is given by $(n - 2)180^\circ$, where n is the number the sides of the polygon. The measure of its exterior angles is given by $360^\circ/n$.

REGULAR POLYHEDRON: A solid figure with all its faces having the shape of congruent regular polygons. There are five possible regular polyhedrons known as the Platonic solids: tetrahedron (four sides), hexahedron or cube (six sides), octahedron (eight sides), dodecahedron (12 sides) and icosahedrons (20 sides).

REGULAR POLYMER: A polymer composed of regular macromolecules, regular star macromolecules or regular comb macromolecules. A polymer consisting of star macromolecules with arms identical with respect to constitution and degree of polymerisation is considered to be regular. Analogously, a polymer consisting of comb macromolecules with the sub-chains between the branch points of the main chain and the terminal sub-chains of the main chain identical with respect to constitution and degree of polymerisation and the side-chains identical with respect to constitution and degree of polymerisation is considered to be regular.

REGULAR REFLECTION: Bouncing back of light from a smooth, even surface such as a clean mirror.

REGULAR SINGULAR POINT: In mathematics, a point a of a second order differential equation $y'' + P(x)y' + Q(x)y = 0$, for which a is a singular point (either P or Q is not real analytic at a and $(x - a)P(x)$ and $(x - a)^2Q(x)$ are real analytic around a).

REGULAR SOLID (PLATONIC SOLID): Any of the five regular polyhedrons, which are solids with regular polygon faces and the same number of faces meeting at each corner that are possible in three dimensions. They are the tetrahedron (a pyramid with triangular faces), the octahedron (an eight-sided figure with triangular faces), the dodecahedron (a 12-sided figure with pentagonal faces), the icosahedron (a 20-sided figure with triangular faces) and the hexahedron or cube. A regular solid with hexagonal faces cannot exist because if it did, the sum of the angles of any three hexagonal corners that met would already equal 360° , so such an object would be planar (of, relating to, or lying in a plane).

REGULARITY CONDITION: In mathematics, any condition imposed on a problem to guarantee that it meets the demands of a theorem or method, as with a constraint qualification.

REGULATION: In embryology, it is the process in which animal embryonic development continues normally, after any injuries or abnormalities to its structure during its developmental stages.

REGULATIVE DEVELOPMENT: A pattern of development, such as that of a mammal, in which the early blastomeres retain the potential to form the entire animal.

REGULATOR: A control device which may act as a feedback, used to maintain the constant value of some systems such as fluid, pressure and temperature.

REGULATORY ENZYME: Any enzyme, either active or inactive that controls the metabolic pathways such as hormone production in the cells. The two types of regulatory enzymes are: **allosteric enzyme**, which has two different binding sites on its surface and **covalently modulated enzyme**, which converts the enzyme into one with more activity and the other with less activity.

REGULATORY GENES (I-GENES; REGULATION OF GENE EXPRESSION): Genes that control development by regulating the expression of structural genes responsible for the formation of body components. They encode transcription factors, which interact with regulatory sites of other genes causing activation or repression of developmental pathways.

REGULATORY PROTEIN (GENE-REGULATORY PROTEIN): 1. Any special protein that controls or regulates the transcription of genes by switching the genes on and off. This may involve switching off the action of RNA polymerase, which initiates transcription. 2. The two proteins, troponin and tropomyosin that regulate the contraction of muscle by controlling the interaction of myosin and actin.

REGULUS: 1. A semi-purified mass of metal that sinks to the bottom of the crucible during smelting. 2. One of the brightest stars at night and located in the constellation Leo. It is 78 light years from the Earth. Its surface temperature is $11,727^\circ\text{C}$ ($12,000^\circ\text{K}$).

REGURGITATION: 1. The return of partially digested or undigested food from the stomach to the mouth. This is common in some bird and animal species, as a means of feeding their young as they are immature to go out to feed by themselves. For example, worker termites have protozoans in their guts which digest cellulose and

lignin in plant cells. After the digestion of these plant parts, the worker termites regurgitate the food to feed the other castes. Ruminants also regurgitate. 2. The circulation in the opposite direction to the normal especially due to defective heart valve.

REIN: A long narrow strap used to control a horse by a rider. Each end of the strap is attached to the bit in the opposite sides of the mouth.

REINFORCEMENT: Increasing the frequency of a particular behaviour through conditioning by arranging for stimuli such as reward or punishment called **reinforcer** to take effect immediately after a response that results in an increase in the future rate or probability of that response. Reinforcement is used as a basic tool to modify behaviour and stimulate learning. There are two types of reinforcement: **positive reinforcement**, which is giving an incentive such as food after a performance; and **negative reinforcement**, which is not giving or removing an unpleasant incentive to improve behaviour.

REISSERT COMPOUNDS: Compounds formed by formal addition of an acyl group and a cyano group to a nitrogen atom and carbon atom, respectively, of a nitrogen-carbon bond in quinolines, isoquinolines and related nitrogen heterocycles.

REISSNER'S MEMBRANE (VESTIBULAR MEMBRANE; VESTIBULAR WALL): Named after a German anatomist Ernst Reissner (1824-1878), it is the membrane in the cochlea of the inner ear that separates the vestibular canal from the cochlea duct.

REJUVENATION: 1. To stimulate a river or stream to erode the land it runs over by uplifting or removing a barrier in the stream bed. 2. To develop youthful topographic features in a previously levelled area by the action of rejuvenated streams.

RELATED ANGLE: In mathematics, the acute angle for which the trigonometric functions have the same absolute values as for some given angle, so that $\phi = n\pi \pm \theta$, where ϕ is the given angle, n is an integer and θ the related angle, in radians.

RELATION: 1. An association between elements of one set and that of another set usually represented in ordered pairs containing one object from each set. In other words, it is an association between elements of one set and that of another set. A relation can be represented on a graph, a table of values or by any rule in words or symbols that determine pairs of values. 2. It is also the rule connecting the association or matching of members of one set to another.

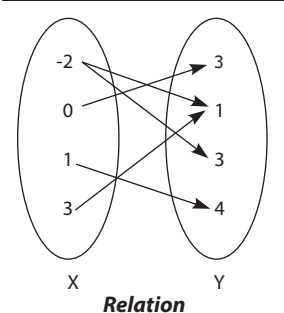
RELATIONSHIP SYMBOLS:

Symbols that show how numbers or terms of an equation relate to each other. The following are some of the commonly used symbols and their meanings, enclosed in parenthesis: = (is equal to), $\neq$ (is not equal to), < (is less than), > (is greater than), $\leq$ (is less than or equal to), $\geq$ (is greater than or equal to) and $\approx$ (is approximately equal to).

RELATIONAL DATABASE: A database structure in which data is organised into related tables, with the rows in the table containing the records.

RELATIONAL THEORY: A type of theory in which absolute space and absolute time do not occur, with the theory depending on relationships between events. In a relational spacetime theory, space does not exist unless there are objects in it; nor does time exist without events. The relational view proposes that space is contained in objects and that an object represents within itself relationships to other objects. Newtonian mechanics is not a relational theory but the general theory of relativity is nearer to being a relational theory. A theory that fully incorporated Mach's principle would be a relational theory.

RELATIVE: 1. Existing or having its specific nature only by relation to external conditions; not absolute or independent. 2. Of a scientific quantity, being measured or stated relative to some chosen value or



subject to some stated assumption. 3. In mathematics, also known as **local**, a relative maximum is a point at which the value of a function is greatest within a neighbourhood of that point. 4. In statistics, proportional to a total; for example, relative frequency or relative error.

RELATIVE AGE: The geologic age of a fossil organism, rock, or geologic feature or event defined relative to other organisms, rocks, or features or events rather than in terms of years.

RELATIVE ADDRESS: A memory address that is some distance from a starting point, called the base address. The absolute address is derived by adding the relative address to the base address.

RELATIVE ANGULAR MOMENTUM (MOMENT OF RELATIVE MOMENTUM): Of a particle about a point with position vector $\vec{p}$, the quantity $m(\vec{x} - \vec{p}) \times (\dot{\vec{x}} - \dot{\vec{p}})$, where m is the mass and $\vec{x}$ is the position vector of the particle.

RELATIVE APERTURE (FOCAL RATIO; F-RATIO, F-NUMBER; F-STOP): A dimensionless number that is a quantitative measure of lens speed, an important concept in photography. It is the ratio of the equivalent focal length of a lens to the effective aperture diameter of the lens; written as f/n , $f:n$ or f_n , where n is the numerical value of this ratio and is equivalent to the f -number.

RELATIVE ATOMIC MASS: Formerly known as **atomic weight**, it is the weighted average of the masses of the constituent isotopes of a chemical element or the ratio of the average mass of a chemical element's atoms to $1/12^{\text{th}}$ the mass of an atom of the carbon-12 isotope. Its symbol is A_r . Measuring the masses of atoms in grams, for example, would imply dealing with extremely small numbers; thus, the mass of an atom is compared with that of an atom of carbon-12. In calculating the relative atomic mass of an element with isotopes, the relative mass and proportion of each is taken into account. For example, naturally occurring chlorine consists of atoms of relative isotopic masses 35 (75%) and 37 (25%). Its relative atomic mass is

35.5, i.e., $A_r = \left(\frac{75}{100} \times 35\right) + \left(\frac{25}{100} \times 37\right) = 35.5$. The relative masses of atoms are measured using an instrument called a mass spectrometer, invented by the English physicist Francis William Aston (1877-1945). The cations formed are brought down on to a detector in turn according to their mass-to-charge ratio. The instrument provides a measure of the relative mass (compared to ^{12}C) and the relative number of each isotope. It was in his use of this instrument that the existence of isotopes of elements was discovered. Relative masses have no units.

RELATIVE AUTOMORPHISM: An automorphism of an extension field that leaves the base field fixed.

RELATIVE BIOLOGICAL EFFECTIVENESS (RBE): The relative damage caused to living tissue by different types of radiation, which affects living organisms differently and defined as the ratio of the absorbed dose of a standard amount of radiation to the same amount required to cause the same biological level of effect.

RELATIVE BIOLOGICAL EFFECTIVENESS (QUALITY FACTOR): The linear-energy-transfer dependent factor by which absorbed dose is multiplied to obtain dose equivalent, or constant of proportionality between dose equivalent and absorbed dose.

RELATIVE COMPACTNESS: Of a set in a topological space, possessing a closure that is compact. Every bounded subset of Euclidean space is relatively compact.

RELATIVE COMPLEMENT: The members of a set lying outside another set. The relative complement of A in B , is written $B \setminus A$.

RELATIVE COUNTING: A measurement in which the activity of a test portion is derived from the ratio between the count rates observed for the test portion and for a radioactive source of known activity.

RELATIVE DATING: A dating method which places events in chronological order without requiring that a specific numerical age be assigned to each event. One such method used in relative dating

of sedimentary and volcanic rocks is studying the layers of strata in a sediment using stratigraphy.

RELATIVE DENSITY: The ratio of the density of a substance to the known density of a reference substance, usually with respect to pure water at a temperature of 4 °C (277 K), which is 1000 kg/m³ or simply 1. Relative density has no unit because it is the ratio of two similar quantities. It was formerly called **specific gravity**, it is without dimension. If the relative density of a substance is in reference to water and less than 1, then the substance can float in water and if greater than 1, then the substance will sink. Relative density is dependent on temperature and the value varies with temperature variations, for example, the relative density of water at 25 °C (298 K) is 0.997074. A spring balance is used to determine it. Relative density (R.D.) of a solid and liquid can be determined by the following formulae:

$$\text{Relative Density (R.D.) of a solid} = \frac{\text{weight of solid in air}}{\text{upthrust of water}}$$

$$\text{R.D. of liquid} = \frac{\text{weight of liquid displaced}}{\text{weight of equal volume of water displaced}}$$

$$= \frac{\text{weight of liquid displaced}}{\text{weight of water displaced by same solid}}$$

$$= \frac{\text{upthrust in liquid}}{\text{upthrust in water}}$$

RELATIVE DISPLACEMENT: The difference between the position vector $\mathbf{x}$ of a particle and a chosen point with position vector $\mathbf{p}$, i.e., $(\mathbf{x} - \mathbf{p})$.

RELATIVE ELECTRODE POTENTIAL: The electrode potential of a single electrode is not amenable to direct experimental determination and can only be measured against a reference system. The measured quantity, E_r is the relative electrode potential.

RELATIVE ENTROPY (KULLBACK LEIBER DISTANCE): In information science, the relative entropy is a measure of the distance between two distributions. It can be viewed as a measure of the inefficiency of assuming that the distribution is q , when the true distribution is p . Mathematically, relative entropy between two probability mass functions $p(x)$ and $q(x)$ is defined by

$$D(p||q) = \sum_{x \in \Omega} p(x) \log \frac{p(x)}{q(x)}$$

RELATIVE ERROR: Error of measurement divided by the true value of the measurand. Since a true value cannot be determined, in practice the conventional true value is used.

RELATIVE FITNESS: One of the two ways by which fitness can be measured, it is biological fitness expressed as the ratio of the absolute fitness of an individual or of a genotype or of a phenotype to the absolute fitness of a reference individual or of genotype or of phenotype. The other way of measuring fitness is called absolute fitness. The more likely that an individual is able to survive and live longer to reproduce, the higher is the fitness of that individual.

RELATIVE FREQUENCY: A way of estimating statistical data if they cannot be accurately interpreted. It is the ratio of the number of times an event has occurred divided by the total number of frequency.

RELATIVE GROWTH RATE (RGR): A measure of the productivity of a plant defined as the mass increase in dry mass per unit of plant mass over a specified period.

Class	Frequency	Relative frequency
1	3	3/15
2	4	4/15
5	6	6/15
7	2	2/15

$$\Sigma f = 15$$

Relative frequency table

RELATIVE HUMIDITY: The ratio of the amount of water moisture/vapour in the air to the saturated or maximum amount that the air could hold at a given temperature, expressed as a percentage.

RELATIVE IDENTITY: A relation of qualitative identity that defines a partition of its domain, so that the elements are equivalent for the purposes of the theory.

RELATIVE INTERIOR: The interior of a convex set in the induced topology determined by the affine span of the set. It is denoted $\text{relint}(A)$ or $\text{ri}A$. In infinite dimensions, one distinguishes this from the interior relative to the closed affine span. Every finite-dimensional convex set A has a nonempty relative interior.

RELATIVE MAXIMUM: A value of a function at a point x_0 which is equal to or greater than the values of the function at all points in some neighbourhood of x_0 .

RELATIVE MINIMUM: A value of a function at a point x_0 which is equal to or less than the values of the function at all points in some neighbourhood of x_0 .

RELATIVE MOLECULAR MASS (MOLECULAR WEIGHT): The ratio of the average mass per molecule of the natural nuclidic composition of an element or compound to one-twelfth of the mass of carbon-12 atom. Its symbol is M_r . For example, the relative molecular mass of sodium hydroxide (NaOH) is 40.

RELATIVE MOMENTUM: Of a particle about a point with position vector $\dot{\mathbf{p}}$, the quantity $m(\dot{\mathbf{x}} - \dot{\mathbf{p}})$, where m is the mass and $\dot{\mathbf{x}}$ is the position vector of the particle.

RELATIVE PERMEABILITY: 1. Also known as **dielectric constant**, it is the ratio of the permeability of a substance to the permeability of a vacuum at the same magnetic field strength. Its symbol is μ_r . It is expressed as $\mu_r = \mu/\mu_0$, where μ is the permeability of the medium and μ_0 is the permeability of free space or vacuum. It is an important parameter in characterising capacitors and materials with high dielectric constants are used in the manufacture of high-value capacitors. If all other factors such as temperature remain constant, the value of the dielectric constant increases as the electric flux density increases. This enables objects of a given size, such as sets of metal plates, to hold their electric charge for long periods of time and/or to hold large quantities of charge. Diamagnetic materials have constant relative permeabilities of slightly less than 1. Paramagnetic materials have constant relative permeabilities of slightly more than 1. The relative permeability of ferromagnetic materials increases as the magnetising field increases, reaches a maximum and then decreases. The reciprocal of magnetic permeability is magnetic reluctivity. 2. The specific permeability of a porous rock formation to a particular phase such as oil, water and gas at a particular saturation and a particular saturation distribution. It is the ratio of the effective permeability of that phase to the absolute permeability.

RELATIVE PERMITTIVITY: The permittivity of a material given relative to that of free space or a vacuum or simply the ratio of the permittivity of a substance to that of free space. It is also the ratio of the capacitance of a capacitor using that material as a dielectric, compared to a similar capacitor which has a vacuum as its dielectric. Its symbol is ϵ_r and can be expressed as the ratio of the permittivity of a substance, ϵ to the permittivity of vacuum: $\epsilon_r = \epsilon/\epsilon_0$.

RELATIVE REFRACTORY PERIOD (RRP): A point of time after a nerve (cell) axon has produced an action potential during which another action potential cannot be produced with the threshold stimulus, but a larger than threshold stimulus can produce an action potential.

RELATIVE SEA LEVEL: Sea level measured by a tide gauge with respect to the land upon which it is situated. Relative sea level changes occur by changes of the land with respect to the sea surface.

RELATIVE SEA LEVEL SECULAR CHANGE: Long-term changes in relative sea level caused by either eustatic changes, brought about by thermal expansion or changes in vertical land movements.

RELATIVE TIME: Geologic time that places events in a sequence relative to each other, especially time based on Law of Superposition.

RELATIVE TOPOLOGY (INDUCED TOPOLOGY): The topology defined on a subset of the underlying set of a given topological space by taking its open sets to be intersections of the open sets of the given topology with the subset. For example, the open sets of the induced topology on a line in the Euclidean plane are the segments of the line in the interior of the open disks.

RELATIVE VELOCITY: The vector difference between the velocities of two objects, as evaluated in terms of a single coordinate system. When two objects are moving with velocities V_C and V_D in a given direction the velocity of C relative to D is ${}_C V_D = V_C - V_D$ in that direction, where ${}_C V_D$ is the velocity C to D .

RELATIVELY COMPLETE (CONDITIONALLY COMPLETE): Of a partially ordered set, such that every set with an upper or lower bound has a supremum or infimum, respectively.

RELATIVELY CONSISTENT: Able to be shown to be consistent with respect to some larger theory, that is, having a model within that larger theory. An example of relative consistency theorems is: If Euclidean geometry is consistent then non-Euclidean geometry is consistent.

RELATIVELY PRIME (COPRIME): A set of numbers which do not have any other common factor other than 1 or numbers whose highest common factor is 1. For example, 8 and 9 have no other common factor other than 1 so they are co-prime numbers.

RELATIVISTIC: 1. A term used to describe any situation involving a speed close to that of light so that the behaviour is described by Einstein's theories of relativity that are significantly different from those of classical (Newtonian) physics. 2. Having to do with or based on the theory of relativity.

RELATIVISTIC BEAMING (DOPPLER BEAMING): An effect in which charged particles moving at a significant fraction of the speed of light will emit electromagnetic radiation in a narrow beam in the direction of motion. The faster the particles, the narrower the beam. In astronomy, relativistic beaming commonly occurs in two oppositely-directed relativistic jets of plasma that originate from a central compact object that is accreting matter.

RELATIVISTIC EFFECTS: Several peculiar effects such as time dilation, mass increase and length contraction, predicted by Einstein's special theory of relativity, which happen when objects such as spacecraft travel at speeds that are a substantial fraction of the speed of light.

RELATIVISTIC ELECTRONICS: A branch of electronics that involves the special theory of relativity. The free electron laser is an example of an application in relativistic electronics.

RELATIVISTIC JETS: Extremely powerful jets of plasma of matter or electromagnetic radiation that move away from some active galaxies at very large speeds. It is thought that the energy of these jets is supplied by the activity around the black holes, either through a Blandford-Znajek process or through a Penrose process.

RELATIVISTIC MASS: The mass of a body in motion at a velocity exceeding about one-tenth the velocity of light, measured by an observer in the same frame of reference as the body. Relativistic mass is significantly larger than the rest mass.

RELATIVISTIC MECHANICS: Any form of mechanics compatible with either the special or the general theory of relativity. It is concerned with the motion of bodies whose relative velocities approach the speed of light c or whose kinetic energies are comparable with the product of their masses m and the square of the velocity of light or mc^2 . Such bodies are said to be relativistic and when their motion is studied, it is necessary to take into account Einstein's special theory of relativity.

RELATIVISTIC QUANTUM THEORY: The formulation of quantum theory that applies to particles that have zero rest and move close to the speed of light.

RELATIVISTIC RUNAWAY THEORY: A theory of the origin of lightning in which it is proposed that lightning is initiated by a cosmic ray hitting a molecule in a cloud with an electric field. This releases electron into the cloud where it is accelerated in the field. This theory explains how natural lightning can originate and propagate with an average electric field of only 200kV/m.

RELATIVISTIC SPEED: A speed that is a significant proportion of the speed of light and makes the mass of a body significantly greater than its rest mass.

RELATIVITY THEORY (RELATIVITY; THEORY OF RELATIVITY): Albert Einstein's theory of space and time, comprising the special theory of relativity, which describes the causal structure of spacetime without gravity and the general theory of relativity, in which the causal structure becomes a dynamical entity that is partly determined by the distribution of matter and energy.

RELATUM: One of the entities by which a relation is related.

RELAXANTS (MUSCLE RELAXANTS): Agents or substances containing properties which relax the body when there is much tension and stress as well as bodily pains such as pain in the back, arms, knees, legs, neck, etc. Certain herbs or their derivations can be effective in treating this condition; such herbs include cannabinoids, peppermint oil, and lemongrass essential oil.

RELAXATION: 1. In physics, it is the return to the equilibrium state of a perturbed or disturbed system due to an external influence. The time for relaxation to take place is called the **relaxation time** (τ), expressed as a function of time t is an exponential law $\exp(-t/\tau)$. An example is the average time that a system remains in the higher energy state before it falls to the lower energy state. 2. In mathematics, it is a way of solving equations using a series of approximate solutions, each of which reduces the number of errors contained in the previous one, until the errors fall within acceptable limits.

RELAXATION OSCILLATOR: An oscillator, whose fundamental frequency is determined by the time of charging or discharging a capacitor or coil through a resistor or an electronic circuit giving a regularly repeating output that is not sinusoidal, producing waveforms that may be rectangular or sawtooth.

RELAXIN: A peptide hormone produced naturally by the corpus luteum and placenta in women during pregnancy that assists in childbirth. It works by relaxing the ligaments, thus widening the pelvic opening, dilates the cervix of the uterus and prevents premature labour by reducing uterine contraction.

RELAY: An electromagnetically operated switch, used to control a low power signal in a circuit, such as operating several circuits by the same source of signal.

RELAY INTERCROPPING: In agriculture, it is a type of growing two types of crops together that involves the planting of a second crop into a first crop that is partly through its growth but is not yet ready for harvesting.

RELAY NEURON (INTERNEURON): A nerve cell found within the central nervous system that transmits messages from the sensory neurons to motor neurons, allowing information to pass directly to the spinal cord, without reaching the brain. Relay neurons are involved in reflex actions, which are extremely rapid, causing the sudden removal of a limb from a painful stimulus.

RELEASE AGENT (ABHERENTS; ABHESIVE; PARTING AGENTS): A chemical substance such as wax, polyvinyl alcohol (PVA) and glycerides that produces a slip effect and prevents sticking when applied to surfaces of materials. They are used in the manufacture of food, glass, paper, plastics, die-casting and moulding.

RELEASE FACTOR (RF): A protein that terminates translation of messenger RNA during protein synthesis by recognising the termination codon in an mRNA sequence and releases the completed polypeptide chain from the ribosome. It is a component of the specialised transport system involved in the transport of cobalamin (vitamin B12) across the wall of the intestine. It dissociates the complex between cobalamin and the extracellular cobalamin binding glycoprotein known as intrinsic factor.

RELEASE-INHIBITING HORMONE, RIH (SOMATOSTATIN; GROWTH HORMONE-INHIBITING HORMONE, GHIH; SOMATOTROPIN RELEASE-INHIBITING FACTOR, SRIF): A hormone produced mainly by the hypothalamus that inhibits the secretion of hormones such as somatotropin, glucagon, insulin, thyrotrophin and gastrin.

RELEASER (SIGN STIMULUS): The essential features of a stimulus, which is necessary to elicit response or a specific external stimulus that initiates certain behavioural sequences that typically occur in a fixed stereotyped fashion. For example, a red belly is the sign stimulus necessary to provoke an attack from a rival male. Even a very crude model fish is attacked if it has a red under surface. Similarly territorial fighting in male robin is triggered by a sign stimulus of a red breast. A bunch of red feathers is enough to induce territorial behaviour.

RELEASING HORMONES (RELEASING FACTOR): A hormone such as thyrotrophin-releasing hormone (TRH) and corticotrophin-releasing hormone (CRH) produced by the hypothalamus that stimulates the release of hormone from the anterior pituitary gland into the bloodstream. Each hormone has a specific releasing hormone.

RELIABILITY: 1. In statistics, consistency and validity of test results determined through statistical methods after repeated trials. 2. A measure of the precision of some quantitative method, for example, by computation of the variance of repeated measurements of the same quantity.

RELIABLE DATA PROTOCOL (RDP): A network transport protocol intended to efficiently provide facilities for the bulk data transfer of images and data, control applications, remote loading and debugging.

RELICT: 1. A species, genera, taxa or population of organisms that existed in a larger area and group but now exists in isolated regions and numbers because of environmental factors. 2. A structure or mineral that did not undergo a change in physical structure when the parent rock metamorphosed. 3. A rock or land forming part of a larger one that survived a destructive process.

RELIEF: In geography, it is the difference in elevations on land surface in an area. It is inequalities of a land between the highest and lowest elevations in an area, including mountains and valleys. Land having no unevenness or differences of elevation is called **level**; gentle relief is called **undulating**; strong relief is **rolling**, and very strong relief is termed **hilly**. A relief map shows the topography of the area.

RELOAD (REFRESH): In computer science, it is a command that is used to reopen a currently displayed window or Web page.

RELUCTANCE (MAGNETIC RELUCTANCE): A measure of magnetic resistance in a closed magnetic circuit. It is the ratio of the magnetomotive force or magnetic potential difference to the total magnetic flux in a magnetic circuit. Its unit is henry and the symbol is R.

REM SLEEP (RAPID-EYE-MOVEMENT SLEEP): One of the stages of sleep that recurs several times in most humans, characterised by rapid eye movements beneath closed lids and is associated with dreaming. It is also characterised by irregular breathing and heart rate; and involuntary muscle jerks. REM occurs at intervals of between 1-2 hours and may last for about 5 minutes or over an hour.

REMAINDER: 1. The quantity that is left over when one quantity cannot be exactly divided by another; for example, the remainder of $5 \div 3$ is 2. In other words, it is the difference between a given

dividend and the largest multiple of the divisor less than the dividend. In algebra, the remainder is the polynomial that remains after dividing one polynomial by another. 2. The difference between an infinite series and one of its partial sums.

REMAINDER THEOREM: 1. Also known as **little Bézout's theorem**, it is an application of polynomial long division. It states that the remainder of a polynomial $f(x)$ divided by a linear divisor $x - a$ is equal to $f(a)$. 2. Also known as **Chinese remainder theorem**, it is a result about congruences in number theory and its generalisations in abstract algebra. In its basic form, the Chinese remainder theorem will determine a number n that when divided by some given divisors leave given remainders.

REMIGES (FLIGHT FEATHERS): Stiff long feathers of a bird that is adapted for flight, consisting of primary, secondary, and tertial feathers used to provide lift and control direction during flight. **Primary flight feathers** are about 9-10 feathers responsible for thrust to propel the bird forward. **Secondary flight feathers** are, innermost, stiff long flight feathers close to the body of the bird, positioned just behind the primary flight feathers on the rear edge of the wings. Many birds have six secondary feathers. **Tertial flight feathers** are three flight feathers that are closest to the bird's body along the wing, located next to the secondary flight feathers.

REMANENCE (RETENTIVITY): The magnetic flux density remaining in a ferromagnetic substance when the external magnetic field is removed and the saturating field is reduced to zero.

REMOTE: In botany, parts that are separated by an interval or distantly spaced, as used for an involucre in flowers.

REMOTE ACCESS: A term used to describe a device capable of accessing a shared resource. For example, a user may have the capability of remotely accessing his or her company network or computer with their portable mobile computer.

REMOTE ACCESS TROJAN (RAT): A malware program usually downloaded by a user that is invisible to the individual when downloaded. This gives the intruder administrative control over the target computer.

REMOTE AUTHENTICATION DIAL-IN USER SERVICE (RADIUS): An authentication program used on computers and networking equipment that enables servers located remotely to communicate with the central server, thereby allowing dial-in users to authenticate themselves to gain access to the service.

REMOTE DESKTOP PROTOCOL (RDP): A Microsoft multi-channel secure network communications protocol that allows a user to connect to a computer running Microsoft Terminal Services. RDP is designed to support many different types of network topologies such as ISDN, POTS and many LAN protocols such as IPX, NetBIOS or TCP/IP.

REMOTE DESKTOP TECHNOLOGY: A technology that makes it possible to view another computer's desktop on your computer. This means you can open folders, move files and even run programs on the remote computer, right from your own desktop. Both Windows and Macintosh computer support remote desktop connections, though they use different implementations.

REMOTE SENSING: Gathering and recording of information concerning the Earth's surface by remote sensors installed in aircraft and satellites without actual contact with the object or area under study.

REMOTE TERMINAL: A terminal that communicates with a computer via a modem or acoustic coupler and a telephone line.

RENAL: Relating to the kidneys. Examples are renal capsule and renal tubule.

RENAL CAPSULE: The transparent, fibrous membrane that encases each kidney and helps protect it against trauma and infection.

RENAL CLEARANCE: The measure of the volume of plasma from which a substance is completely removed by kidneys per unit time. The higher the renal clearance, the more plasma that is cleared of the substance.

RENAL DIALYSIS (KIDNEY DIALYSIS; DIALYSIS): A medical treatment for removing harmful wastes and surplus salt and water from the blood following kidney failure. When in good working order kidneys clean the blood. They also make hormones that keep bones strong and blood healthy.

RENAL TUBULE (KIDNEY TUBULE): Any nephron of the kidneys, which extends from the glomerulus in the Bowman's capsule that reabsorbs water and solute from the glomerular filtrate.

RENATURATION: The restoration of the physical and chemical properties of a denatured protein or nucleic acid to its original state and function. Proteins can be renatured by disulfide reducing agents and urea. Nucleic acids are renatured by heat or salts.

RENDERING: The process of extracting fat from fatty animal tissue by slow heating. The fat is converted into products such as lard, tallow and others.

RENDZINA: Dark rich soil that develops beneath grassland above a layer of limestone or chalk. It is rich in humus and calcium carbonate.

RENEWABLE ENERGY SOURCE: A source of energy that cannot be used up and replenishes itself in a short period. Examples are hydroelectric energy, geothermal energy, solar energy, wind energy and biofuels. The Sun is the primary energy source. Coal, oil or natural gas are not considered as renewable energy source as they are finite or limited in size or extent.

RENIFORM: Having the shape of a kidney with the petiole attached at the notch of the leaf, as found in Geranium (*Pelargonium* sp.).

RENIN: An enzyme produced by the kidney in response to low blood pressure, reduced blood sodium levels and reduced blood flow. Renin produces angiotensin in response to low blood pressure, which constricts the arteries and raises the blood pressure. It also sends signals to the adrenal cortex to release aldosterone to increase the reabsorption of sodium from the kidney tubule into the blood.

RENNET (CHYMASE; RENNIN): A natural complex of enzymes produced in the stomach of calves which contains rennin, a coagulant. Rennet is useful in the digestion of milk in calves and extracted for the production of some milk products such as cheese.

RENNIN: A protease enzyme found in the stomach, which cuddles milk for action by another protease enzyme, pepsin.

RENORM: In mathematics, to construct an equivalent norm (called a renorm) for a given norm on a normed space. Every separable space can be renormed so that the renormed unit ball is simultaneously (Gateaux) smooth and strictly convex.

RENORMALISATION (RENORMALIZATION): A mathematical technique used in quantum physics that eliminates infinite terms by carefully defining fundamental quantities such as mass and charge. Theories, for which finite results for all perturbation-theory calculations exist, by taking a finite number of parameters from experiment and using renormalisation, are called **renormalisable**. Only certain types of quantum field theories are renormalisable. Theories that need an infinite number of parameters are said to be **nonrenormalisable** and are regarded as incomplete physical theories. The gauge theories that describe the strong, weak and electromagnetic interactions are renormalisable. The quantum theory of gravitational interactions is a nonrenormalisable theory, which perhaps indicates that gravity needs to be unified with other fundamental interactions before one can have a consistent quantum theory of gravity.

RENORMALISATION/RENORMALIZATION GROUP (RG): A mathematical technique used to understand the changes of a physical system as one views it at different distance scales. Such systems

include phase transitions, turbulence, polymers, many-electron systems and the localisation of electrons in disordered systems. In particle physics, it reflects the changes in the underlying force laws as one varies the energy scale at which physical processes occur. The renormalisation group has its origin in quantum field theory, in which it is used to calculate change with energy.

RENORMALON: A type of divergent behaviour in a quantum field theory such as quantum electrodynamics (QED) that is associated with the perturbation series for the theory not having a Borel sum. The existence of renormalons in a theory is associated with Landau ghost and may indicate a mathematical inconsistency in the theory.

RENSHAW CELL: A type of inhibitory interneuron in the grey matter of the spinal cord in the central nervous system that controls the level of excitation of motor neurons. The Renshaw cells receive input from side branches of motor axon and in turn form inhibitory synapses with the motor-neuron cell bodies.

REORGANISATION (REORGANIZATION): The molecular process by which (i) amorphous or poorly ordered regions of a polymer specimen become incorporated into crystals or (ii) a change to a more stable crystal structure takes place or (iii) defects within the crystal decrease.

REP-UNIT (REPEATED UNIT): An integer such as 11, 111 or 1111 consisting only of repetitions of the single digit 1.

REPAND: Having an undulating or slightly wavy margin, as seen in some leaves.

REPDIGIT: An integer of repeated digits, for example, 88888.

REPEAT-UNTIL LOOP: A programming language used in a programme called Pascal, which requires that a set of statements repeat until some condition is satisfied.

REPEATABILITY: The closeness of agreement between independent results obtained with the same method on identical test material, under the same conditions (same operator, same apparatus, same laboratory and after short intervals of time). The measure of repeatability is the standard deviation qualified with the term: 'repeatability' as repeatability standard deviation. In some contexts repeatability may be defined as the value below which the absolute difference between two single test results obtained under the above conditions, may be expected to lie with a specified probability.

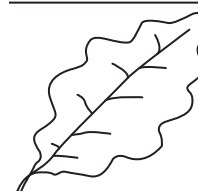
REPEATED ADDITION STRATEGY: Adding the same addend a given number of times in order to solve multiplication problems.

REPEATED INTEGRAL (MULTIPLE INTEGRAL): An expression requiring more than one integration. **Multiple integration** or **repeated integration** is the integration of functions of more than one real variable. Integrals of a function of two variables over a region in R^2 are called **double integrals** and integrals of a function of three variables over a region of R^3 are called **triple integrals**.

REPEATED MEASURES DESIGN (WITHIN-SUBJECTS DESIGN): In statistics, it is a type of an experiment, which repeatedly measures the values of the dependent variable for the same subjects under the various experimental conditions and the values are compared.

REPEATED ROOT (MULTIPLE ROOT): One of a number of equal roots of the same polynomial or equation that occurs when the polynomial has $(x - a)^n$ as a factor, for $n > 1$. The largest such n is the multiplicity or order of a . A root of order 1 is a simple root. Equivalently, it is a root of one or more derivatives of the polynomial.

REPEATER: A network device that is used to regenerate or replicate incoming analogue or digital signals that are weakened or distorted by transmission loss over long distances and through areas with high levels of electromagnetic interference.



A repand leaf

REPEATING DECIMAL (RECURRING DECIMAL; CIRCULATING DECIMAL): Decimal numbers that repeat indefinitely in a pattern after the decimal point, for example, 0.3333 or 2.1818181... The repeating figure or figures of a recurring decimal fraction is called repetend(s). Recurring decimals can also be written by drawing a horizontal line called a vinculum above the repetend, placing dots above the outermost numerals of the repetend or enclosing the repetend in parentheses. For example, 0.3333 can be written as $0.\overline{3}$, $0.\dot{3}$ or 0.(3); and 2.18, can be written as $2.\overline{18}$, $2.\dot{1}8$ or 2.(18).

REPEL: In physics, to push back or away by a force, such as two similar magnetic fields opposing the force of attraction or coming together. For example, the north pole of one magnet will repel the north pole of another, likewise the south poles. This is expressed as like poles (north and north or south and south) repel each other; whilst unlike poles (north and south) attract each other.

REPELLENT (REPELLANT): 1. Any substance used to keep organisms away from destroying or feeding on other organisms. An example is insect repellent that keeps mosquitoes at bay. 2. A substance that is applied to a surface of something to resist water, mould or mildew.

REPENT (REPTANT): In botany, it refers to lying or creeping along the ground, for example, repent stems.

REPETEND: In mathematics, the digit(s) that repeat in a recurring decimal.

REPETITIVE DNA: DNA whose nucleotide sequence is repeated many times throughout the genome of an organism. It is responsible for about half of the total DNA in mammals. Repetitive DNA sequences are either closely linked or dispersed throughout the genome or parts of the genome.

REPETITIVE STRAIN INJURY (RSI): Inflammation of tendon sheaths, mainly in the hands and wrists due to recurrent overuse or improper use. It occurs mainly in those who do highly repetitive work such as that of factory workers involved in constant repetitive movements and those who work with computer keyboards such as typists, word processors, accountants and writers. The symptoms include aching muscles, weak wrists, tingling fingers and in severe cases, pain and paralysis.

REPLACEMENT MILK: Milk used to feed young animals, which cannot be fed by their mothers.

REPLACEMENT NAME: A name in which the replacement of an atom or a group of a parent structure by another atom or group is indicated by affixes attached to or inserted into the name of the parent structure. There are two main types of replacement names: (i) **Skeletal replacement name**, in which the replacement of skeletal atoms and their associated hydrogen atoms is indicated by non-detachable prefixes. When carbon atoms are replaced by heteroatoms this method has been called 'a' nomenclature since the prefixes end in 'a'. Certain names in which the prefix 'thio-', 'seleno-' or 'telluro-' indicates replacement of a skeletal oxygen atom by a sulfur, selenium or tellurium atom, respectively, are also skeletal replacement names. (ii) **Functional replacement name**, which is a name containing prefixes or infixes which indicate the replacement of an oxygen atom or hydroxy group of a characteristic group, functional parent or class name by other atoms or groups.

REPLACEMENT OPERATION: The exchange of one group of atoms or a single non-hydrogen atom for another.

REPLACEMENT RATE: The rate at which heifers are introduced into a dairy herd to replace aging cows with no milk yield.

REPLANT DISEASE: A disease characterised by weak and blackened roots affecting apples, pears, plums, cherries and roses planted into soil previously occupied by a related species.

REPLICASE ENZYME: An enzyme that catalyses the synthesis of DNA and RNA within living cells.

REPLICATE: 1. In botany, it refers to folding over or backward. 2. To reproduce or make an exact copy, as in DNA.

REPLICATE SAMPLE (DUPLICATE SAMPLE): Multiple (or two) samples taken under comparable conditions. This selection may be accomplished by taking units adjacent in time or space. Although the replicate samples are expected to be identical, often the only thing replicated is the act of taking the physical sample. A **duplicate sample** is a replicate sample consisting of two portions. The **umpire sample** is usually used to settle a dispute; the replicate sample is usually used to estimate sample variability.

REPLICATION: The duplication of the genetic information of DNA. During replication the two DNA strands in the double helix separate and each strand then acts as a template specifying the base sequence of newly synthesised complementary strands. In humans and other eukaryotes, replication occurs in the cell nucleus.

REPLICATION FORK: Y-shaped point on a replicating DNA molecule where new strands grow.

REPLICON: A DNA or RNA molecule that replicates as a unit for single initiation site. The genome of plasmids and bacteriophages contain a single replicon, whilst eukaryotes contain a number of replicons on each chromosome.

REPLUM: A thin wall or septum that partitions the two valves in some fruits into compartments. This is seen in siliques and some legumes.

REPOLARISATION (REPOLARIZATION): The restoration of the normal electrical polarity of a muscle cell membrane potential to a negative value or nerve following reversal of its polarity to a positive value.

REPORT: A combination of specimen information and results. The report should contain information about unequivocal identification of the source and type of material analysed and the requesting agency. It may contain such other information that is pertinent to the correct interpretation of a result (e.g. confidence interval, reference data and interpretative information).

REPORTER GENE: A gene that is easily identified and monitored and used to tag a regulatory sequence of another gene or DNA sequence of interest such as a promoter. It is often used as an indication of whether a certain gene has been taken up by or expressed in the cell or organism population.

REPOSITORY: In software development, it is a central file storage location that contains stored information and data about users, people, computers or other important information. It is used by version control systems to store multiple versions of files. While a repository can be configured on a local machine for a single user, it is often stored on a server, which can be accessed by multiple users.

REPOT: To take a plant out of its pot and plant it in another, changing or adding to the soil at the same time.

REPPE PROCESSES: A family of high-pressure, catalytic acetylene-reaction processes, which yield butadiene, allyl alcohol, acrylonitrile, vinyl ethers and derivatives, acrylic acid esters, cyclooctatetraene and resins depending upon what the acetylene reacts with.

REPRECIPITATION: The planned repetition of a precipitation in order to remove chemically different species from a previous precipitate or to improve its stoichiometry.

REPRESENT: In mathematics, of a series of functions, to converge to the value of the given function at each point of a specified set.

REPRESENTATIVE ELEMENTS: The entire elements in Group 1 (IA) to 17 (VIIA) of the Periodic Table, with partially filled s or p subshell of principal quantum number, n .

REPRESENTATIVE SAMPLE: A sample resulting from a sampling plan that can be expected to adequately reflect the properties of interest of the parent population. A representative sample may be a

random sample or, for example, a stratified sample, depending upon the objective of sampling and the characteristics of the population. The degree of representativeness of the sample may be limited by cost or convenience.

REPRESENTING FUNCTION: In mathematics, it is the function with the same domain as a given predicate that for each domain value takes the value 1 if the predicate holds and 0 if it fails. It is the characteristic function of the extension of the predicate.

REPRESSIBLE ENZYME: An enzyme whose synthesis is generally continuous but can be halted if a particular substance is present in concentrations greater than normal.

REPRESSION: Binding of a protein (repressor) to the operator sequence in an operon thus preventing the transcription of the following structural gene(s) into mRNA and consequently synthesis of protein.

REPRESSOR: A DNA-binding protein that can prevent the expression of a gene or genes by binding to the operator and blocking the attachment of RNA polymerase to the promoter, preventing transcription of the genes. Repressor genes can detect the absence or presence of lactose.

REPRODUCIBILITY: The closeness of agreement between independent results obtained with the same method on identical test material but under different conditions (different operators, different apparatus, different laboratories and/or after different intervals of time). The measure of reproducibility is the standard deviation qualified with the term 'reproducibility' as reproducibility standard deviation.

REPRODUCTION: The process by which organisms replicate ensuring continuation of their species. The two main types of reproduction are asexual and sexual reproduction. Most animals, including all vertebrates, reproduce sexually through the fusion of special sex cells called gametes, usually from different parents. Asexual reproduction produces an offspring genetically identical to its single parent. It includes fission, spore formation, regeneration and vegetative reproduction.

REPRODUCTIVE ISOLATING MECHANISM (MECHANISMS OF REPRODUCTIVE ISOLATION; HYBRIDISATION BARRIERS): Biological or behavioural characteristics that prevent interbreeding within two different species and ensures that even if they do any offspring that may be produced is a sterile hybrid.

REPRODUCTIVE SYSTEM (GENITAL SYSTEM): The organs of an organism that are involved in the process of sexual reproduction. Examples in male mammals include the penis, testes, vas deferens, seminal vesicle and accessory glands; in female mammals, they include ovaries, oviducts (fallopian tubes), uterus, cervix, vagina and vulva. In flowering plants it is the stamen and carpels. The reproductive systems of human male and female and a hen are shown in the following diagrams:

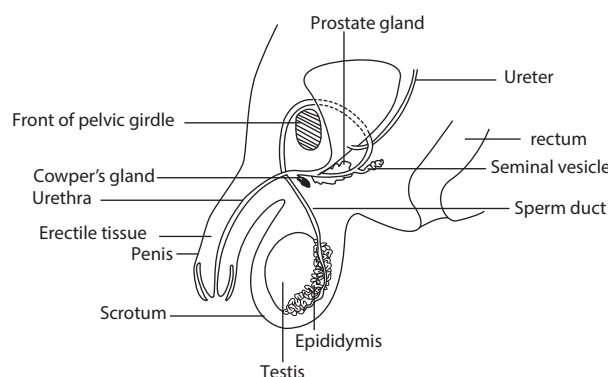


Fig. 1: The human male reproductive organs

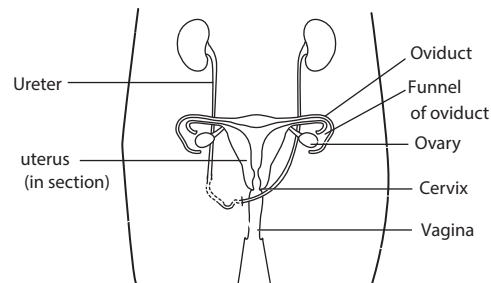


Fig. II: The female (human) reproductive organs

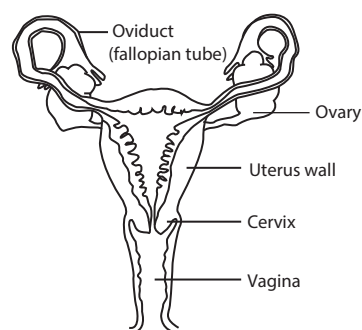


Fig. III: The female (human) reproductive system

REPTATION: A thermal motion of polymer chains in a highly entangled state found in concentrated solutions of polymers used to explain the viscous flow in an amorphous polymer.

REPTILE: A type of cold-blooded vertebrates of the class Reptilia. They have tough, dry skin covered with horny scales to prevent water loss through perspiration. Their sizes vary between 5 cm to 9 m. Reptiles include lizards, crocodiles, tortoises and snakes. They are found in the tropics and temperate regions in many habitats and some are adapted to live in water as well as on land.

REPTILIA: A class of terrestrial vertebrates of four orders: Chelonia (tortoises, turtles and terapins), Squamata (the lizards and snakes), Sphenodonta (tuataras) and Crocodylia (crocodiles, alligators and caimans). They are cold-blooded vertebrates characterised by tough, dry skin covered with horny scales to prevent water loss through perspiration.

REPULSION: A force between two objects of the same electric charge or magnetic polarity that tends to repel or separate them. This force between atoms and molecules at very short distances that tends to keep them separated.

RESEARCH: A systematic gathering, investigation or examination of data, reports and observations in a search for facts or principles that may include a combination of theory and experimentation directed towards finding a scientific explanation of phenomena. It is commonly classified into two types: pure research and applied research. Many empirical sciences, especially the social sciences, use mathematical tools borrowed from probability theory and statistics, together with such outgrowths of these as decision theory, game theory, utility theory and operations research.

RESEARCH, DEVELOPMENT, AND DEMONSTRATION: Scientific and/or technical research and development of new production processes or products, together with analysis and measures that provide information to potential users regarding the application of

the new product or process; demonstration tests; and feasibility of applying these products processes via pilot plants and other pre-commercial applications.

RESEED: To re-plant seeds on an area of land.

RESERVE: In ecology, it is an area of land set aside for the welfare of plant or animal life devoid of any farming and commercial activities. Examples are forest and wild life reserves.

RESERVED WORD: A reserved word that has a meaning special to a programming language. For example, 'if' and 'for' are reserved words in most high-level languages because they have special meanings.

RESERVOIR: An artificial or natural area of water, for capturing and storing water and making it available for domestic or industrial use.

RESIDENCE TIME: 1. In instrumentation, it is the time required for air or reagent parcel to pass from the entrance to the exit of an instrument. Often this is approximated as the ratio of the interior volume of the device to the flow rate. 2. The average time a molecule or aerosol spends in the atmosphere after it is released or generated there. For compounds with well defined sources and emission rates, this is estimated by the ratio of the average global concentration of a substance to its production rate on a global scale. It is a function of not only the emission rates but the loss rates by chemical and physical removal processes. 3. In biotechnology, it is also known as **hydraulic retention time** and it is the average time a particle or volume element of the culture resides in a bioreactor (or other device) through which a liquid medium continuously flows. Residence time of the liquid is the reciprocal of dilution rate.

RESIDENT: A non-migratory species of birds or other animals that do not migrate, but stay in a given geographical area throughout their life.

RESIDUAL: 1. In statistics, the difference between an actual, observed value and a value predicted by a regression. A positive residual is where the observed value exceeds the computed value; a negative residual is the opposite. 2. Pertaining to the portion of something that remains after an activity that removes the bulk of the substance. An example is unconsolidated and partly weathered mineral materials formed by the disintegration of consolidated rock in place. 3. Of a set, having complement of first category (Baire category).

RESIDUAL CURRENT: The current that flows, at any particular value of the applied potential, in the absence of the substance whose behaviour is being investigated (i.e. in a blank solution).

RESIDUAL FUEL/OIL: The liquid or semi-liquid, high-boiling fraction of residue from the distillation of petroleum which is used as a fuel. After removal of the lower boiling fraction of crude oil, sold as petroleum gas, the somewhat higher boiling fraction becomes gasoline and diesel oil. A portion of the higher boiling fraction is 'cracked' to yield additional gasoline. Still heavier oils become lubricants. Paraffin, asphalt, etc, are also extracted from the crude oil. It is the remaining residual oil, not economically usable for other industrial purposes, which is then sold as a relatively low cost fuel for burning.

RESIDUAL HERBICIDE: A chemical preparation applied to the surface of soil that remains active in the soil so that weeds can be controlled over an extended period of time.

RESIDUAL LIQUID JUNCTION ERROR (POTENTIAL ERROR): Error arising from breakdown in the assumption that the liquid junction potential remains constant when solution X is substituted for solution S in the operational cell.

RESIDUAL SHRINKAGE: The decrease in volume after the proportionality between water loss and volume change ceases.

RESIDUAL SOIL: Soil formed from, or resting on, consolidated rock of the same kind as that from which it was formed and in the same location. Residual soils are products of chemical weathering and thus their characteristics are dependent upon environmental factors of climate, parent material, topography, drainage, and age.

RESIDUAL VOLUME: The amount of air remaining in the lungs after the maximum expiration possible and which cannot be expelled from the lungs voluntarily. An average person has a residual volume of one litre.

RESIDUE: 1. The waste material that remains after the completion of a chemical or physical process such as evaporation, combustion, distillation or filtration. 2. In modulus arithmetic, it is the equivalence classes for an integer n of integer partitioned by their remainders when they are divided by m .

RESIDUE CLASS RING (FACTOR RING; QUOTIENT RING): The ring, denoted R/K , whose members are the cosets of an ideal K in a ring R ; these cosets are called residue classes and are equivalence classes whose members differ by a member of K . Since K is an ideal, the sums and products of elements of the factor ring are unique and the cosets of a sum or product of ring elements equals the sum or product (in the same order) of their individual cosets. K is the zero of the factor ring and it has a unit element if R has.

RESIDUE THEOREM OF CAUCHY (CAUCHY'S RESIDUE THEOREM): A tool used to evaluate line integrals of analytic functions over closed curves. It generalises the Cauchy integral theorem and Cauchy's integral formula that if a function f is analytic in a simply connected domain D except for finitely many isolated singularities, then the contour integral of f over any simple closed curve that misses the singularities is equal to $2\pi i$ times the sum of the residues of f at singularities inside the contour. An equivalent definition is that the integral of a complex function round a Jordan contour is the sum of residues of that function inside the curve multiplied by $2\pi i w$, where w is the winding number of the curve about that point. This is of great use for evaluating curvilinear integrals of meromorphic functions and so certain classes of definite integrals of real functions.

RESILIENCE: The elastic property of a material to absorb energy when it is stretched, deformed or bent and upon unloading to have this energy recovered and return to its original shape. It is the maximum energy per unit volume that can be elastically stored.

RESIN: 1. A sticky solid or semisolid, viscous organic substance exuded from trees such as pines and firs that hardens in air. Resins are insoluble in water, but soluble in organic solvents such as alcohol and ether. All resins contain carbon, hydrogen and oxygen, but have other different chemical constituents. Examples of resins are copal and mastic, which are hard resins used in making varnishes. Mastic is also used in making lacquer, adhesives, etc. Amber, a hard translucent fossil resin varying in colour from yellow to light brown is used in making jewellery and ornaments. Another resin which is formed by the insect, *Laccifer lacca* on trees is called lac, used in making shellac. 2. Sticky, amorphous semisolid residue such as rosin and balsam of distilled oleoresin turpentine. Rosin is a hard resin used in making soap, varnish, paint, etc. 3. Synthetic products manufactured by organic polymers used in adhesive, plastics and varnishes. Examples of synthetic resins are polyvinyl, polystyrene, polyethylene, polyester, epoxy and silicone. **Resiniferous** or **Resinous** means producing resin and therefore sticky.

RESISTANCE: 1. A property of an electrical conductor to oppose the free flow of current, resulting in the conversion of electrical energy to heat energy. The SI unit of resistance is the ohm, with symbol Ω . 2. The resistance to antibiotics and other drugs by a pathogen. 3. The degree to which a pest can withstand the effects of a pesticide, which is normally lethal to the organisms of that species. 4. The natural resistance to infectious agents that an organism possesses.

RESISTANCE ARM (OUT-LEVER ARM): The part of a lever that moves against weight or resistance. It is the resistance distance in a lever that works together with the force arm to move an object. Ideally, it is the distance between the axis and the point of weight or resistance. The length of a resistance arm is similar to the distance between the load and the fulcrum. A lever can either have a long resistance arm or a short resistance arm. A long resistance arm implies speed and range of movement, while a short resistance arm implies greater force.

RESISTANCE RESPONSE: A response lasting for a long time to any stimulus that potentially threatens the wellbeing of an organism. An example is the release of adrenocorticotrophic hormone (ACTH) in response to stress.

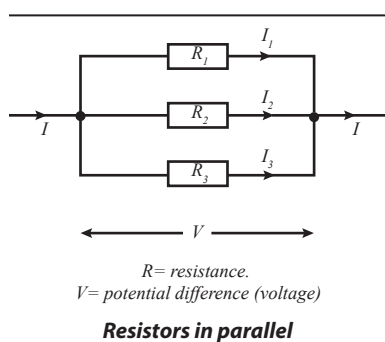
RESISTANCE THERMOMETER (RESISTANCE PYROMETER; RESISTIVE THERMAL DEVICES, RTDS): A sensor used for measuring temperature by the property of some materials that increase electrical resistance with rising temperature. Platinum is the material often used, other materials also used are nickel and copper. When a constant voltage is applied across a resistor, which consists of a coil of platinum wire or the other materials, it induces a certain resistance according to the specific temperature that it senses. This resistance is measured by a galvanometer as temperature.

RESISTIVITY: The resistance of a wire per unit cross sectional area. It is the measure of a material's ability to resist the flow of an electric current. A high value resistivity denotes higher resistance to current flow. Its symbol is ρ (rho) and the SI unit is ohm metre [Ωm]. The resistivity of a material is given by RA/l , where R is the resistance of the uniform specimen of the material having a length l and A is the cross sectional area of the wire.

RESISTOR: A component in an electrical or electronic circuit which resists the free flow of current. Its symbol is Ω and its unit is ohms. Resistors control current, voltage, and protect an electric circuit. One important characteristic of a resistor is that it converts electrical energy into other forms of heat energy, for example, heaters or electric lights. Examples of resistors are rheostats and potentiometers. Resistors are related with current I and voltage or potential difference V across an electrical circuit by Ohm's law, expressed as $R = V/I$. Resistors are either wire-wound or consist of carbon particles in a ceramic binder, with colour coding indicating the value of the resistance. Resistors may be fixed or variable. The resistance of a fixed resistor is set at a specific resistance which changes slightly with temperature, time or operating voltage. Variable resistors can regulate current flows at and below a specific level and can be used, for example, as rheostats to adjust brightness of lamps; as thermistors for temperature or voltage regulation, volume control, time delays, and circuit protection; and for potentiometers as a voltage divider, in radio and television (TV) receiver, for volume control, frequency attenuation, etc.

RESISTORS IN PARALLEL: Resistors in parallel have the same potential difference (p.d).

The reason is that each coulomb of charge passing through the combination delivers the same amount of energy whatever path the charge takes. For three resistors in parallel as shown in the resistor diagram, the current passing through each resistor is given as:



$$I_1 = \frac{V}{R_1},$$

$$I_2 = \frac{V}{R_2},$$

$$I_3 = \frac{V}{R_3},$$

combination.

where V is the p.d across the
The total current $I = \frac{V}{R}$. The

total resistance $R = \frac{V}{I}$.

RESISTORS IN SERIES: Resistors in series give the same current

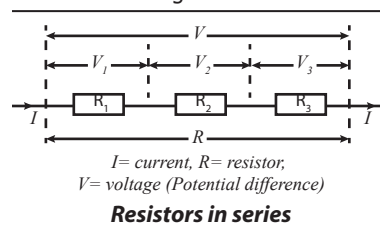
(I). The reason is that equal amounts of charges pass through each resistor (R) every second. For three resistors of resistance R_1 , R_2 , R_3 in series as shown in the figure, the potential difference (p.d.) across R_1 is $V_1 = IR_1$;

p.d across R_2 is $V_2 = IR_2$;

p.d across R_3 is $V_3 = IR_3$. The total p.d, V across the three resistors = $V_1 + V_2 + V_3$. This is because each coulomb of charge passing through delivers energy V_1 to R_1 , V_2 to R_2 and V_3 to R_3 . So the total energy delivered per coulomb is: $V_1 + V_2 + V_3$.

Hence, $V = IR_1 + IR_2 + IR_3$. The total resistance R is given by:

$$R = \frac{V}{I} = \frac{IR_1}{I} + \frac{IR_2}{I} + \frac{IR_3}{I} = R_1 + R_2 + R_3$$



RESOLUTION: 1. The separation of a vector quantity into two components which are usually at right angles to each other. Thus, we can have a force, F acting on a body in a vertical plane at an angle θ and a vertical component $F\sin\theta$, both to the same plane as the original force. 2. The precision with which any instrument measures and records (in an image or spectrum) any variable in the specimen or sample under study. 3. The number of pixels per unit of measurement such as centimetre or inch on a video monitor, used to denote the total number of pixels displayed horizontally or vertically on the video monitor. 4. The ability of a photographic emulsion to produce a picture containing fine detail. 5. The process of separating something such as a chemical compound or a source of light into its constituent parts. 6. In terms of electron density, resolution is a measure of the resolvability in the electron density map of a molecule. 7. In mass spectrometry, resolution is a measure of the ability to distinguish two peaks of slightly different mass-to-charge ratios in a mass spectrum. 8. In x-ray crystallography, resolution is the highest resolvable peak in the diffraction pattern. 9. Also known as **minimum resolvable distance**, it is the minimum distance between distinguishable objects in an image. The term is also loosely used by many users of microscopes and telescopes to describe resolving power. 10. **Angular resolution** or **spatial resolution** is the ability of any image-forming device such as an optical or radio telescope, a microscope, a camera or an eye, to distinguish small details of an object. 11. In number storage, resolution is the reciprocal of the unit in the last place. 12. In computer science, it is the clarity or fineness of detail attained by a monitor or a printer in producing an image. 13. In printing, it is the number of individual dots a printer can produce within a unit of distance, for example, dots per centimetre.

RESOLVENT: In mathematics, the inverse of the matrix $\iota I - A$, where A is a given matrix or operator and ι is not in the spectrum of the matrix A . When operators in normed space are considered, it is required that $\iota I - A$ have dense range and possess a bounded inverse on that range. The set of such ι is the resolvent set of A .

RESOLVING POWER: A measure of the ability of an optical instrument to form separate images of close objects or to separate close wavelengths of radiation. The spectral resolution or resolving power of a spectrograph or, more generally, of a frequency spectrum, is a measure of its ability to resolve features in the electromagnetic spectrum. It is usually defined by $R = \frac{\lambda}{\Delta\lambda}$ where $\Delta\lambda$ is the smallest difference in wavelengths that can be distinguished, at a wavelength of λ . For example, the Space Telescope Imaging Spectrograph (STIS) can distinguish features 0.17 nm apart at a wavelength of 1000 nm, giving it a resolving power of about 5,900. An example of a high resolution spectrograph is the Cryogenic High-Resolution IREchelle Spectrograph (CRIRES) installed at European Southern Observatory's (ESO) Very Large Telescope, which has a spectral resolution of up to 100,000.

RESOLVING TIME: The smallest time interval which must elapse between the occurrence of two consecutive ionising events or signal pulses, in order that the measuring device be capable of fulfilling its function of each of them separately.

RESONANCE: 1. The increased amplitude of oscillation of an electric or mechanical system when it is subjected to a periodic force such as vibration from another source whose frequency is near or equal to the natural frequency of the system. 2. A subatomic particle lasting too short a time to be observed directly. The existence of such particles is usually inferred from a peak in the energy distribution of its decay products. 3. The sound heard during diagnostic percussion of a normal abdomen or chest. 4. The property of a chemical compound having simultaneously the characteristics of two or more limiting structures that differ in the arrangement of electrons. Such compounds are highly stable and cannot be properly represented by a single structural formula.

RESONANCE ABSORPTION TECHNIQUE: The monitoring of atoms or radicals generated in the gas phase by observing the attenuation of the radiation from a lamp emitting the characteristic resonance radiation of the observed species.

RESONANCE EFFECT MAGNITUDE: Ratio of the difference in the transmitted or scattered intensity at resonance maximum and off-resonance, relative to the intensity off-resonance.

RESONANCE ENERGY: The difference in potential energy between the actual molecular entity and the contributing structure of lowest potential energy. The resonance energy cannot be measured, but only estimated, since contributing structures are not observable molecular entities.

RESONANCE FLUORESCENCE: Fluorescence from the primary excited atomic or molecular species at the wavelength of the exciting radiation (no relaxation within the excited manifold). This term is also used to designate the radiation emitted by an atom of the same wavelength as the longest one capable of exciting its fluorescence, for example, 122.6 nm in the case of the hydrogen atom and 253.7 nm in the case of the mercury atom.

RESONANCE FLUORESCENCE TECHNIQUE: The monitoring of atoms or radicals generated in the gas phase by observing the intensity of fluorescence (exitance) emitted by the species after excitation with radiation of the same wavelength.

RESONANCE HYBRID: A localised valence bond representation of a molecule. This describes pictorially an electronic wavefunction that makes some contribution to the state wavefunction of the system. Resonance hybrids are needed to describe the state of a system when that system is not adequately described by a single Lewis structure.

RESONANCE INTEGRAL: The integral, over all or some specified portion of the resonance energy range, of the cross-section divided by the energy of a radiation.

RESONANCE LAMP: Lamp emitting resonance radiation of atoms and their ions. Depending on the requirements, the lamp is filled either with pure vapour of the element or with a mixture of it and other gases.

RESONANCE LINE: The longest wavelength capable of exciting fluorescence in an atom.

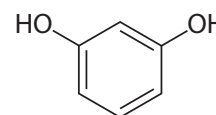
RESONANCE NEUTRONS: Neutrons whose energy corresponds to the resonance energy of a specified nuclide or element. If the nuclide is not specified, the term refers to resonance neutrons of ^{239}U (Uranium-239).

RESONANT CAVITY (CAVITY RESONATOR): An enclosure within a conductor in which an electromagnetic field can be made to oscillate at frequencies above those of which a resonant circuit will operate thereby intensifying sound tones or electromagnetic waves. The resonant frequency of the oscillation depends on the dimensions such as length and the shape of the cavity. An example is cavity pipes used as resonators in musical instruments such as pipe organs and flutes to increase their sonority.

RESONANT CIRCUIT: A reactive circuit, arranged in such a way that it is capable of resonance in a resistor, inductor and capacitor. In an electrical network there is oscillation between the potential energy of charge on capacitance and the kinetic energy of current in inductance. Resonance occurs when the impedance is minimum (capacitance and inductance are connected in series) and the current amplitude is a maximum. This occurs when the capacitive reactance is equal to the inductive reactance. In parallel resonant circuit, the impedance and capacitance are in parallel and resonance (with minimal current amplitude) occurs at maximum impedances. There are three kinds of resonant frequency: (i) **Phase resonance** is the frequency at which the phase angle between sinusoidal current entering a circuit and sinusoidal voltage applied to the terminals of the circuit is zero. (ii) **Amplitude resonance** is the frequency at which a given sinusoidal excitation (voltage or current) produces the maximum oscillation of electric charge in the resonant circuit. (iii) **Natural resonance** is the natural frequency of oscillation of the resonant circuit in the absence of any forcing excitation. Resonant circuits are used as filters in communications, permitting certain frequencies to be selected or rejected and used to tune radio and television receivers to the transmission station in order to receive signals. Also, a pair of telephone wires can carry many messages at the same time, each modulating a different carrier frequency and each being separated from the others at the receiving end of the line by an appropriate arrangement of resonant filters.

RESONATING VALENCE BOND THEORY: An extension of valence bond theory to solids in which there is quantum mechanical resonance between different structures throughout the solid. It focuses on how the atomic orbitals of the dissociated atoms combine to give individual chemical bonds when a molecule is formed. In contrast, molecular orbital theory has orbitals that cover the whole molecule.

RESORCINOL (1,3-BENZENEDIOL; RESORCIN): Crystalline dihydric phenol, the 1,3-isomer of benzenediol with the chemical formula $\text{C}_6\text{H}_4(\text{OH})_2$. It is produced when resins such as galbanum and asafoetida are melted with potassium hydroxide or by the distillation of Brazilwood extract. Synthetically it is prepared by melting 3-iodophenol, phenol-3-sulfonic acid or benzene-1,3-disulfonic acid with potassium carbonate; by the action of nitrous acid on 3-aminophenol; or by the action of 10% hydrochloric acid on 1,3-diaminobenzene. Its uses include in the synthesis of drugs such as antiseptic and disinfectant, in ointments in the treatment of some chronic skin diseases and dyes.



Chemical Structure
of Resorcinol

RESOURCE DESCRIPTION FRAMEWORK (RDF): A specification that defines how metadata or descriptive information, should be formatted. The RDF model uses a subject-predicate-object format, which is a standardised way of describing something.

RESOURCE PARTITIONING: The evolutionary process of species living in the same ecosystem coexisting such that the niche of each species differs by one or more significant factors from the niches of all coexisting species so that each species develops dissimilar resource requirements to avoid competition.

RESPIRATION: The metabolic process in animals and plants in which organic substances derived from foodstuffs are broken down to simpler products with the release of energy, which is incorporated into energy carrying molecules adenosine triphosphate (ATP) and subsequently used for other metabolic processes. External respiration moves oxygen from the space outside an organism to the cells inside its body, then carbon dioxide moves from the cells to the outside. In most multicellular organisms and nearly all vertebrates external respiration takes place in specialised structures termed respiratory organs, such as gills and lungs. The exception is a few vertebrates such as salamanders lacking both lungs and gills.

RESPIRATORY CHAIN: A series of membrane-linked oxidation-reduction reactions in which electrons are transferred from reduced cofactors, formed in TCA cycle and other dehydrogenase reactions, to oxygen, which then combines with hydrogen ions to form water.

RESPIRATORY MOVEMENT: The muscular movement of the diaphragm and the rib cage (in humans) that enables the passage of air to and from lungs or other respiratory organs of an organism. The movement mechanism varies with the species. For example, in humans, during inspiration, the lateral dimensions of the thoracic cavity are increased by the 7-10th ribs moving laterally and during expiration, the intercostal muscles and the diaphragm relax and the elastic fibres of the lung and the costal cartilages recoil to their original state before inspiration. In insects, the abdominal muscles relax and contract rhythmically to encourage the flow of air through the tracheae. In amphibians, air is drawn into the lungs by the pumping action of the muscle in the floor of the mouth.

RESPIRATORY ORGAN: Any organ such as a lung or gill in vertebrates and lung books in arachnid across which exchange of oxygen and carbon dioxide takes place.

RESPIRATORY PIGMENT: A coloured compound, such as haemoglobin in humans and haemocyanin in invertebrates that is capable of reversibly binding with oxygen at high oxygen concentration and releasing it at low oxygen concentration, thus increasing the oxygen-carrying capacity of the blood. Most of such pigments are present in the blood, transporting oxygen within the circulatory system from the respiratory organs to the tissues of the body.

RESPIRATORY QUOTIENT, RQ (RESPIRATORY COEFFICIENT): The unitless number, which is the ratio of the volume of carbon dioxide produced by an organism during respiration to the volume of oxygen consumed. It varies with type of the respiratory substrate. It is measured using Ganong's Respirometer. The respiratory quotient

(RQ) is calculated from the ratio: $RQ = \frac{CO_2 \text{ eliminated}}{O_2 \text{ consumed}}$, where the term

"eliminated" refers to carbon dioxide (CO_2) removed from the body.

RESPIRATORY SURFACE: Surface used by an organism for gas exchange. Examples in animals are the lungs, gills and body surface. In plants it is the plasma membrane of parenchymatous cells.

RESPIRATORY SYSTEM: The system of organs in the body responsible for the intake of oxygen and the expiration of carbon dioxide. In mammals it consists of the nasal cavity, larynx, trachea, bronchi, bronchioles, alveoli and lungs.

RESPIROMETER: Any device that measures an organism's oxygen uptake. An example is spirometer used in measuring oxygen uptake of man. Simple ones consist of a chamber connected to a manometer. They allow investigation into how factors such as age, chemicals or the effect of light affect the rate of respiration. Carbon dioxide is chemically removed from the chamber so that only oxygen uptake is measured.

RESPONSE: The physiological, muscular or behavioural activity that can be elicited by a stimulus or any change in an organism made in a reaction to a stimulus.

RESPONSE CONSTANT: A quantity whose expression includes a current, whose value is characteristic of a charge-transfer process and the experimental conditions under which it is investigated and whose nature depends on the technique that is employed. Some typical response constants are the diffusion current constant in polarography, the voltammetric constant in linear-sweep voltammetry and the chronopotentiometric constant in chronopotentiometry.

RESPONSE CURVE: A graph that is used to plot the results of many kinds of experiments showing the yield (or some associated factor) against fertiliser input, level of feed, antibiotics, etc. The X-axis plots concentration of a drug or hormone. The Y-axis plots response, which could be almost anything. For example, the response might be enzyme activity, accumulation of an intracellular second messenger, membrane potential, secretion of a hormone, heart rate or contraction of a muscle.

RESPONSE FUNCTION: The symmetric tensor-valued function which defines the stress in an elastic body.

RESPONSE GENERALISATION: The act or process by which an organism makes a different but similar response to the same stimulus.

RESPONSE TIME: 1. The interval between entering a command and seeing its effect in computing. 2. The time required for the output of a control system or element to reach a specified fraction of its new value after application of a step input or disturbance. 3. The time it takes for the pointer of an electrical or electronic instrument to come to rest at a new value, after the quantity it measures has been abruptly changed. 4. Also called **adjustment time** in climate change, it is the time needed for the climate system or its components to re-equilibrate to a new state, following a forcing resulting from external and internal processes or feedbacks. It is very different for various components of the climate system. The response time of the troposphere is relatively short, from days to weeks, whereas the stratosphere comes into equilibrium on a time scale of typically a few months. Due to their large heat capacity, the oceans have a much longer response time, typically decades, or up to centuries or millennia. The response time of the strongly coupled surface troposphere system is, therefore, slow compared to that of the stratosphere, and mainly determined by the oceans. The biosphere may respond fast (e.g., to droughts), but also very slowly to changes that are imposed.

RESPONSE VARIABLE (DEPENDENT VARIABLE; PREDICTED VARIABLE): In statistics, it is the variable whose values are measured or observed in an experiment for different values of the independent variable.

RESPONSIBLE CARE: A voluntary initiative of the chemical industry run in 52 countries, which requires member firms to follow codes of conducts on such matters as toxic materials, waste reduction, chemical accidents minimisation, worker safety and community consultation.

REST: The state of an object when it is not in motion relative to a particular frame of reference or another object. An object at rest will remain at rest until a force makes it move due to inertia.

REST LENGTH (PROPER LENGTH): The length of an object measured by an observer in a frame of reference where the object is at rest. This is the greatest possible length that could be recorded for an object.

REST MASS: The mass of a body when its velocity is zero, as opposed to the additional mass it gains as a result of its movement (according to the theory of relativity). For a subatomic particle it is the mass at rest or at velocities considerably below that of light.

REST POINT: The position of the pointer with respect to the pointer scale when the motion of the beam has ceased.

RESTING: 1. Describing or referring to organisms that are dormant or inactive. 2. A term describing seeds, eggs and spores that is dormant for a prolonged period before germination.

RESTING POTENTIAL: The potential difference between the electrical charges that exists across the membrane of nerve cells that is not in the process of transmitting a nerve impulse. It is established by an unequal distribution of charged ions on either side of the cell surface membrane. It is maintained by means of the sodium pump exchange of sodium ions for potassium ions across a cell membrane or the sodium-potassium exchange process. In a motor neurone, the inside of the cell is negatively charged relative to the positively charged outside and the resting membrane potential is about 60 mV.

RESTITUTION COEFFICIENT: A measure of the constant elasticity of colliding bodies. It is the ratio of the relative velocity of two elastic spheres after direct impact to that before impact. It can vary from 0 to 1, with 1 equivalent to an elastic collision and 0 equivalent to a perfectly elastic collision.

RESTORE: 1. In computing, also called **system restore**, it is to return a system such as a file or data that has been lost, stolen or damaged to a previous state. This is achieved without losing personal

data files. Restore points are automatically/manually (such as backing up documents) created, which can be used to revert the system to the way it was at a previous time. 2. Restore is a term used to describe the process of recovering lost or old data from a backup. 3. Restoring is the process of bringing back a window that has been minimised and enlarging it back to maximise mode or its "Normal" size.

RESTRICTED QUANTIFIER: In logic, a quantifier regarded as ranging over the extension of a predicate rather than over the entire domain of a logical theory.

RESTRICTING FRAGMENT LENGTH POLYMORPHISM (RFLP):

The occurrence of different cleavage sites for the restriction enzymes in the DNA from different individuals of the same species thus producing differing sets of restriction fragments. In RFLP analysis, the DNA sample is broken into pieces or digested by restriction enzymes and the resulting restriction fragments are separated according to their lengths by gel electrophoresis. RFLP have provided geneticists with a powerful set of genetic markers for mapping the genome, localisation of genes for genetic disorders, determination of risk for disease and paternity testing.

RESTRICTION: 1. A condition that imposes a constraint on the possible values of a variable, on the domain of definition of an expression or on the range of arguments of a function. 2. A function defined on a subset of the domain of a given function and taking the same values as the given function for those arguments. The restriction $f(x)$ of the function $f(x)$ to the set E is the set of pairs (x, y) such that $y = f(x)$ and x is a member of E . the restriction itself is sometimes written $f|E$.

RESTRICTION ENZYME (RESTRICTION ENDONUCLEASE): A bacterial enzyme that recognises a short specific sequence within a deoxyribonucleic acid (DNA) molecule and then catalyses double-strand cleavage of that molecule. It is produced by many bacteria and protect the cell by cleaving the DNA of invading viruses. There are three known types of restriction enzymes: (i) Type I enzymes recognise a specific sequence on DNA, but cleave the DNA chain at random locations with respect to this sequence. (ii) Type II enzymes also recognise a specific nucleotide sequence but differ from the type I enzymes in that they do not require cofactors and they cleave specifically within or close to the recognition sequence, thus generating a specific set of fragments. (iii) Type III enzymes have properties intermediate between those of the type I and type II enzymes. They recognise a specific sequence and cleave specifically a short distance away from the recognition sequence.

RESTRICTION MAPPING: A technique for determining the restriction sites within a sequence of DNA using restriction enzymes. By cleaving the DNA with various such enzymes both individually and in combination and analysing the resultant number and size of fragments by electrophoresis, the DNA can be deduced.

RESTRICTION RECOGNITION SITE (CLEAVAGE SITE; RECOGNITION SITE): The site in a polynucleotide chain as to where the restriction enzyme cleaves nucleotides by hydrolysing the phosphodiester bond between them. The restriction enzyme recognises the site as to where it cuts via a specific sequence of nucleotides along the polynucleotide chain. For example, when the restriction enzyme, Eco RI, recognises the palindromic sequence GAATTC, it cuts between G and A on the top and bottom strands.

RESULT: 1. In chemical analysis, it is the final value reported for a measured or computed quantity, after performing a measuring procedure including all sub-procedures and evaluations. 2. Any

information obtained by experiment or through some other scientific method; or the quantity or expression obtained from performing a mathematical operation or solving a mathematical problem.

RESULTANT: 1. A single vector that is equivalent to two or more other vectors or is the sum of two or more vectors. For

example, referring to the diagram,

$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$, therefore $\overrightarrow{AC}$ is the resultant vector.

2. Also called **eliminant**, it is the outcome of eliminating n variables between n homogeneous of any degree.

RESUPINATE: Inverted or turned upside down as a result of twisting of the pedicel, as seen in the flowers of most orchids.

RETAINED PLACENTA: A condition after childbirth in which all or part of the placenta or membranes have not been expelled within one hour after the birth of a baby.

RETARDATION: 1. Also known as **deceleration**, it is the rate of reduction of velocity. Deceleration is acceleration that causes reduction in velocity. Deceleration includes a negative velocity and a positive acceleration and a positive velocity and a negative acceleration. Acceleration includes positive velocity and a positive acceleration and a negative velocity and a negative acceleration. 2. The slowing down of mental functioning and bodily movement. Mental retardation can be defined as the subaverage intellectual ability equivalent to or less than an IQ of 70 that is accompanied by significant deficits in abilities such as in communication or self-care necessary for independent daily functioning, is present from birth or infancy and is manifested especially by delayed or abnormal development, by learning difficulties and by problems in social adjustment.

RETARDATION FACTOR (RF VALUE; RETENTION FACTOR):

The ratio of time spent in the stationary phase relative to time spent in the mobile phase or the distance travelled by a given component (spot) divided by the distance travelled by the solvent front in a chromatographic experiment. For example, the photosynthetic pigments of plants and the metabolites of drugs excreted in the urine can be identified by their R_f values in paper or thin-layer chromatography. The R_f can be described by the following ratio: $R_f = \text{migration distance of substance} / \text{migration distance of solvent front}$. An R_f value will always be in the range 0 to 1.

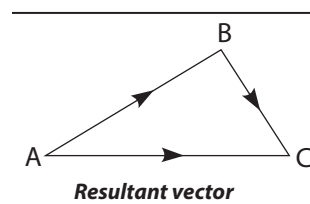
RETARDATION PLATE: A transparent plate of a birefringent material, such as quartz, with a carefully chosen orientation and thickness cut parallel to the optic axis. It works by shifting the phase between two perpendicular polarisation components of the light wave. Light falling on the plate at 90° to the optic axis is split into an ordinary ray and an extraordinary ray which travels through the plate at different speeds.

RETENTION: 1. The abnormal holding within the body of waste such as urine and faeces, which are normally excreted. 2. Ability to retain things in memory. 3. Of atoms undergoing a nuclear transformation, it is that fraction which remains in or reverts to its initial chemical form.

RETENTION EFFICIENCY: The ratio of the quantity of particles retained by a separator to the quantity entering it, generally expressed as a percentage.

RETENTION FACTOR: A measure of the time the sample component resides in the stationary phase relative to the time it resides in the mobile phase. It expresses how much longer a sample component is retarded by the stationary phase than it would take to travel through the column with the velocity of the mobile phase.

RETENTIVITY: 1. In a ferromagnetic material, it is the ability to retain its magnetisation after an external inducing field is removed. It is measured as the strength of the magnetic field that remains in a sample after removal of the external inducing field. 2. **Retentivity profile of soil** is a graph showing the retaining capacity of a soil as



a function of depth. The retaining capacity may be for water at any given tension, for cations, or for any other substances held by soils.

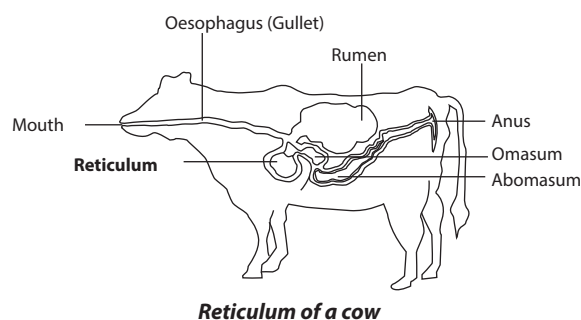
RETICULAR FORMATION: A network of nerve pathways and nuclei throughout the brainstem, connecting motor and sensory nerves to and from the spinal cord, the cerebellum and the cerebrum. It is estimated that a single neuron in this network may have synapses with as many as 25,000 other neurons.

RETICULATE CHLOROPLAST: A complex plastid that forms a cylindrical network in the peripheral region of the cells of some green algae, for example, *Oedogonium*.

RETICULATE THICKENING: A type of secondary wall patterning in tracheary elements in which the secondary cell wall is laid down to form a network. Unlike annular thickening and spiral thickening, reticulate thickening permits little further elongation and therefore occurs in tissues such as metaxylem, which have already completed most or all of their elongation.

RETICULATED: 1. Resembling a network or a net, as seen in some leaves. 2. Referring to an arrangement of veins or nerves that are like threads of a net, arranged in a network as seen in leaves.

RETICULUM: 1. Anything resembling a network in structure, as shown in the diagram. 2. Also known as **tripe**, it is the second stomach in the alimentary canal of a ruminant animal such as sheep, cattle and other ruminants. It receives coarse regurgitated food from the rumen and transfers it to the omasum. **Reticulate** means in the form of a network or net-veined. 3. A small constellation in the Southern Hemisphere lying between Dorado and Hydrus. Dorado is a constellation in the southern sky at about 5 hours right ascension and 60° south in declination; Hydrus constellation is located in the first quadrant of the southern hemisphere (SQ1) between latitudes +8° and -90°.



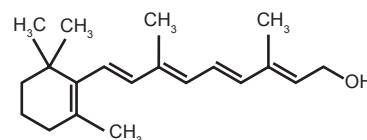
RETINA: The light sensitive tissue that lines the interior of the vertebrate eye and responsible for forming images on the retina. Because it is connected to the brain by the optic nerve, it is considered as part of the central nervous system (CNS). Light striking the retina initiates a cascade of chemical and electrical events that ultimately trigger nerve impulses. These are sent to various visual centres of the brain through the fibres of the optic nerve.

RETINA DISPLAY: A term coined by Apple in June, 2010 to describe display screens on the iPad, iPhone, iPod Touch and MacBook Pro with resolution of over 300 dpi. The iPhone 4 has a screen resolution of 326 dpi and was the first Apple product to include a retina display. The iPad, having a larger screen than the iPhone, has 264 pixels per inch and the Macbook Pro, having the largest screen, has 220 pixels per inch.

RETINACULUM: 1. One of the annular ligaments which hold the tendons close to the bones at the large joints, as at the wrist and ankle. 2. One of the retractor muscles of the proboscis of certain worms or a loop on the underside of the forewing of some moths. 3. A small gland to which bodies are attached, as the gland attached to the pollinia of orchidaceae or the hooks that support the seeds in many Acanthaceae plants.

RETINAL (RETINENE): A crystalline retinal pigment, with the chemical formula $C_{19}H_{27}CHO$, which is the aldehyde of vitamin A. It is a component of rhodopsin.

RETINOL (VITAMIN A): A fat-soluble chemical with the chemical formula $C_{20}H_{30}O$. It is a diterpenoid and an alcohol derived from carotene. It is one of the animal forms of vitamin A. Retinol is produced in the body from the hydrolysis of retinyl esters and from the reduction of retinal. It is found in milk, butter, cheese, egg, yolk and liver. Lack of retinol in the diet leads to an eye disease called **xerophthalmia**.



Chemical Structure of Retinol

RETORT: A spherical vessel with a long downward-pointing neck vessel used for distillation, sublimation or decomposition by heat. The neck acts as a condenser, allowing the evaporated vapours to condense and flow along the neck to a collection vessel placed underneath.

RETRACTION: In mathematics, it is a continuous mapping from a topological space into a subspace which preserves the position of all points in that subspace. The subspace then becomes the **retract** of the space. The retract is an **absolute retract** if, whenever B is a closed subspace of a normal space S and B is homeomorphic to A, then B is actually a retract of S. An example is the *n*-cube.

RETRACTOR: 1. A muscle that retracts an organ or a protruded body part. An example is a flexor whose retraction bends a joint or limb in the body. 2. An instrument or appliance used for drawing back a body part that obstructs another, for a procedure such as a surgery to be performed.

RETRO-: 1. A prefix which indicates a shift by one position of all single and double bonds in a conjugated polyene system. It has been applied only to carotenoids and retinoids, for example, *retro*-carotenoids, *retro*-retinoids. 2. A prefix which indicates the reverse sequence of a peptide.

RETROCURVED: To be curved or bent downwards.

RETROFLEXED: Bent backward or downward.

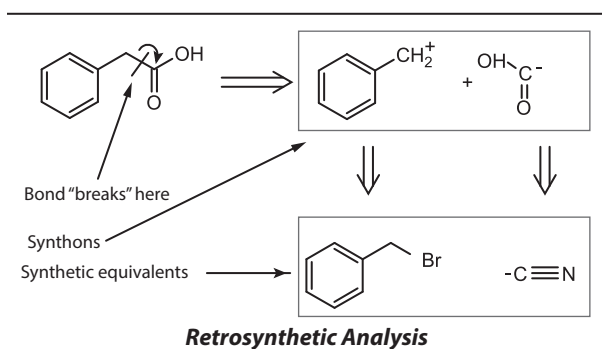
RETROGRADE MOTION: The apparent motion of a planet from the East to West as seen from the Earth against the background of the stars or a real east-to-west orbital motion of a comet about the Sun or of a satellite about its primary. The majority of the objects in the solar system revolve from west to east about their primaries.

RETROLENTAL FIBROPLASIA: A form of blindness occurring in premature babies exposed to high oxygen concentrations during the treatment of respiratory distress syndrome.

RETROROCKET (RETROGRADE ROCKET): A backward facing rocket fired to slow the speed of a capsule on its return to Earth or a rocket engine providing thrust opposing the motion of a spacecraft, thereby causing it to decelerate.

RETROORSE: In botany, it describes a plant part that is directed or turned downward or backward. Example is Glochids, which bear retrorse barbs.

RETROSYNTHETIC ANALYSIS: A technique for planning the series of reactions required to synthesise a given molecule that works backward from the known structure. It was pioneered by Elias James Corey, American chemist and Nobel Prize winner. The molecule to be synthesised called the **target molecule** is considered to be split into two parts, a process called disconnection. The parts are not themselves molecules, but are fragments called **synthons** that correspond to actual reagents.



RETROTRANSPONON: A type of genetic element that can amplify itself in a genome, found in the DNA of various organisms including yeast, drosophila and a mammal that forms copies of itself using a mechanism similar to that of retrovirus. They are abundant in plants such as maize, where they are often a principal component of nuclear DNA. About 42% of the human genome is made up of retrotransposons. It undergoes transcription to RNA and creates a DNA copy of the transcript with the aid of the enzyme reverse transcriptase. This DNA copy can then reintegrate into the cells' genome. Retrotransposons, like retroviruses can cause cancer in vertebrates by inserting in the genome near growth-promoting genes, causing the latter to be abnormally expressed.

RETROVIRUS: Any of a family of viruses containing the genetic material RNA rather than DNA and reproduces by transcribing itself into DNA. Some retroviruses destroy the cells of the DNA that they alter causing diseases such as AIDS and cancer.

RETTING: The process of soaking plant fibres such as flax or hemp so that they become easier to separate. The fibre is spread on the ground in order to subject it to the action of rain, dew and microorganisms and to cyclic wetting, drying, freezing and thawing. Another process, called water-retting consists of immersing the straw in ponds, streams or special tanks in which water and microorganisms promote decomposition.

RETURN KEY (ENTER KEY): The keyboard key that is pressed to move the cursor to the beginning of the next line or to start a new paragraph. It usually signals the computer to act on the information just entered on the keyboard, for example, to submit requests online or search an engine without using the cursor to point to the search button.

REUSE FACTOR: The multiple use of a product by returning it to the manufacturer or processor each time or to use something again, often for a different purpose and usually as an alternative to throwing it out. Many such returnable items are sold with a deposit which is reimbursed if the item is returned.

REVERBERATION: The multiple reflections or echoes of sounds inside in a particular space that merge and persist for a short while before fading away as the sound is absorbed by the walls and air. At each reflection some of the sound energy is absorbed causing the amplitude of the sound wave and the intensity of the sound to reduce a little. This is most noticeable when the sound source stops but the reflections continue, decreasing in amplitude, until they can no longer be heard.

REVERBERATION FURNACE: A metallurgical furnace used for smelting, refining or melting in which the fuel is not in direct contact with the contents. It consists of a shallow hearth on which the charge is heated by flames that pass over it and by radiation reflected onto it from a low roof. It is used in copper, tin and nickel production, in the production of certain concretes and cements and in aluminium recycling.

REVERBERATION TIME: The time taken for the average sound-energy density at a given frequency to decay to one-millionth of its initial steady-state value after the sound source has been stopped, which is a decrease of 60 dB. It is an important characteristic of an auditorium. The optimum value is proportional to the linear dimensions of the auditorium.

REVERSE: In mathematics, the reverse of a module, ring or other structure is having a non-commutative multiplication operator that is the reverse multiplication of the operator of a given structure. A reverse ring is also called an **opposite ring**.

REVERSE FAULT: A type of fault (fracture) which occurs where two blocks of rock are forced together by compression and the rock below the fault is forced upward over the rock below the fault.

REVERSE GENETICS: Any approach to genetic investigation that aims to discover the function for some known protein or gene by analysing the phenotypic effects of specific gene sequences obtained by DNA sequencing. It proceeds in the opposite direction of forward genetic screens of classical genetics. For example, an analysis of gene sequences reveals open reading frames, which are the hallmarks of functional genes. Reverse genetics methods can be used to discover the function of such genes.

REVERSE MIGRATION: A phenomenon in which the migrating organism migrates in the opposite direction of the correct and normal migratory route.

REVERSE MULTIPLICATION: In mathematics, the non-commutative operator defined by reversing the order of multiplication in some given non-commutative setting; i.e., defining $a * b = b * a$.

REVERSE OSMOSIS: A filtration process that removes many types of large molecules and ions from solutions by applying pressure to the solution when it is on one side of a selective membrane, retaining a solute on one side and allowing the pure solvent to pass to the other side. It is the reverse of the normal **osmotic process** which is the natural movement of solvent from an area of high solute concentration, when no pressure is applied. Reverse osmosis is used for desalination of drinking water.

REVERSE POLISH NOTATION (POSTFIX NOTATION): A notation that does not use parenthesis and write operators such as logical constants after their arguments. For example, P or Q is written pqa and $(1 + 2) * 3$ is written as $12+3*$ or $3\ 1\ 2\ +\ *$.

REVERSE TRANSCRIPTASE: The replication enzyme of retroviruses, that catalyses the formation of double stranded DNA using the single RNA strand of the viral genome as template or an enzyme that synthesises DNA from an RNA template, using deoxyribonucleotide triphosphates. It is called reverse transcriptase because it reverses the flow of genetic information from DNA to RNA, rather than from RNA to DNA found in transcription and therefore an RNA-directed DNA polymerase. The enzyme is used in genetic engineering for producing complementary DNA from messenger RNA.

REVERSED (RADIOCHEMICAL) ISOTOPE DILUTION ANALYSIS: Isotope dilution analysis used for the determination of the isotopic carrier in a solution of a radionuclide with the aid of one of its stable isotopes.

REVERSED-PHASE CHROMATOGRAPHY: An elution procedure used in liquid chromatography in which the mobile phase is significantly more polar than the stationary phase, for example, a micro porous silica-based material with chemically bonded alkyl chains.

REVERSIBLE PLOUGH: A plough with two mouldboard ploughs mounted back-to-back, left and right-handed which makes it possible to plough up and down the same furrow.

REVERSIBLE PROCESS: Any process in which the variables that define the state of the system can be made to reverse by means of infinitesimal changes in some property of the system without loss or dissipation of energy. Due to these infinitesimal changes, the system is in thermodynamic equilibrium throughout the entire process. Since it would take an infinite amount of time for the reversible process to finish, perfectly reversible processes are impossible. It is also a condition of a reversible process that any exchanges of energy, work or matter with the surroundings should be reversed in direction and order when the process is reversed. Any process that does not comply with these conditions when it is reversed is said to be an **irreversible process**. All natural processes are irreversible, although some processes can be made to approach closely to a reversible process.

REVERSIBLE REACTION: A chemical reaction in which the product(s) formed from the combination of the reactant(s) also recombine to reform the reactants.



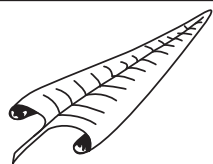
Reversible Reaction

Equilibrium is reached when the rate of formation of the products is equal to the rate of formation of the reactants. Reversible reactions are represented by a double arrow; one pointing to the product and the other pointing to the reactant. If the system attains equilibrium, then the full arrows are replaced by half-arrows.

REVERSIBLE TRANSITION: A transition that returns a system directly to its original state by reversing the process that changed it.

REVERSION: 1. The restoration of mutation to the wild-type genotype or phenotype by a second mutation. The gene or organism affected in this way is called a **revertant**. 2. The return of individuals organs, etc., to a more primitive condition or type. 3. In agricultural science, it is the changing of essential plant nutrient elements from soluble to less soluble forms by interaction with or reactions in the soil; especially the conversion of monocalcium phosphate to the less soluble dicalcium phosphate. 4. In mathematics, the formal process of computing the power series of the inverse function to that represented by a given power series. Thus one reverts the power series for arctan to obtain a recurrence for the coefficients of the power series for tan. 5. In statistics, **reversion to the mean**, also known as **regression to the mean**, is the statistical phenomenon stating that the greater the deviation of a random variate from its mean, the greater the probability that the next measured variate will deviate less far.

REVOLUTE: Having margins or tip rolled backwards towards the underside, as seen in some leaves.



Revolute leaf

REVOLUTION: 1. A complete circular movement of 360° made by a circular or cylindrical object, such as a wheel, around a fixed point or an axis. In other words, it is a complete circle made around something, such as the orbit made by a planet or satellite around another body. Revolve is to rotate around a fixed point or an axis. 2. A period during which the Earth's crust changes considerably and major features such as mountain ranges may emerge.

REYNOLDS NUMBER: A number used in fluid mechanics to determine whether fluid flow in a particular situation will be turbulent or smooth. The Reynolds number is calculated using the flow velocity, viscosity of the fluid and the dimensions of the flow channel. It is important in the design of a model of any system in which the effect of viscosity is important in controlling the velocities or the flow pattern.

RF VALUE (RETARDATION FACTOR; RETENTION FACTOR): The ratio of time spent in the stationary phase relative to time spent in the mobile phase or the distance travelled by a given component (spot) divided by the distance travelled by the solvent front in a chromatographic experiment. For example, the photosynthetic pigments of plants and the metabolites of drugs excreted in the urine can be identified by their R_f values in paper or thin-layer chromatography. The R_f can be described by the following ratio: $R_f = \text{migration distance of substance} / \text{migration distance of solvent front}$. An R_f value will always be in the range 0 to 1.

RH INCOMPATIBILITY: A condition that occurs during pregnancy if a woman has Rh-negative blood and her baby has Rh-positive blood. Rh-negative and Rh-positive refer to whether the blood has Rhesus factor (Rh factor). Rh factor is a protein on red blood cells.

RHABDOCARPOUS: Having long, rod-shaped fruits.

RHACHILLA: 1. The secondary axis of a pinnately compound leaf, such as that of a fern. 2. The main axis of a sedge or grass spikelet that bears the floret.

RHE: 1. A unit of dynamic fluidity, which is equal to the dynamic fluidity of a fluid whose dynamic viscosity is 1 centipoise. 2. A unit of kinematic fluidity, which is equal to the kinematic fluidity of a fluid whose kinematic viscosity is 1 centistoke.

RHEA: 1. The second largest moon of Saturn and the ninth largest moon in the Solar System. It was discovered on December 23, 1672, by Giovanni Cassini. Rhea appears to resemble Dione in composition, albedo features and variety of terrain. It probably has a rocky core extending out to about one third of the way from the centre, the rest of the moon being composed primarily of water ice. 2. A large ostrich-like flightless bird native to South America with grey-brown plumage, which can reach up to 1.7 metres in height and weigh up to 40 kilograms. There are two species: *Rhea americana* (greater rhea) and *Pterocnemia pennata* (Darwin's rhea). It has large legs, a long neck and three-toed feet. Rheas are omnivorous and feed mostly on broad-leaved plants, but also eat seeds, roots, fruit, lizards, beetles, grasshoppers and carrion. Rheas are commercially farmed to provide meat and eggs for consumption, feathers for use in feather dusters, hide for leather and rhea oil for use in cosmetics and soaps.

RHENIUM: A heavy, silvery-white metallic element with the symbol Re, atomic number 75, relative atomic mass 186.2, relative density 21.02, melting point 3186 °C (3459 K) and boiling point 5596 °C (5870 K). It is a rare metallic element of Group 7 (VIIB) of the Periodic Table (a transition element) and has chemical properties similar to those of manganese. Because of its high melting point, it is used in making thermocouples, electrical filaments, welding rods, thermocouples, cryogenic magnets, airplane engines, photographic flashbulb filaments and as a catalyst.

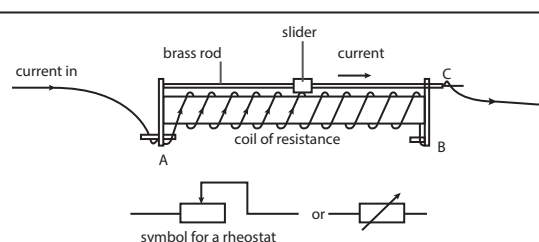
RHEO DISCOLOUR (TRADESCANTIA SPATHACEA; MOSES-IN-THE-CRADLE; OYSTER PLANT): A herb belonging to the Rhoeco genus in the Commelinaceae family that is cultivated as an ornamental houseplant. It is drought-resistant and reaches up to 30 cm in height. It has fleshy rhizomes and sword-shaped leaves that are crowded, with dark green above and purple underneath.

RHEOLOGY: The study of the deformation and flow of matter. It applies to substances which have complex molecular structures such as muds, sludges, suspensions, polymers and other glass formers such as silicates, as well as many foods and additives, bodily fluids such as blood and other biological materials.

RHEOPEXIC GEL: Gel for which the time of solidification after discontinuation of a relatively high shear rate, is reduced by applying a small shear rate.

RHEOPEXY (RHEOPECTY): The property exhibited by certain sols to become fluid when shaken or stirred and returning to a gel state when allowed to stand. Gypsum in water and printing inks are examples of rheopectic substances.

RHEOSTAT: A type of variable resistor that is used to vary the value of electric current in a circuit without breaking the circuit, used for example in controlling the brightness of a light.



A rheostat and its symbol in a circuit

RHESUS DISEASE (RH DISEASE): A serious condition affecting some babies as a result of incompatibility between a pregnant woman's blood type and the baby's, causing antibodies in the blood to destroy the baby's blood cells.

RHESUS FACTOR (RH FACTOR): A protein substance antigen on the surface of red blood cells of most humans, capable of inducing intense antigenic reactions. Most individuals possess the main Rhesus factor and termed RH positive (Rh+). Those without this factor are RH negative (Rh-) and produce antibodies if they come into contact with it. If a baby is Rh+ and the mother is Rh-, the antibodies produced upon contact with the mother's blood will attack the baby's Rh+ blood, producing disorders which may be so serious as to require blood transfusion. Rh compatibility is important in matching blood for transfusions and between pregnant women and their foetuses. The name comes from rhesus monkeys, in whose blood rhesus factors were first found.

RHESUS MONKEY: A common brownish macaque monkey (*Macaca mulatta*) family common in the forests of south and southeast Asia. It is used often in medical research for the determination of the Rh (rhesus) factor in human blood involving reaction with the blood of the rhesus monkey.

RHEUMATOID ARTHRITIS (RA): A chronic, systemic inflammatory that mainly affects synovial joints but may also affect many tissues and organs. It begins with an inflammation and thickening of the synovial membrane, followed by bone degeneration and disfigurement.

RHINOCEROS (RHINO): Large odd-toed ungulate mammal belonging to the order Perissodactyla and native to Africa and southern Asia. There are five species, with two living in Africa: the black rhino (*Diceros bicornis*) and the white rhino (*Ceratotherium simum*). Three species live in Asia: Sumatran rhino (*Dicerorhinus sumatrensis*), Javan rhino (*Rhinoceros sondaicus*) and Indian rhino (*Rhinoceros unicornis*). White rhino (*Ceratotherium simum*) is the largest and can grow to 3.7 - 4 metres long and up to 1.8 metres from hoof to shoulder and weighs about 2,300 kilograms. The Sumatran rhino is the smallest, about 2 - 4 metres long, 1 - 1.5 metres high and weighs 600 - 950 kg. Rhinos usually have grey wrinkled skin and almost entirely without hair. African species have two horns, with the one toward the end of the nose being the largest and most prominent. Sumatran rhinos also have two horns and the Javan and Indian rhinos have only one horn. Rhinos live in grasslands and open savannas and are herbivores, feeding on grass or leaves, depending on the species. They are currently under threat of overhunting by humans for their horns, which are valued by some cultures for medicinal purposes.

RHIPIDIUM: A flattened, fan-shaped cymose inflorescence in which the branches lie in the same plane and are suppressed alternatively on each side. Examples are seen in some Iris species and some Sedge.

RHIZANTHOUS: Producing flowers from a rootstock or having flowers that grow so close to the ground such that they appear to be arising from the root.

RHIZARIA: An assemblage of unicellular eukaryotic protists comprising amoeboids with filose, reticulose or microtubule-supported pseudopods. There are three main groups of rhizaria: (i) **Cercozoa**, which comprises various amoebae and flagellates, usually with filose pseudopods, common in soil. (ii) **Foraminifera** comprising amoeboids with reticulose pseudopods, common as marine benthos. (iii) **Radiolaria** comprising amoeboids with axopods, common as marine plankton.

RHIZOBIUM: A soil bacterium found in the nodules of roots of legume species that takes up nitrogen from the atmosphere and converts it to ammonia and then provides organic nitrogenous compounds such as glutamine or ureides (acyl derivative of urea) to the plant. The plant in turn provides the bacteria with organic compounds made by photosynthesis.

RHIZOBIUM RADIOBACTER (AGROBACTERIUM TUMEFACIENS): A gram-negative bacterium which is of significance for its use in genetic engineering because of its possession of tumour inducing plasmid (Ti plasmid) used in transferring cells of host plants. It also infects a wide range of plants and causes tumorous growth known as crown gall especially at the root-stem junction. It is found in soil and is parasitic to plants unlike other Rhizobium species that dwell in root nodules of legumes and are mutualistic.

RHIZOCARPOUS (RHIZOCARPIC): Having the stem dying each year, but the leaves living for several years.

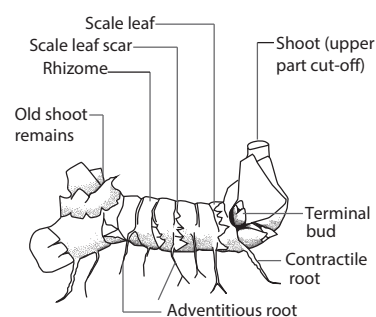
RHIZOCTONIA ROOT ROT: Rotting of roots affecting some plants such as diseases of soybean caused by the soil fungus *Rhizoctonia solani*. This fungus can be found in most soils and survives as sclerotia (very resistant fungal survival structures) in soil.

RHIZOGENIC: Producing root.

RHIZOID: Miniature root-like structures formed by the gametophytes of ferns, mosses and liverworts and by fungi in the substratum, which absorbs water and nutrients. They may be composed of many cells as in mosses, where they are usually brownish or may be unicellular as in liverworts, where they are usually colourless. In vascular plants they are often called root hairs and may be unicellular or multicellular.

RHIZOMANIA: A notifiable virus disease affecting sugar beets, in which hairs grow on the root and the leaves turn yellow.

RHIZOME (ROOTSTOCK): A horizontal underground stem having nodes, buds and tiny leaves or an organ of vegetative reproduction and food storage in flowering plants, consisting of a horizontal underground stem, growing from a parent plant. The tip of the rhizome is a bud from which grows a new plant. Most types of grasses and ginger have rhizomes. A rhizome stem helps the plant to survive from one growing season to the next and in some species it helps in vegetative propagation of the plants. If rhizomes are cut or broken into pieces they do not die, each piece may be able to give rise to a new plant, when replanted.



Rhizome

Rhizomatous means having rhizomes.

RHIZOMORPH: Mass of fungal hyphae that are organised in long, thick strands with a darkly pigmented outer rind that contains specialised tissues for absorption and water transport.

RHIZOPLANE: The external surface of roots that strongly adheres to soil particles.

RHIZOPODA: A phylum of the Protoctista free-living species found in the soil, in fresh and salt water that contains amoebas and cellular slime-mould. They possess pseudopodia, used for locomotion and engulfing food particles. Rhizopodea include five subclasses: Lobosia, Filosia, Granuloreticulosa, Mycetozoa and Labyrinthula.

RHIZOPUS: A widespread genus of saprobic fungi of the family Mucoraceae that includes several species. Examples include *R. arrhizus* (which causes fruit rot on apples), *R. artocarp* (which causes fruit drop of jack fruit), *R. nigricans* or bread mould (which grows on spoiled food), *R. oligosporus* (used to make tempeh, a fermented food from soyabeans), *R. oryzae* (which is used in the production of alcoholic beverages) and *R. stolonifer* or black bread mould (which causes fruit rot on strawberry, tomato and sweet potato). Rhizopus species develop as filamentous, branching hyphae that lack cross-walls. Reproduction is by sexual and asexual spores. Some Rhizopus species are opportunistic agents of human zygomycosis (a fungal infection) that can be fatal.

RHIZOSPHERE: The soil surrounding the roots of plant containing exudates of the root. It is characterised by a zone of increased microbiological activity. It is an example of the relationship of soil microbes to higher plants.

RHIZOSPHERE COMPETENCE: Ability of an organism to colonise the rhizosphere.

RHIZOTAXIS (RHIZOTAXY): The type of root arrangement on a plant.

RHODIUM: A hard, durable, silvery-white metallic element with the symbol Rh, atomic number 45, relative density 12.4, relative atomic mass 102.905, melting point 1,966 °C (2239 K) and boiling point 3,727 °C (4000 K). It is a member of the platinum groups of metals and is resistant to tarnish, corrosion and acid. It occurs as a free metal in the natural alloy osmiridium. It is used in jewellery, electroplating and thermocouples.

RHODOPHYTA (RED ALGAE): A large diverse phylum or division of plants of mostly multicellular, marine algae including many notable types of seaweed distinguished by having an abundance of the pigments phycoerythrin and phycocyanin. Members of the Rhodophyta may be unicellular or multicellular. They are commonly found along the coastal areas of tropical regions. Sexual reproduction is by means of carpospores. Some characteristics of rhodophyta are eukaryotic cells without flagella and centrioles, using floridean starch as food reserve and chloroplasts lacking external endoplasmic reticulum containing unstacked thylakoids.

RHODOPSIN (VISUAL PURPLE): The light sensitive purple-red organic pigment found in the rods of the vertebrate retina. It consists of opsin (a glycoprotein), linked to retinal (a non-protein chromophore derived from vitamin A). Photons of light falling on the rod is absorbed by the retinal, which changes its form and separates it from the opsin component, starting a biochemical chain of events that ends with impulses being sent along the optic nerve to the brain. Lack of rhodopsin results in night blindness. Vitamin A is the primary source of rhodopsin.

RHOMBIC: 1. In botany, a 2-dimensional shape or diamond shaped in outline with the broadest axis in the middle and with a length:breadth ratio between 3:2 and 2:1. 2. Shaped like a rhombus or resembling a rhombus.



Rhombic leaf

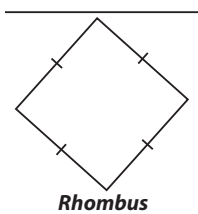
RHOMBIC SULFUR (OCTAHEDRAL SULFUR; ALPHA-SULFUR): A transparent pale yellow crystalline solid, which crystallises from a solution in carbon disulfide. It has a density of 2.06 g/cm³ at 20°C (293 K). It consists of rings of 8 atoms of sulfur. It is insoluble in water and soluble in carbon disulfide, benzene, chloroform, etc. It is the most stable of all the allotropes of sulfur. It is a non-conductor of heat and electricity.

RHOBOMBOHEDRON: A six-sided prism of which the sides are parallelograms or a solid bounded by six rhombic planes.

RHOMBOID: The two rhomboid muscles extend as parallel bands from the spines of the vertebrae to the scapula. They pull the scapula upward and backward and help to rotate it. The rhomboideus minor runs from the ligamentum nuchae and the seventh cervical spine to the medial margin of the scapula opposite its spine. The rhomboideus major, about twice the width of the minor, extends from the upper thoracic spines and supraspinous ligaments to the medial margin of the scapula from the spine to the inferior angle.

RHOMBOIDAL: Shaped nearly like a rhomboid, with obtuse lateral angles.

RHOMBUS: A parallelogram with four equal sides and oblique angles or oblique-angled equilateral parallelogram with diagonals bisecting each other at right angles. Its area is half the product of the lengths of the two diagonals. A rhombus whose internal angles are 90° is called a **square**.



Rhombus

RHUBARB: Any of several species of the genus Rheum (family Polygonaceae), with edible leaf stalks, used mainly as dessert, grown mainly in the temperate zones. It has high oxalate content and the leaves are toxic to cattle and humans.

RHYNIOPHYTES: An extinct division of the subkingdom Embryobionta of vascular land plants that flourished in the early Devonian, 400-380 million years ago. Three classes of Rhyniophyta recognised are Rhyniopsida, Zosterophyllopsida and Trimerophytosida.

RHYOLITE: A fine-grained light-coloured volcanic rock, occurring principally in volcanic lava. It is chemically identical to granite and composed primarily of quartz and orthoclase feldspar with subordinate amounts of plagioclase feldspar, biotite mica, amphiboles and pyroxenes rhyolite.

RHYTHM METHOD (NATURAL FAMILY PLANNING, NFP):

A method of natural contraception that works by avoiding sexual intercourse when the woman is producing egg cell (ovulating). Sexual intercourse is also permitted during naturally infertile periods (portions of the menstrual cycle, during pregnancy or after menopause). The time of ovulation can be worked out by the calendar, by temperature changes or by inspection of the cervical mucus. During each menstrual cycle, a woman's fertile period begins around five days before ovulation (the release of an egg from one of the ovaries) and continues for about two days after ovulation.

RIA: 1. A long narrow sea inlet, usually branching and surrounded by hills, which are almost always estuaries. A RIA is deeper and wider towards its mouth. It is formed by the flooding of a river valley due to either eustatic sea level change or isostatic sea level change. 2. The abbreviation for **Radioimmunoassay**, a sensitive method for measuring the level of antigens such as enzymes, hormones and steroids in blood, in which a radioactively labelled form of the antigen is mixed with a known amount of antibody for that antigen, with the two binding to each other.

RIB: 1. A long, slender bone attached to the backbone that curves around the chest cavity or thorax. The front of each rib ends in a short, flexible section of cartilage. Ribs occur in pairs and are found in almost all vertebrates. Humans have 12 pairs extending from the spine to or toward the sternum that form a cage and serve to enclose, support and protect the heart and lungs in the thoracic cavity. In some animals, especially snakes, ribs may provide support and protection for the entire body. 2. The main vein or any of the prominent veins of a leaf or other plant organ. 3. One of many beams extending from the keel to the top of the hull of a ship, forming the framework of the hull. 4. A part of an aircraft's wing crossing from the leading to the trailing edge of the wing.

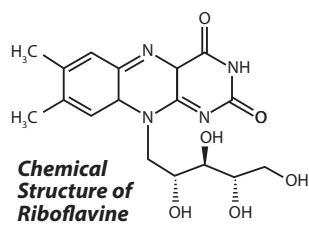
RIB GRASS (ENGLISH PLANTAIN; RIPLEGRASS): A palatable, deep-rooting weedy Eurasian plant (*Plantago lanceolata*) with a high mineral content. It has lance-shaped, ribbed leaves and a dense spike of small whitish flowers.

RIBBON: A new feature of Microsoft Office 2007 Window that replaces the Menu Bar. It is a user interface element created by Microsoft, which was introduced with Microsoft Office 2007. It combines the menu bar and toolbar into a single floating pane. By default, the ribbon is located at the top of the screen in Office applications, such as Access, Excel, PowerPoint, Word and Outlook. The purpose of the ribbon is to provide quick access to commonly used tasks within each program. The Microsoft Word ribbon includes Home, Insert, Page Layout, References and other tabs that each displays a different set of commands when selected.

RIBBON CABLE: A flat cable in which all the wires are parallel to one another in a single plane, often used in computers.

RIBBON LAKE: A long, narrow finger-shaped lake found on the floor of a glacial trough. It often forms in an elongated hollow carved out by a glacier. It can also form when water ponds up behind a terminal moraine or a landslide.

RIBOFLAVINE (E101 FOOD COLOUR; VITAMIN B2): The yellow component of the B complex group with the chemical formula $C_{17}H_{20}N_4O_6$. It is an easily absorbed micronutrient with a key role in maintaining health in humans and animals and an important coenzyme in many biochemical processes such as energy metabolism and for the metabolism of fats, ketone bodies, carbohydrates and proteins. Exposure to light destroys riboflavin. Examples of sources of riboflavin are milk, liver, kidneys, legumes, tomatoes, yeast, mushrooms, cheese and leafy green vegetables.



RIBONUCLEIC ACID (RNA): A molecule occurring in certain viruses (RNA viruses) and in cellular organisms that directs the middle steps of protein production. It is one of the two biologically important types of molecule, the other being DNA. It consists of a long chain of chemical compounds called **nucleotides**. Each nucleotide is made up of a sugar molecule called ribose, a phosphate group and one of four different nitrogen-containing compounds called bases. The four bases are adenine, guanine, uracil and cytosine. These components are joined together in the same manner as in a deoxyribonucleic acid (DNA) molecule. RNA differs chemically from DNA in two ways: the RNA sugar molecule contains an oxygen atom not found in DNA and RNA contains the base uracil in the place of the base thymine in DNA. In RNA viruses, the RNA directs two processes: **protein synthesis** (production of the virus's protein coat) and **replication** (the process by which RNA copies itself). There are three kinds of RNA: (i) **Messenger (mRNA)** that carries the genetic code from the DNA to the ribosomes; (ii) **Ribosomal RNA (rRNA)** that combines to form ribosomes; (iii) **Transfer RNA (tRNA)** carries amino acid from the cytoplasm to the ribosome for protein synthesis. RNA is transcribed from DNA by enzymes called **RNA polymerases** and is generally further processed by other enzymes. RNA is central to protein synthesis.

RIBONUCLEOPROTEIN (RNP): Any nucleoprotein that contains RNA that forms during the synthesis of RNA in eukaryotes. The protein is removed in the packaging and condensation of the RNA. The most important RNP occurring in the nucleus is heterogeneous nuclear RNP which consists of protein bound to the primary transcript of DNA. RNPs play an important role in influenza virus replication. The viral RNA is transcribed into mRNAs by the RNA polymerase attached to the RNPs.

RIBOSE (D-RIBOSE): An organic compound of aldopentose, which is a monosaccharide containing five carbon atoms that, in its acyclic form, has an aldehyde functional group at one end with the chemical formula $C_5H_{10}O_5$. It is also related to deoxyribose found in DNA, by the removal of one hydroxyl group. It is found especially in the levorotatory D-form as a constituent of a number of nucleosides (as adenosine, cytidine and guanosine) especially in RNA. Once phosphorylated, ribose can become a subunit of adenosine triphosphate (ATP), nicotinamide adenine dinucleotide (NADH) and several other compounds that are critical to metabolism. D-Ribose is often marketed as supplement for body builders (5g per day) and also used to reduce fatigue in fibromyalgia and chronic fatigue syndrome.

RIBOSOMAL DNA (rDNA): Genes that code for the various forms of ribosomal RNA and the segments of DNA separating the individual ribosomal RNA genes, referred to as ribosomal spacer DNA.

RIBOSOMAL RNA (rRNA): One of the three types of RNA, which is found in the cell's ribosomes, the specialised structures that are the sites of protein synthesis. Ribosomal RNA provides a mechanism for decoding mRNA into amino acids and interacts with tRNAs during translation by providing peptidyl transferase activity.

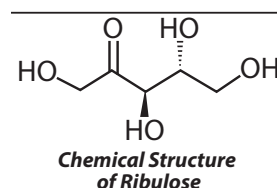
RIBOSOMAL SUBUNITS: Two units that combine with mRNA to form the ribosomal mRNA complex at which protein synthesis occurs.

RIBOSOMES: A submicroscopic cluster of proteins and RNA found in the cytoplasm of living cells, on which protein synthesis from amino acids takes place. Ribosomes consist of two sub-units: the **large subunit (LSU)** and **small subunit (SSU)**, each of which comprises a type of RNA and protein. A ribosome also has three binding sites called A, P and E. The A site in the ribosome binds to an aminoacyl-tRNA (a tRNA bound to an amino acid). The amino (NH_2) group of the aminoacyl-tRNA, which contains the new amino acid, attacks the ester linkage of peptidyl-tRNA (contained within the P site), which contains the last amino acid of the growing chain, forming a new peptide bond. This reaction is catalysed by peptidyl transferase. The tRNA that was holding on the last amino acid is moved to the E site and what used to be the aminoacyl-tRNA is the peptidyl-tRNA.

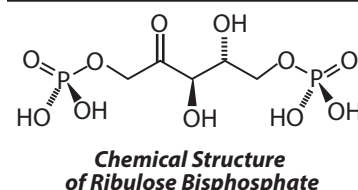
RIBOSWITCH: A region of an RNA molecule, usually a messenger RNA (mRNA), that can directly bind a small target molecule and 'switch off' further transcription or translation of the mRNA. It acts as a mechanism in bacterial cells for regulating the concentration of a metabolite by controlling the synthesis of, for example, an enzyme or transport protein involved in synthesis or uptake of the metabolite.

RIBOZYME (CATALYTIC RNA; RNAENZYME): Any RNA molecule that can catalyse a chemical reaction, which changes its own molecular structure. Self-splicing introns such as glucose are examples of ribozymes. Ribozymes have properties very similar to viroid (an infectious RNA particle that resembles a virus, but smaller). Many natural ribozymes catalyse either the hydrolysis of one of their own phosphodiester bonds or the hydrolysis of bonds in other RNAs. They have also been found to catalyse the aminotransferase activity of the ribosome.

RIBULOSE: A ketopentose sugar with the chemical formula, $C_5H_{10}O_5$ which is a monosaccharide containing five carbon atoms, including a ketone functional group. It occurs in plants and is involved in carbon dioxide fixation in photosynthesis as a component of ribulose biphosphate. It is also an intermediate in the fungal pathway for D-arabitol production. A synthetic form of ribulose known as **sucro ribulose** is found in many brands of artificial sweeteners.



RIBULOSE BISPHTHOSPHATE (RuBP; RIBULOSE-1,5-BISPHTHOSPHATE): A five-carbon sugar that is combined with carbon dioxide to form three-carbon intermediates in photosynthetic carbon fixation. RuBP is both used and created during the Calvin Cycle. It is a product of the phosphorylation of ribulose-5-phosphate by adenosine triphosphate (ATP). It is the enzyme that mediates the carboxylation of ribulose biphosphate, ribulose biphosphate carboxylase/oxygenase and is also involved in photorespiration.



(ATP). It is the enzyme that mediates the carboxylation of ribulose biphosphate, ribulose biphosphate carboxylase/oxygenase and is also involved in photorespiration.

RICCATI EQUATION: In mathematics, a non-linear differential equation of the form $y' = f(x) + P(x)y + Q(x)y^2$ whose solutions are not generally obtainable by integration. It is named after Jacopo Riccati (1676–1754), Italian mathematician.

RICE: An edible starchy cereal grain and the cereal grass *oryza sativa*, of family Poaceae of South-Asian origin grown in warm wet places. It can grow to a height of about 1.2 m and has long, flat leaves and an inflorescence made up of spikelets bearing flowers that produce a brownish grain covered by bran and enclosed by a husk. Brown rice contains 8% protein and iron, calcium and B vitamins. Removal of the bran layer leaves white rice, greatly diminished in nutrients.

RICE-HERZFELD MECHANISM: A mechanism enabling complex chain reactions to give simple rate laws in chemical kinetics. An example of a Rice-Herzfeld mechanism occurs with the pyrolysis of acetaldehyde, the mechanism consisting of initiation, propagation (in two steps) and termination. This leads to the experimental result that the overall reaction is three-half order in CH_3CHO . To ascertain that such a mechanism is correct, either the steady-state approximation is used or the differential equations for the reaction rates are solved numerically.

RICE-RAMSPERGER-KASSEL-MARCUS THEORY (RRKM THEORY): A theory of unimolecular chemical reactions that enables the computation of simple estimates of the unimolecular reaction rates from a few characteristics of the potential energy surface. The theory improved the Rice-Ramsperger-Kassel theory by the US chemists, O. K. Rice and R. A. Marcus in 1905. The RRK theory was generalised into the RRKM theory in 1952 by R. A. Marcus who took the transition state theory of the individual vibrational frequencies, rotations and zero-point energies of the energised species and activated complexes developed by Eyring in 1935 into account. The RRKM theory is very successful in explaining the results of experiments for unimolecular reactions for a variety of reactions.

RICH TEXT FORMAT (RTF): A file format standardised by Microsoft for creating formatted text files. Unlike text files, RTF files may offer such formatting features as bold, italic, underline, bullets, different fonts and text justification. However, it might not allow all the features that may be found in a full word processor. RTF format can be read by nearly all word processors. For example, you can create a file using Microsoft Word in Windows 98, save it as an RTF file (it will have a «.rtf» file name suffix) and send it to someone who uses WordPerfect 6.0 on Windows 3.1 and he will be able to open the file and read it. The RTF Specification uses the ANSI, PC-8, Macintosh and IBM PC character sets. It defines control words and symbols that serve as "common denominator" formatting commands. When saving a file in the Rich Text Format, the file is processed by an RTF writer that converts the word processor's markup to the RTF language. When being read, the control words and symbols are processed by an RTF reader that converts the RTF language into formatting for the word processor that will display the document.

RICHARDSON EXTRAPOLATION (RICHARDSON IMPROVEMENT; DEFERRED APPROACH TO THE LIMIT): An extrapolation method for improving the rate of convergence of a sequence, using two computed values, h_{LARGER} and h , of the form

$$E(h) = \frac{F(h_{\text{LARGER}}) - F(h)r^n}{1 - r^n}, \text{ where } r \text{ is the ratio of } h_{\text{LARGER}} \text{ to } h \text{ and } F$$

is an $O(h^n)$ approximation to some quantity. If the truncation error for F is of higher order $O(h^n)$, then the error in the extrapolation $E(h)$ will be of this order. The process can then be repeated with m replacing n . It is named after Lewis Fry Richardson (1881-1953), English mathematician and physicist.

RICHTER SCALE: A measure of the magnitude of seismic waves from an earthquake at its epicentre based on a scale from 1 to 10 according to the amount of energy released by the earthquake, with a higher number indicating stronger tremors. It was developed in 1935 by Charles F. Richter of the California Institute of Technology. It is logarithmic and whole-number jumps indicate a tenfold increase. In this case, the increase is in wave amplitude. That is, the wave amplitude in a level 6 earthquake is 10 times greater than in a level 5 earthquake and the amplitude increases 100 times between a level 7 earthquake and a level 9 earthquake. The amount of energy released increases 31.7 times between whole number values. A magnitude of less than 1.0 to 2.9 is termed micro and generally not felt by people, though recorded on local instruments. A magnitude of 8.0 and higher is termed great and involves severe destruction and loss of life over large areas.

RICCIA: A genus of liverworts in the order Marchantiales, family Ricciaceae that is widely distributed throughout the world and found floating in ponds, forming rosettes on moist soils or growing in bare patches in lawns during the wet season.. The plant body possesses two types of rhizoids on the ventral surface for absorption and support. An example is *Riccia fluitans*, known commonly as slender riccia or crystalwort, which floats in shallow ponds forming branching green ribbons about 0.1 centimetre wide and about 1.3 to 5 centimetres long. The ribbons often become tangled in large masses and often forms thick mats on and under the water surface. It has a dichotomously branched thallus plant body which is not distinguished into true root, stem and leaves. It is sometimes used in aquariums.

RICIN: A highly toxic substance that is extracted from the castor bean (*Ricinus communis*). It is one of the most poisonous substances present in plants. The lethal dose (LD50) of ricin is around 22 mg per kg, which is 1.76 mg per kilogram of body weight for an average adult. Ricin is poisonous if inhaled, injected or ingested, acting as a toxin by the inhibition of protein synthesis. It agglutinates red blood cells.

RICKETS: A nutritional disorder in children characterised by skeletal deformities caused by lack of vitamin D. This results in insufficient calcium phosphate deposits in the bones, causing the bones not to harden adequately and become soft, curved and stunted. If medical attention is not sought early, effects may include bowlegs, knock-knees and beading where the ribs meet the breastbone.

RICKETTSIA: A very small pathogenic coccoid-shaped or rod-shaped gram-negative bacterium belonging to the phylum Proteobacteria that live in vertebrates and are transmitted by bloodsucking parasitic arthropods such as fleas, lice and ticks. Rickettsias are mostly obligate parasites, which cannot produce outside the cell of their host.

RICTAL BRISTLES: A stiff bristle-like feather, with little or no vane, which grows at the base of many insectivorous birds' bill that may serve to capture food in flight.

RICTUS: 1. The cleft of an open mouth or beak. 2. The mouth of a bilabiate corolla.

RIDDLE: 1. A coarse sieve, for sieving soil. 2. Grading and sorting produce according to size using a sieve.

RIDGE: 1. A long narrow protuberance or crest, such as on a bone or the backbone of an animal such as a whale. 2. A large narrow raised part of a surface especially a high edge along a mountain or a long narrow hilltop or range of hills or an elevation on the ocean floor resembling a ridge on land that results from volcanic eruption along the fissures between tectonic plates. 3. A long area of high pressure in a weather system.

RIDGE TILLAGE SYSTEM: A type of reduced tillage system where the crop is grown on top of permanent ridges. A ridge is built and the crop is then planted on top of this shallow ridge. Residue is left on the surface between ridges. Weed control is accomplished with herbicides and/or cultivation. The ridges are preserved and rows are planted in the same location each year.

RIEMANN CONDITION: In mathematics, the condition that a function is integrable on an interval if, for every $\epsilon > 0$, there is a partition of the interval for which the upper sum and lower sum differ by less than ϵ .

RIEMANN HYPOTHESIS (RIEMANN ZETA HYPOTHESIS): The conjecture that the zeta function, defined by

$$\zeta(z) = \sum_{k=1}^{\infty} \left(\frac{1}{k^z} \right), \text{ has no non-trivial zeros except on the line}$$

with $\text{re}(z) = \frac{1}{2}$. The trivial zeros occur at negative even integers and known to be true for the first several million zeros.

RIEMANN INTEGRAL: Mathematically, it is the definite integral of a real-valued bounded function, defined on a bounded interval as the value

to which all Riemann sums, converge as their mesh-fineness tends to zero. It is a process of integration of a function on an interval that can be evaluated by the fundamental theorem of calculus or approximated by numerical integration, which amounts to finding the area under a curve.

RIEMANN MAPPING THEOREM: The theorem that any simply-connected complex domain, whose boundary contains at least two points, can be mapped in a conformal fashion onto the open unit disk.

RIEMANN SPHERE: A topological sphere consisting of the complex plane and the point at infinity. It is an example of a Riemann surface.

RIEMANN SUM: A method for approximating the total area underneath a curve or lines on a graph and may also be used to define the integration operation. For a real function f on an interval

$[a, b]$, it is any sum of the form $\sum_{i=0}^n f(c_{i+1})\Delta_i$, where $\Delta_i = t_{i+1} - t_i$, for any partition with $b = t_n > t_{n-1} > \dots > t_1 > t_0 = a$, where $t_{i+1} \geq c_i, t_i$.

RIEMANN SURFACE (COMPLEX CURVE): A complex manifold with one complex dimension. The Riemann surface is a conformal structure.

RIEMANN-LEBESGUE LEMMA: The lemma that is of importance in harmonic analysis and asymptotic analysis that

$$\lim_{t \rightarrow \infty} \int_I f(x) \exp(itx) dx = 0 \quad \text{for any interval } I \text{ of the real line, real variable}$$

t and any Lebesgue integrable function f .

RIEMANN-STIELTJES INTEGRATION: A generalisation of the Riemann integral allowing integration of a function, f , with respect to a function, g , defined as the limit of the sums $m_f(i) [g(t_{i+1}) - g(t_i)]$, where $m_f(i)$ is the supremum for the upper sums or the infimum for the lower sums of f on the subinterval $[t_i, t_{i+1}]$; these limits are equal and the integral exists, when f is continuous and g is of bounded.

RIEMANN-STIELTJES MEASURE (RADON MEASURE): A measure on sigma algebra on a general topological space such that every point has a neighbourhood of finite measure and the measure of any set equals the supremum of the inner measure of all its compact subsets. It may also be defined in terms of the infimum of the outer measure of all open sets containing the given set.

RIEMANN-ZETA FUNCTION (EULER-RIEMANN ZETA FUNCTION; ZETA FUNCTION): In mathematics, it is the special function, denoted $\zeta(s)$ that is defined over the complex plane for one

complex variable, s . The sum of the infinite series $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, is

infinite when $s = 1$; when s is greater than 1, then it converges to a finite number as successive terms are added; if s is less than 1, the sum is again infinite. It is a very important function in number theory.

RIEMANN, GEORG FRIEDRICH BERNHARD (1826-66): German mathematician, who studied mathematics under Gauss at Göttingen, where he later became Professor in 1859. His major achievements include work in function theory, the development of differential geometry from its origins in the work of Gauss and founded non-Euclidean Riemannian geometry and introduced the Riemann integral. He also formulated the Riemann hypothesis; and was elected a Fellow of the Royal Society.

RIEMANNIAN GEOMETRY: 1. In mathematics, a type of non-Euclidean geometry ideally suited to describing the curvature of space-time in general relativity theory. In other words, it is a generalisation of the intrinsic geometry of a surface in which an n -dimensional manifold is endowed with a differential quadratic form that is taken as its element of arc, in terms of which analogues of length, angle, curvature, etc., can be defined. Riemannian geometry in which the quadratic form need not be positive definite has applications in relativity theory and it can be regarded as a distortion of Euclidean geometry. 2. Also known as **elliptic geometry**, it is one of the two most important types of non-Euclidean geometry, the other is hyperbolic geometry. In elliptic geometry, no line is parallel to any

other line through any given point; lines intersect in 1 point and the angles of a triangle always add up to more than 180 degrees. In spherical geometry, all lines intersect in 2 points.

RIEMANNIAN MANIFOLD: A manifold endowed with a metric tensor. In other words, it is a differentiable manifold in which each tangent space is equipped with an inner product $\langle \cdot, \cdot \rangle$ in a manner which varies smoothly from point to point. It is useful in measuring distances and angles on manifolds.

RIESZ-FISCHER THEOREM: In mathematics, it is the theorem that every square summable sequence is the sequence of Fourier coefficients of a unique Hilbert function which is square integrable.

RIESZ REPRESENTATION THEOREM (RIESZ-KAKUTANI THEOREM): In mathematics, it is the theorem that all continuous linear functionals on the space $C(S)$ of real valued continuous functions, where S is a compact Hausdorff space, may be identified isometrically with differences of regular Borel measures on S .

RIFFLING: The separation of a free-flowing sample into (usually) equal parts by means of a mechanical device composed of diverter chutes.

RIFT VALLEY: A valley formed between highlands or mountain by the subsidence of a block of the Earth's crust between two or more parallel faults or a geologic rift or fault.

RIFT ZONE: An elongate system of crustal fractures associated with an area that has undergone extension (ground has spread apart). On the great shield volcanoes in Hawaii, a rift zone consists of many different features associated with the rise and eruption of magma from narrow dikes, including eruptive fissures, cinder and spatter cones, spatter ramparts, pit craters, lava flows, ground cracks and normal faults.

RIG: 1. A condition in a male animal, especially pigs and horses in which one or both testicles has not descended into the scrotum at the usual time. 2. Also called **oil rig** or **drill rig**, it is the equipment used in drilling for oil or gas. 3. To fit a ship with the necessary sails, shrouds and yards; or the arrangement of masts, spars and sails on a sailing vessel.

RIGEL (BETA ORIONUS): The seventh brightest star in the sky in the constellation Orion. It is a blue-white super giant, with an estimated diameter of 50 times that of the Sun and an intrinsic brightness about 40,000 times as luminous as that of the Sun. Its distance from the Earth is nearly 1,000 light-years (1 light year is about 9.46 trillion km).

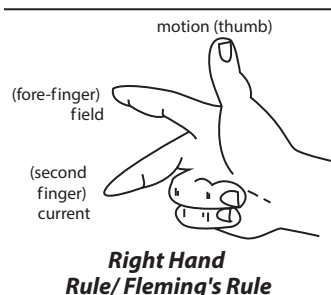
RIGESCENT: Becoming rigid or numb.

RIGHT: 1. In mathematics, also called **right-angled**, possessing a right angle; for example, a right circular cone, has a right angle between the axis and any diameter of the base. 2. Of an operator in a non-commutative theory, acting on the right; I_r is a right identity if $xI_r = x$ for all x ; r_x is a right inverse of x if $xr_x = I_r$.

RIGHT ASCENSION: The coordinate on the celestial sphere or equatorial coordinate system that corresponds to longitudes on the surface of the Earth. The right ascension of a celestial body is the angular distance measured eastward from the vernal equinox along the celestial equator to its intersection with the body's hour circle. It is measured in hours, minutes and seconds eastwards from the point where the Sun's path, once a year intersects the celestial equator. This point of intersection is called the **vernal equinox**.

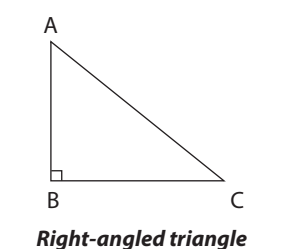
RIGHT CLICK: In computing, to press the right button on the mouse and release it, which opens a drop-down menu of ancillary operations such as copying and pasting, viewing a file's properties, change case, new folder, etc., depending on the location of the screen cursor at that moment and the system being used.

RIGHT HAND RULE (FLEMING'S RULE): A relative direction of motion, magnetic field and current flow in a moving wire in a magnetic field, such as the wire on the armature of a generator. It states that if the right thumb is held at right angles to the first and second fingers, then the second finger will be pointing in the direction of the current, the first finger in the direction of the magnetic field or force and the thumb will be pointing in the direction of the motion.



RIGHT OF ACCESS: 1. The right of someone to be able to get to his land by passing over someone else's property. 2. The right of the public to walk in areas of the country side, providing they do not harm crops or farm animals.

RIGHT-ANGLE: An angle whose size or measure is ninety degrees (90°) or $\pi/2$ radians. Perpendicular lines form right angles at their intersections. A **right-angled triangle** is a triangle in which there are two complementary angles (the sum of the angles is 90°) or one of the angles is a right angle (90°). It can be inscribed in a special way into a circle. It is the basic form of triangle for defining trigonometric ratios (sine, cosine and tangent) and for which Pythagoras' theorem holds true.



RIGHT-HAND LIMIT: In mathematics, the one-sided limit of a function defined on an interval from above or from the right; the limit where x is restricted to values greater than a and expressed as $\lim_{x \rightarrow a+} f(x) = f(a+)$.

RIGHT-HANDED COORDINATE SYSTEM: In mathematics, a three-dimensional coordinate system, with the x -, y - and z -axes satisfying the right-hand rule of the cross product: if the thumb is placed in the positive direction of the first line (x -axis) then the angle between the positive directions of the other two fingers (middle finger for y -axis and forefinger for z -axis) is less than π and the triple product is positive. A **right-handed trihedral** is a configuration of three non-planar directed lines also has the same characteristics.

RIGID BODY: In physics, it is a solid body that is not deformable, thus the distance between two points or particles remain fixed by internal forces of constraint whatever the force applied to it.

RIGID BODY MOTION: In physics, it is a motion of a body consisting only of **rotations** described by angular motion; and **translations** defined as any motion in which every line in the body remains parallel to its original position at all times.

RIGID MOTION: In physics, any transformation, consisting of translation, rotation, reflection and glide reflection of a plane such that the distance between points is preserved.

RIGIDITY: The ability of a body or substance to oppose or withstand forces directed toward changing the body's shape, while keeping its volume constant. Rigidity refers to a purely static situation. A related quantity is viscosity.

RIGIDITY MODULUS (COEFFICIENT OF RIGIDITY; MODULUS OF RIGIDITY; RIGIDITY MODULUS; SHEAR MODULUS): The modulus of elasticity for an object exposed to shearing stress. It is expressed as the ratio between F/A (tangential force per unit area) and ϕ , where ϕ (the Greek letter phi) stands for the angle of shear or tangential force per unit area/angular deformation (ϕ).

RIGOR MORTIS: The recognisable sign of death, caused by a chemical change in the muscles after death, resulting in a temporary rigidity of the muscles. It arises because adenosine triphosphate (ATP), which is required to break the bridges that form between actin and

myosin filaments in muscle tissue during contraction, is no longer synthesised after death. Factors such as physical activity immediately prior to death affect the onset and duration of rigor mortis. In humans it starts about 3 hours, reaches maximum rigidity after 12 hours and gradually dissipates approximately 72 hours (3 days) after death.

RIGOROUS: In mathematical logic, of a proof, making completely explicit the validity of the successive steps, usually with reference to some underpinning formal system.

RILL: 1. An artificial channel, often a straight and narrow one, along which water is made to flow, used in landscaping a garden. 2. A narrow and shallow channel eroded in soil by rain water. It can be removed during ordinary cultivation.

RIMOUS (RIMOSE): Having fissures, cracks or crevices, as in the bark of a tree.

RIND: The tough outer protective layer or covering of some fruits such as orange or vegetables and foods such as cheese. A fruit with a thick rind, such as a citrus fruit, is called a **hesperidium**. The rind of some fruits, for example, pomegranate, which is high in tannins and other polyphenols, is employed in the production of dyes.

RINDERPEST: An acute and highly infectious viral disease affecting mostly cattle and also seen in sheep and goats. It occurs mainly in northeast Africa and southwest Asia, marked by symptoms such as fever, haemorrhage, inflammation of the mucous membranes, particularly the intestines, discharges from the eyes and nose, dehydration and diarrhoea. It can be controlled by destroying infected animals. Animals can be immunised against rinderpest and importation of animals from affected regions is strictly controlled.

RING: 1. A set with the following properties: (i) it has two binary operations, addition and multiplication ($+$ and $\times$); (ii) it is an Abelian group under addition; (iii) multiplication is associative and distributive over addition; and (iv) a and b are members of the set, $a \times b$ has a unique value c , also a member of the set. If there is a member n of the set such that $a \times n = a$ for every member a of the set, the ring is a ring with unity. A ring in which multiplication is commutative is a commutative ring. 2. **Planetary ring** is an annulus of material, in the form of countless numbers of small objects, in orbit around a planet. 3. A circular band of metal or other durable material such as copper, diamond, gold or steel. An example is a wedding ring.

RING ASSEMBLY: Two or more cyclic systems (single rings or fused systems) which are directly joined to each other by double or single bonds are named ring assemblies when the number of such direct ring junctions is one less than the number of cyclic systems involved.

RING MAIN: A domestic wiring system in which all the socket outlets have their own individual fuses. The outlets are connected in parallel to a ring circuit, which starts and ends at a mains supply point. Therefore, if a fault in the ring occurs the section with the fault can be repaired without disrupting the supply of electricity to all consumers.

RING NETWORK: A local area network (LAN) in which each device is connected to two other devices to form a closed loop configuration.

RING OF FIRE: A major geographical zone, with a length of about 40,000 kilometres where earth-quakes and volcanoes are common because of tectonic plate boundaries and move-ments. The zone extends nearly all the way around the edge of the Pacific Ocean.

RING ROT: 1. A fungus disease that affects sweet potatoes caused by *Rhizopus stolonifer* and marked by rings of dry rot. 2. A bacterial disease of potatoes caused by *Corynebacterium sepedonicum* and marked by brown discolouration of the annular vascular tissue.

RING SPECIES: A connected series of neighbouring species that can interbreed with relatively closely related species, but for which there exist at least two end species in the series that are too distantly related to interbreed. The populations which coexist in the same region and are genetically connected but do not inter breed create a "ring". Ring species consequently demonstrate how the evolution of differences among the different populations or subspecies leads eventually to the formation of new species.

RING OF SETS: In mathematics, it is a Boolean ring of sets that is closed under finite union and relative complementation.

RING-BARKING (GIRDLING): The complete removal of a strip of bark, which consist of secondary phloem tissue, cork cambium and cork from a tree in order to destroy it and as a means of thinning forests and by farmers to yield larger fruits.

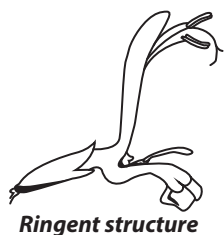
RING-BONE: A growth of bony tissue in the joint of a horse's foot.

RING-SECTOR: A sequence of ring atoms, linked together as an unbranched chain and connecting two bridgeheads or junctions, but not itself containing any bridgehead or junction, is termed an **atomic ring-sector**. A bond linking together directly two bridgeheads or junctions is termed a **bond ring-sector**.

RINGELMANN CHART: A chart which has been used in air pollution evaluation for assigning the degree of blackness of smoke emanating from a source. The observer compares the shades of grey (white to black) with a series of shade diagrams formed by horizontal and vertical black grid lines on a white background. A corresponding number, the Ringelmann number, is then assigned to describe the best match; numbers range from 0 (white) to 5 (black). This method of pollution evaluation, although relevant to the enforcement of legislation still in force in many countries, is not recommended for use today since more quantitative indicators of the efficiency of the combustion are now available.

RINGENT: Having lips or lip-like parts widely spreading, as seen in some corollas or the shells of certain bivalves.

RINGWORM (TINEA): A fungal disease of the superficial skin, scalp or nails caused by various kinds of fungi that belong to the genera *Trichophyton*, *Epidermophyton* and *Microsporum*. Infection is transmitted to humans from pets or livestock or from infected bedding. The affected area usually becomes inflamed and itchy and may result in an acute infection that produces sores on the scalp or painful blisters on the feet. On the feet, ringworm results into athlete's foot. Ringworm of the scalp occurs primarily in children causing itching and the hair may break off or fall out.



Ringent structure

RIP CURRENTS: Strong surface currents that move like small rivers perpendicular to the shoreline, caused by water returning from the beach to the ocean. Rip currents can move at about 8 kph or faster, which is strong enough to carry inexperienced swimmers to sea.

RIPARIAN (RIPARIOUS): Growing or situated along banks of rivers, streams, springs or seeps.

RIPE: 1. Fully grown and developed, matured such as fruit or grain that is ready for harvesting. 2. Used for cheese or beer, it is advanced to the point of being in the best condition for use.

RIPPER: 1. A programme used to copy digital music from a compact disk onto a computer before converting it into a format storable as a computer file. 2. A heavy cultivator consisting of a strong frame with long spikes attached to it. It is used to break up compacted soil to allow free passage of air and water.

RIPPING: In computer terminology, it is the process of copying data from a CD or DVD to a hard disk. This is done using a software utility, often referred to as a CD or DVD "ripper," which can extract audio or video data from optical media.

RIPPLE: 1. Also known as **capillary wave** and **capillary ripple**, it is a surface wave occurring at the interface between two fluids, such as the interface between air and water on oceans and lakes having so short a wavelength that the motion of the wave is dominated by surface tension. The maximum wavelength of a capillary wave is 1.73 centimetres. Longer waves are controlled by gravity and are appropriately termed **gravity waves**. Unlike the velocity of gravity waves, the velocity of capillary waves increases with decreasing wavelength, the minimum velocity being 23.1 centimetres per second, where the wavelength is the maximum 1.73 cm. 2. In electricity, it is the small unwanted residual periodic variation of the direct current (dc) output of a power supply which has been derived from an alternating current (ac) source, which arises in some classes of filter and other

signal processing networks. It is due to incomplete suppression of the alternating waveform within the power supply. 3. In geology, it is an area of shallow water in a river broken by rocks or sand bars.

RIPPLE TANK: A shallow water-filled glass tank used to demonstrate various properties of waves such as reflection, refraction and interference and diffraction.

RISE: In mathematics, the vertical distance between a pair of points on a graph or the difference between the values of the ordinates of a pair of points. The ratio of rise to run (rise/run) for a given pair of points gives the slope of the line segment joining the points.

RISER CARD (DAUGHTERBOARD; PIGGYBACK BOARD; DAUGHTER CARD; MEZZANINE BOARD; RISER BOARD): A circuit board that connects directly into the computer motherboard and provides the ability for additional expansion cards to be added to the computer. They were used with LPX motherboards and are now largely replaced by ATX motherboards that allow expansion cards to be connected directly to the computer.

RISK: 1. Probability of adverse effects caused under specified circumstances by an agent in an organism, a population or an ecological system. 2. Expected frequency of occurrence of a harmful event arising from such an exposure.

RISK ASSESSMENT: Identification and quantification of the risk resulting from a specific use or occurrence of an agent, taking into account possible harmful effects on individuals exposed to the agent in the amount and manner proposed and all the possible routes of exposure.

RISK ESTIMATION: Assessment with or without mathematical modelling, of the probability that organisms exposed to a specific dose of a chemical or other pollutant will develop an adverse response, for example, cancer.

RITUALISATION: A behaviour that occurs typically in a member of a given species in a highly stereotyped fashion, which is independent of any direct physiological significance, especially when these are incorporated into displays or the process in which different forms of behaviour are modified and combined to form a ritual.

RIVER: A natural stream of water that flows through a channel with more or less defined banks and empties into a body of water such as an ocean or lake. It originates at a point called the source such as a lake and is fed by overland runoff, groundwater seepage and meltwater released along the edges of snowfields and glaciers and enters a sea or lake at its mouth. They play a major role in shaping the surface features of the Earth. For example, River Nile originates from Lake Victoria in east central Africa, flows north over a distance of 5584 km and enters the Mediterranean Sea in Egypt. Economically, it has provided irrigation to otherwise arid regions for the cultivation of crops, provides fishing and transportation for the communities along the bank, has a dam built on it at Aswan providing hydroelectrical power, etc.

RIVER BLINDNESS (ONCHOCERCIASIS; ROBLES' DISEASE): A parasitic disease transmitted by the blackfly of the genus *Simulium*. It is characterised by lumps under the skin, gradual wrinkling of skin, intense skin irritation and inflammation of the eyes leading to blindness. Onchocerciasis is the world's second-leading infectious cause of blindness after Trachoma, a contagious disease of the conjunctiva and cornea, caused by the gram-negative bacterium *Chlamydia trachomatis* is the leading infectious cause of blindness.

RIVER CAPTURE (STREAM CAPTURE; STREAM PIRACY): The diversion of the head water of one river from its own bed that flows instead down the bed of a neighbouring stream. Some of the reasons of river capture are: (i) Rapid erosion caused by head-ward erosion of one stream valley upwards into another or lateral erosion of a meander through the higher ground dividing the adjacent streams or within an area of karst topography, where streams may sink or flow underground (a sinking or losing stream) and then reappear in a nearby stream valley; (ii) Tectonic earth movements, where the slope of the land changes and the stream is tipped out of its former course; (iii) Natural damming, such as by a landslide or ice sheet. Eventually the stream will cut into the course of a neighbouring river, causing the headwaters of that river to be diverted or captured.

RIVER CONTINUUM CONCEPT (RCC): The concept of a river as an ecosystem whose character changes continually between its source and mouth according to the nature of the energy inputs. It was first developed in 1980 by Robin L. Vannote, with fellow researchers at the Stroud Water Research Centre. The theory is based on the concept of dynamic equilibrium in which streamforms balance between physical parameters, such as width, depth, velocity and sediment load, also taking into account biological factors. The headwaters are likely to be narrow and fast flowing and shaded by trees and other vegetations so that virtually all the energy enters in the form of leaves, etc. from the surroundings.

RIVER TERRACE: Part of an old flooded plain that has been lifted and perched on the side of a river valley, forming a bench-like feature running along a valley side, roughly parallel with the valley walls. It results from rejuvenation or increases in the erosional capacity of a river so that it cuts down through its flood plain. River terraces are fertile and are often used for farming and are safe from flooding.

RIVETING: A method of mechanically joining metal plates by a metal pin called a **rivet**, with an oval or flat head called the **factory head** at one end with a pin-like body ending in a point called **shop head** or **buck-tail**. This end (shop head or buck-tail) is inserted into a punched or pre-drilled matching holes in two overlapping plates and then struck and formed into another head, holding the plates together.

RNA: Abbreviation for ribonucleic acid, a molecule occurring in certain viruses (RNA viruses) and in cellular organisms that directs the middle steps of protein production.

RNA INTERFERENCE (RNAi; TRANSCRIPTIONAL GENE SILENCING): A system within living cells that helps to control which genes are active and how active they are making it possible for double-stranded RNA to interfere with or suppress the expression of a gene with a corresponding base sequence. Two types of small RNA molecules called microRNA (miRNA) and small interfering RNA (siRNA) are central to RNA interference. The phenomenon is initiated by the enzyme Dicer, which cleaves long double-stranded RNA (dsRNA) molecules into short fragments of ~20 nucleotides. It occurs in many kinds of organisms including plants, fungi and animals.

RNA POLYMERASE (RNAP; RNAPol): An enzyme that attaches to the promoter region of the DNA template during transcription. It joins nucleotides to form the synthesised strand of RNA and detaches from the template when it reaches the terminator region. RNA polymerase enzymes are essential to life and are found in all organisms and many viruses.

RNA PROCESSING (POST TRANSCRIPTIONAL MODIFICATION): The modification of newly formed RNA transcripts to produce functional messenger RNA molecules, which occurs in the nucleus of eukaryotic cells. It involves removal of the encoding stretches from the primary transcript and splicing together of the discontinuous coding sequence. An example is the conversion of precursor messenger RNA into mature messenger (mRNA), which includes splicing and occurs prior to protein synthesis.

RNA SPLICING: The removal of non-coding portions (introns) of the RNA molecule after initial synthesis. It is a two step reaction in which introns are removed from a primary RNA transcript and exons are joined together to form a mature mRNA.

RNA TRANSCRIPT: The creation of an equivalent RNA copy of a sequence of DNA. Both RNA and DNA are nucleic acids, which use base pairs of nucleotides as a complementary language that can be converted back and forth from DNA to RNA in the presence of the

correct enzymes. During transcription, a DNA sequence is read by RNA polymerase, which produces a complementary, antiparallel RNA strand. In DNA replication, transcription results in an RNA complement that includes uracil (U) in all instances where thymine (T) would have occurred in a DNA complement.

RNASE (RIBONUCLEASE; RNAase): Any of a group of enzymes, widely distributed in nature that catalyses hydrolysis of the internucleotide phosphodiester bonds in ribonucleic acid (RNA) or an enzyme that splits or degrades RNA. Each RNase has specificity for a different cleavage site depending upon the specificity of the particular enzyme leading to the use of these various ribonucleases as tools in determining the structure and chemistry of RNA. For example, RNase A is a digestive enzyme secreted by the enzyme that hydrolyses phosphodiester bonds in the nucleotide chain. Other RNases are active at the cellular level, for example, in modifying transfer RNA and ribosomal RNA after transcription.

ROAN: The coat of an animal in which the main colour is mixed with another such as reddish brown, brown or black coat speckled with white or grey hairs.

ROASTING: 1. A cooking method that uses dry heat, on an open flame, oven or other heat source. 2. Heating a compound especially metal ore in air to convert it to an oxide.

ROBINSON PROJECTION: A world map projection showing the poles as lines rather than points that has accurate continent shapes and accurate land areas. It has parallel latitude lines and curved longitude lines.

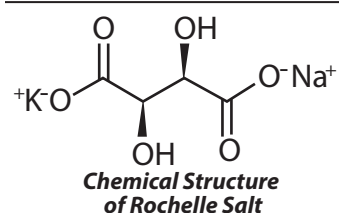
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ROBOT: A mechanical device that sometimes resembles, in whole or in part, a human being and is capable of carrying out a variety of often complex human tasks on command or by being programmed in advance. More generally, a robot or automaton, is a machine or device that operates automatically or by remote control. The simplest form of mobile robots, used to deliver mail in office buildings or to gather and deliver parts in manufacturing, follow the path of a buried cable or a painted line, stopping whenever their sensors detect an object or person in their path. Robots are used in industry to perform tasks that are either repetitive or in dangerous environments. As computers develop robots will increasingly assist humans in the manufacture of new products, the maintenance of the world's infrastructure and the care of homes and businesses.

ROBOT MILKER: A system in an automated milking parlour using lasers, mirrors and cameras to put all four caps on the teats simultaneously.

ROBOTICS: The study of the design, manufacture and operation of general purpose, programmable machine systems to perform mechanical tasks and to move by automatic control (robots).

ROCHELLE SALT (POTASSIUM SODIUM TARTRATE; SEIGNETTE'S SALT): A colourless crystalline double salt with the chemical formula $\text{KNaC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$. It is tartaric acid, in either the dextrorotary or racemic form. It is used for its piezoelectric properties and applied in the early electronic industry in gramophone pick-ups, microphones and earpieces. They are now used as a food



additive (E337), in photography, in tanning, as a laxative and in the process of silvering mirrors. It was first prepared in 1675 by a French pharmacist, Pierre Seignette.

ROCHON PRISM: An optical device consisting of two quartz prisms of a birefringent material such as calcite, which are cemented together, used to produce plane-polarised light and also used with ultraviolet radiation. One of the prisms is cut parallel to the optic axis and receives the light. The second prism with the optic axis at right angles transmits the ordinary ray without deviation but the extraordinary ray is deflected and can be absorbed by a screen. It was invented by Abbé Alexis Marie Rochon and named after him.

ROCK: A naturally occurring and coherent composed of mineral particles and/or organic materials consolidated into a hard mass. Rocks are found everywhere, in the Earth's crust or outer layer and at the bottom of the oceans, forming mountains. Types of rocks include igneous, sedimentary or metamorphic. They are further subdivided on the basis of various factors, especially chemical, mineralogical and textural attributes. Some common rocks include granite (a type of igneous rock) and basalt (black volcanic rock).

ROCK CRYSTAL: A colourless transparent, flawless variety of the silica mineral quartz. It has good optical properties and used in lenses and prisms, to control the oscillation of electrical circuits, etc. Examples are rose quartz, coloured rose red and amethyst, coloured purple or violet used as gems.

ROCK CYCLE: The system of the three main rock types (sedimentary, metamorphic, and igneous) changing from one into another in a never-ending series of processes as each type is altered or destroyed when it is forced out of its equilibrium conditions. For example, metamorphic rock is igneous or sedimentary rock that has been heated and squeezed.

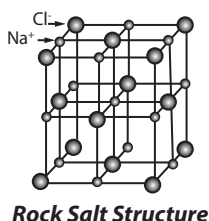
ROCK PHOSPHATE: A natural sedimentary rock composed largely of phosphate minerals, especially apatite. It is used as fertiliser and as a source of phosphorous compounds.

ROCK SALT (HALITE): A naturally occurring crystalline mineral consisting of sodium chloride (NaCl) with a hardness of 2.5 and a relative density of 2.16. It crystallises in the isometric system usually in the form of cubes and shows perfect cubic cleavage. It is colourless and transparent when pure but is often tinted yellow, red, blue or purple by impurities. It occurs in vast beds of sedimentary evaporite minerals that result from the drying up of enclosed lakes, playas and seas. It is an important raw material in the chemical industry and a major source of chlorine and table salt.

ROCK SALT STRUCTURE (SODIUM CHLORIDE STRUCTURE):

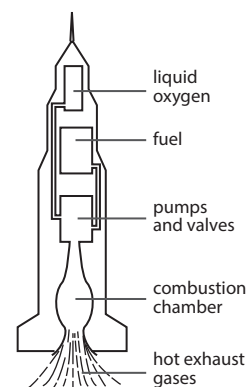
A type of ionic crystal structure in which the cations have a face-centred cubic arrangement, with anions occupying all the octahedral holes. Each ion is surrounded by six ions of the other kind; the surrounding ions are located at the vertices of a regular octahedron. It can equally be described as a face-centred cubic (fcc) lattice with a two atom basis or as two interpenetrating face centred cubic lattices.

Examples of compounds that have rock salt structure are sodium chloride (NaCl), potassium bromide (KBr), silver chloride (AgCl), silver bromide (AgBr), mercuric oxide (HgO), calcium oxide (CaO), ferrous oxide (FeO) and nickel monoxide (NiO).



ROCKET (ROCKET VEHICLE): A projectile such as a missile, spacecraft, aircraft or other vehicle containing its own fuel, as well as the oxygen or other chemical agent, needed to burn its fuel which depends on a stream of expanding hot gases. Rocket engines work by action and reaction and push rockets forward by throwing their exhaust backwards extremely fast. Rockets are very lightweight and powerful, capable of generating large accelerations and of attaining extremely high speeds with reasonable efficiency and not efficient for low speeds. Rockets are used for fireworks, weaponry, ejection seats, as launch vehicles for artificial satellites, human spaceflight and exploration of other planets.

ROCKET ENGINE: A reaction engine that obtains thrust based on the Newtonian principle that "to every action there is an equal and opposite reaction". A rocket engine can be a liquid-propellant or solid propellant engine, usually referred to as **rocket motor**. A rocket engine throws a mass of high-pressure gas in one direction and benefits from the reaction that occurs in the other direction as a result. The mass comes from the weight of the fuel that the rocket engine burns. In most liquid-propellant rocket engines, a fuel and an oxidiser such as petrol and liquid oxygen are pumped into a combustion chamber. There, they burn to create a high-pressure and high-velocity stream of hot gases, which flow through a nozzle and



are accelerated and ejected at high velocity through a nozzle, thereby imparting momentum to the engine. Examples of types of fuel combinations used in liquid propellant rocket engines include the following: Liquid hydrogen and liquid oxygen, petrol and liquid oxygen, kerosene and liquid and alcohol and liquid oxygen. The thrust is the force that propels a rocket or spacecraft and is measured in kilograms or Newtons. It is actually the resultant forces due to the pressures exerted on the inner and outer walls of the combustion chamber by the combustion gases and the surrounding atmosphere.

ROCKET PLANE: An aircraft that uses one or more rocket engines for propulsion, possibly in addition to jet engines. Rocket planes are able to reach much higher speeds than similarly-sized jet aircraft, but for much shorter periods of operation, typically only a few minutes.

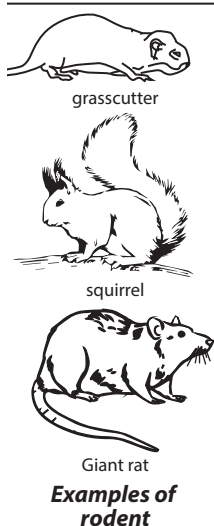
ROCKET PROPULSION: A type of reaction propulsion in which the propulsive force is generated by accelerating and discharging matter contained in the vehicle (as distinct from jet propulsion).

ROCKING-CHAIR CELL: A type of secondary cell in which layered electrodes usually made of metal oxides or graphite, store positive ions between the crystal layers of an electrode. In one type, lithium ions form an intercalation compound with a graphite electrode when the cell is charged. During discharge, the ions move through an electrolyte to the other electrode, made of manganese oxide, where they are more tightly bound. When the cell is being charged, the ions move back to their positions in the graphite.

ROD: A unit of length, now obsolete, which is equal to 5.03 m or a unit of area equal to 25.3 m².

ROD CELL: Any of various cylindrically-shaped photoreceptor cells in the retina of the eye that is not sensitive to colour and responds to dim light. Rods contain the pigment rhodopsin and are essential for vision in dim light. There are about 120 million rods in one retina of a human eye – about 20 times the number of cones. Rods, however, are not found at all in the fovea.

RODENT: Any mammal such as rat (*Rattus*), mouse (*Mus*), squirrel (*Sciuridae*), cane rat or grasscutter (*Thryonomys swinderianus*), giant rat (*Cricetomys*) and guinea pig (*Cavia*) of the order Rodentia forming the largest mammalian order and making up about half of all mammal species. They are characterised by front teeth adapted for gnawing and cheek teeth adapted for chewing and also a single front pair of incisor teeth in both their upper and lower jaw, which continue to grow into chisel shaped cutting edges as they are worn down. Some of them are found worldwide in almost every terrestrial and freshwater habitat, from the shores of the Arctic Ocean to the hottest deserts. Examples of rodents are grasscutter also known as cane rat, squirrel and giant rat (*Cricetomys*) are shown in the diagrams.



RODENT ULCER (BASAL CELL

CARCINOMA): A malignant tumour that spreads slowly usually occurring on areas of the skin that are often exposed to the Sun, such as the face and neck, at the edge of the eyelids, lips or nostrils. It is caused mainly by long-term exposure to ultraviolet (UV) radiation from the sunlight damaging the DNA in the skin cells.

RODENTIA: An order of mammals consisting of the rodents, characterised by a single pair of curved incisors in each jaw adapted for gnawing and cheek teeth adapted for chewing.

ROENTGEN EQUIVALENT MAN (REM): Named after Wilhelm Röntgen (1845-1923), German scientist, it is a measure of absorbed dose of radiation in biological tissue. Its symbol is **rem**, a unit of radiation dose equivalent. It is the product of the absorbed dose in rads and a weighting factor, W_R , which accounts for the effectiveness of the radiation to cause biological damage.

ROENTGEN-HOUR-METRE (RHM): The unit of effective strength of a radioactive source that produces gamma rays, a form of high-energy radiation emitted by some radioactive substances. A source of strength 1 rhm produces ionisation at the rate of 1 roentgen per hour at a distance of 1 metre from the source. It is used for substances for which it is difficult to establish their radioactive disintegration rates.

ROENTGEN, WILLIAM KONRAD (1845-1923): German physicist, who made many contributions to physics, the best known, being his discovery of x-rays in 1895, for which he was awarded the first Noble Prize for Physics in 1901.

ROENTGENIUM: A radioactive transactinide in Group 11 (IB) on the Periodic Table with the symbol Rg, atomic number 111 and mass number of most stable isotope 272. It was named after the German physicist, Wilhelm Conrad Röntgen (1845-1923). It has a large number of particles in the nucleus making it unstable. It is produced artificially by nuclear fusion in which a chemical element with larger atoms is produced by fusing together smaller atoms from other elements. Roentgenium is of research interest only.

ROGERS-RAMANUJAN IDENTITIES: In mathematics, they are two identities related to basic hypergeometric series.

ROGUE: 1. A plant of different variety found growing in a crop. 2. An organism, especially a plant, which shows an undesirable variation from a standard.

ROGUE VALUE (DATA TERMINATOR; FLAG VALUE; SENTINEL VALUE; SIGNAL VALUE; TRIP VALUE): In computing, it is a special value in the context of an algorithm that is used to mark the end of a list of input data items. The computer must be able to detect that

the data terminator is different from the input data in some way, for example, a negative number such as -1 might be used as the sentinel value as the computation will never encounter that value as a legitimate processing output; otherwise the presence of such values would prematurely signal the end of the data. A lot of old programs used data terminators and many programmers used 00 as a year data terminator, leading to problems in the year 2000.

ROLL: A tractor-drawn implement used for breaking clods, firming the soil, pushing stones into the soil and providing a smooth firm surface for drilling.

ROLLE'S THEOREM: In mathematics, the theorem that any real-valued differentiable function that attains equal values at two distinct points must have a point where the first derivative (the slope of the tangent line to the graph of the function) is zero.

ROLLED GRAIN: Grain rolled into flat plates by passing it through flat rollers before it is fed to livestock. Rolled grain, usually barley is more easily digested.

ROLLER MILL: A piece of equipment that uses cylindrical rollers to grind all sorts of materials such as flour and animal feed. It has two smooth steel rollers which crush the material. It is useful in industry in the crushing of various substances.

ROLLER TABLE: A machine, consisting of a horizontal line for rotating rollers. It is used for removing stones and clods from crops such as potatoes.

ROLLING: 1. Moving by revolving or turning over and over as a ball or a wheel. 2. Rising and falling in gentle slopes, as land. 3. Moving in undulating billows, as clouds or waves. 4. A common method of shaping metal by the compressive forces exerted by rotating rolls in a mangle, consisting of several sets or stands of heavy rollers positioned one above the other.

ROLLING FRICTION (ROLLING RESISTANCE; ROLLING DRAG): The friction or force that hinders the motion of an object such as a ball or tire rolling along a surface. As a result of any small distortion of the two surfaces, there is a frictional force with a component, $F(r)$ that opposes the motion. If N is the normal force, $F(r) = N\mu(r)$, where $\mu(r)$ is the coefficient of rolling friction. The coefficient of rolling friction is usually about 0.01 that of sliding friction. Wheels and other round objects will roll along the ground much more easily than they will slide along it.

ROMAN NUMERALS: A numeration system, created and originally used by the ancient Romans more than 2000 years ago, based on seven letters that can express all numbers from 1 to 1,000,000. The representations are: **I** for 1, **V** for 5, **X** for 10, **L** for 50, **C** for 100, **D** for 500 and **M** for 1000. Roman numerals are read from left to right. The symbols representing the largest quantities are placed at the left; immediately to the right of those are the symbols representing the next largest quantities and so on. The symbols are usually added together. For example, LX = 60 and MMCIII = 2103. When a numeral is smaller than the numeral to the right, however, the numeral on the left should be subtracted from the numeral on the right. For example, XIV = 14 and IX = 9. A small bar placed over a numeral multiplies the numeral by 1000. For example, a small horizontal bar placed over the letter M (which represents 1000) means that M multiplied by 1000 equals 1,000,000. In long documents, lowercase Roman numerals may be used to number the pages of an introduction, a table of contents and any other material that precedes the main text. Its disadvantage is that it is not suitable for rapid written calculations.

ROOST: A place where birds rest at night, such as a perch.

ROOSTER: A name used especially in the United States for cock or male domestic fowl.

ROOT: 1. In anatomy, it is the end of a nerve that is nearer to the centre of the body or the portion of a body part such as hair or a tooth that is embedded in tissue. 2. In botany, it is the absorbing and anchoring organ of a vascular plant that bears no leaves or buds and

usually grows down (positive geotropism) and spreads underground. It anchors the plant and absorbs water and nutrients from the soil. The term is also used for an underground plant part such as yam, cassava or carrot that is used as a vegetable. 3. In mathematics, a number that when multiplied by itself a particular number of times equals another number, for example, $3 \times 3 = 9$. Three (3) is therefore the square root of 9. Even roots are roots that can be extracted easily and yield two results, for example, $\sqrt{9} = \pm 3$ and $\sqrt{25} = \pm 5$. Odd roots are roots that when extracted gives only one answer. The sign of the root is the same as the number whose root is being found, for example, $\sqrt[3]{8} = 2$, $\sqrt[3]{27} = 3$ and $\sqrt[3]{-8} = -2$. Examples of mixed roots are: $\sqrt{12} = \sqrt{12} \times 3 = \sqrt{4} \times \sqrt{3} = \pm 2\sqrt{3}$.

4. A number that can take the place of the variable in an equation and help solve the equation. For example, the root of a polynomial $f(x) = 0$ is a value of x for which the value of the polynomial $f(x)$ is zero.

ROOT CAP: A cone-shaped structure that covers the root tip and develops as a result of cell division by meristem underneath it. Root caps contain statoliths which are involved in gravity perception in plants. It protects the root tip as it grows through the soil particles. The cells are constantly worn away and replaced by the meristem.

ROOT CROP: In agriculture, a crop grown for the thick, fleshy storage root that it produces. Examples are cassava, yam, etc.

ROOT CUTTING: A section of a root with adventitious buds cut and used for propagation purposes.

ROOT DIRECTORY: In computer science, it is the top-level directory of a tree-and-branch filing system that contains all the other directories and sub directories. All file systems have a root directory but are labelled differently. For example, in Windows, the root directory is labelled C:\.

ROOT HAIR: A tubular outgrowth from an epidermal cell of a plant root. Root hairs form the principal absorbing tissue of the plant of most of their water and nutrients.

ROOT HARVESTER: A machine for lifting underground crops such as sugar beet out of the ground.

ROOT MEAN SQUARE (RMS): 1. In mathematics, it is also known as **quadratic mean** and is a statistical measure of the magnitude of a varying quantity. The RMS value of a set of values or a continuous-time waveform is the square root of the arithmetic mean (average) of the squares of the original values or the square of the function that defines the continuous waveform. For example, a typical value of a number (n) of values of a quantity ($x_1, x_2, x_3, \dots, x_n$) is equal to the square root of the sum of the squares of the values divided by n and the RMS value $= \sqrt{(x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2)/n}$. 2. In electrical engineering, it is the effective value of a varying voltage or current. It is the equivalent steady DC (constant) value which gives the same effect. The root mean square voltage (V_{RMS}) is 0.7 of the peak voltage (V_{peak}). $V_{RMS} = 0.7 \times V_{peak}$ and

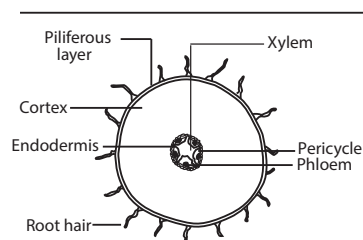


Fig. I: Transverse section of a dicotyledonous root

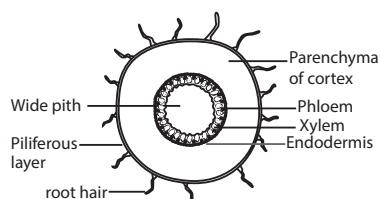


Fig. II: Transverse section of a Monocotyledonous root

$V_{peak} = 1.4 \times V_{RMS}$. These equations also apply to current. They are only true for sine waves because the 0.7 and 1.4 are different values for other shapes.

ROOT NODULE: A swelling on the root of certain plants especially those of the family Fabaceae (formerly leguminosae) including peanuts, beans and acacia. It is associated with symbiotic nitrogen-fixing bacteria known as rhizobia and is capable of fixing atmospheric nitrogen into ammonia which is subsequently converted to nitrate and amino acid.

ROOT OF UNITY (de MOIVRE NUMBERS): A solution of the equation $x^n = 1$, where n is a positive integer. They are located on the unit circle of the complex plane and in that plane they form the vertices of an n -sided regular polygon with one vertex on 1.

ROOT PRESSURE: The osmotic pressure within the cells of a root system that forces water absorbed from the soil to move through the roots and up the stem of a plant. Root pressure occurs in the xylem of some vascular plants due to both the osmosis of water from the soil into the root cells and the active pumping of salt into the xylem tissue which maintains a concentration gradient along which the water moves.

ROOT PRUNING: A systematic cutting back of the roots by hand or machine to prepare them for future transplanting, to encourage the roots to develop within the root ball range or to send out further feeder roots within a circumscribed area. In some cases it stimulates the flowering of barren fruit trees or ornamentals.

ROOTBOUND: Roots of a plant grown for too long in a container which have become restricted by the container. With no room for additional growth, roots become tangled, matted and grow in circles. Root-bound plants placed in the ground without having their roots untangled often fail to overcome their choked condition. This results in stunting the plants' growth and potential.

ROOTING COMPOUND (ROOTING HORMONE): A natural or synthetic compound containing plant hormones (auxin), occurring in either powder or liquid form, into which plant cuttings can be dipped before planting. This hastens the formation of roots and also increases the number and quality of the roots.

ROOTING DEPTHS: The depth of soil from which plant roots take up water or the depths of soil to which roots reach.

ROOTKIT: A software program designed to provide a user with administrator access to a computer without being detected. Rootkits have been created for Unix variants, Linux variants, Microsoft Windows and are also available for most other major operating systems. Rootkits are considered one of the most serious types of malware since they may be used to gain unauthorised access to remote systems and perform malicious operations.

ROOTLET: A little root which grows from a main root.

ROOTSTOCK: 1. Also known as **rhizome**, it is a horizontal underground stem having nodes, buds and tiny leaves or an organ of vegetative reproduction and food storage in flowering plants, consisting of a horizontal underground stem, growing from a parent plant. The tip of the rhizome is a bud from which grows a new plant. Most types of grasses and ginger have rhizomes. A rhizome stem helps the plant to survive from one growing season to the next and in some species it helps in vegetative propagation of the plants. If rhizomes are cut or broken into pieces they do not die, each piece may be able to give rise to a new plant, when replanted. **Rhizomatous** means having rhizomes. **Rhizomorphous** means having a root-like appearance. 2. A plant or stump, which has an already established and healthy root onto which a cutting or bud is grafted. The plant part which is grafted onto the rootstock is called the **scion**.

RORIDULATE: Having a covering of waxy platelets that appears moist.

ROSDAL (REPRESENTATION OF ORGANIC STRUCTURE DESCRIPTIONS ARRANGED LINEARLY): A type of line notation used in databases based on connection table for the molecule written in a single text line.

ROSE CURVE: Also known as **rose** in mathematics, it is a curve plotted in polar coordinates and shaped like a petalled flower with a common origin. Its polar equation is $r = a \sin n\theta$ or $r = a \cos n\theta$; if n is odd it has n petals, if n is even it has $2n$.

ROSE'S METAL (ROSE'S ALLOY; ROSE METAL): An alloy of low melting point about 100°C (273 K) consisting of 50% bismuth, 25-28% lead and 22-25% tin, which does not contract on cooling and used as a solder.

ROSEATE: 1. Having the colour of rose; tinged with red. 2. A term used to describe birds which possess partly pink plumage such as the Roseate tern and Roseate spoonbill.

ROSEMARY: A small perennial evergreen upright to trailing shrub (*Rosmarinus officinalis*) of the mint family whose leaves are used to flavour a wide variety of food and also used as a source of oil used in soaps and cosmetics. It flourishes in well-drained, alkaline soil. It prefers sunny condition and needs protection shelter from gusty winds. The plant reaches about 1.5-3 metres in height and has short linear leaves that resemble curved pine needles.

ROSETTE: In botany, it is a circular arrangement of dense radiating cluster of leaves or other organs, which usually sit near the soil. Its structure is an example of a modified stem.

ROSIN: A hard, brittle, colourless, yellow, brown or black natural resin, insoluble in water, but soluble in most organic solvents that softens at between 80-90°C (353 K-363 K). Rosin is obtained by making incisions on living trees and collecting the exudate (gum rosin), by extraction of pine stumps (wood rosin) and as a by-product from the kraft pulping process (sulfate or tall oil rosin). It is used as a flotation agent, solder flux, sizing compound, adhesives, printing inks and chewing gum and in lacquers and plasticisers.

ROSTELLATE: Having a tiny, short, stout, terminal beak.

ROSTELLUM: A small beak-like part of a plant or an animal that such as the hooked projection from the head of a tapeworm, a projection on the stigma of an orchid and a tubular mouthpart on some insects.

ROSTRAL SCALE: A scale present on the tip of the upper jaw of a snout, usually seen in snakes.

ROSTRUM: A set of mouthparts that look like a beak. Many insects have a rostrum and so do some vertebrates, such as beaked whales and dolphins.

ROSULATE: Having leaves or bracts arranged in basal rosette and either lacks a stem or possesses very short ones.

ROT (DECAY; DECOMPOSE): 1. The breakdown of organic matter from a complex to a simpler form, mainly through the action of fungi and bacteria. 2. Any of several plant diseases caused by various bacteria or fungi. It is characterised by the breakdown of tissue.

ROTAMER: One of a set of conformational isomers that differ from each other by restricted rotation about one or more single bonds.

ROTARY CONVERTER: A type of motor-generator, in which the two machines share a single rotating armature and a set of field coils. It is used as an inverter to convert direct current (DC) to alternating current or as a rectifier to convert alternating current (AC) to direct current (DC). They were used to provide DC power for commercial, industrial and railway electrification from an AC power source.

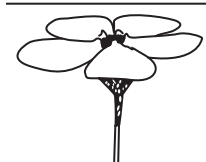
ROTARY CULTIVATOR (ROTARY TILLER; ROTOTILLER; ROTAVATOR; ROTARY HOE; POWER TILLER; ROTARY PLOUGH): A motorised cultivator mounted on or trailed by a machine with a shaft bearing and a number of L-shaped blades that works the soil by means of rotating tines or blades. They are used for stubble-clearing, seedbed work and general land reclamation and cleaning.

ROTARY MOWER: A machine used for cutting grass and other upright crops. Rotary mowers have a rapidly rotating bar or disks mounted on a bar, with sharpened edges that cut the crop. The rotors rotate in opposite directions and leave a single swath of cut grass. In rough cutting conditions the blades attached to the disks are swivelled to absorb blows from obstructions.

ROTARY PARLOUR: The most expensive and complex of the four milking systems, used to milk large herds, where the cow stands on a rotating platform to be milked and then step off, after a full rotation. The operator may work on the inner or the outer side of the circle.

ROTARY SPRINKLER: A machine used for irrigation purposes. It can be fitted with fine spray nozzle for protection of fruit crops and potatoes against frost damage.

ROTATE: 1. To cause to turn around an axis or centre point or to revolve. 2. To cause to go through a cycle of changes. 3. To rotate farm crops. 4. In botany, it refers to a disc-shaped object; or having flat, circular and radiating parts, as seen in sympetalous corolla with widely spreading lobes and little or no tube.



Rotate Corolla

ROTATING FLAILS: Parts used on manure spreaders to distribute materials on a machine for cutting crops or grass verges. It is also used in mixing machines such as composter.

ROTATING FRAME OF REFERENCE: In physics, it is a frame of reference without inertia that rotates about an axis relative to an inertial reference frame. An example is the surface of the Earth.

ROTATING TINES: Spikes used on machines such as rotavators and power harrows for cultivation purposes. It is also used on machine pick-ups.

ROTATION: 1. A mathematical transformation of an object in a circular motion around an imaginary line called an **axis** that does not change the object's shape. The object is turned about a given point, known as the **centre of rotation** at a fixed angle; effected by a change of variables of the form $x' = x \cos \theta + y \sin \theta$ and $y' = x \sin \theta + y \cos \theta$. If the axis of rotation is within the body, the rotation is called a spin. This implies relative speed and perhaps free-movement with angular momentum. A circular motion about an external point such as the Earth about the Sun is called an orbit or an orbital revolution. 2. In agriculture, crop rotation is the system of varying successive crops in a definite order on the same ground, especially to avoid depleting the soil and to control weeds, diseases and pests.

ROTATION DESIGN: A system of conserving soil nutrients in organic farming by planting different crops in different years.

ROTATION GROUP: A group formed by the set of all rotations about the origin of three-dimensional Euclidean space under the operation of composition. The rotation group associated with the angular momentum of a system is important in the theory of the rotational motion of molecules. The group representations of the rotation group are closely associated with the quantum theory of angular momentum.

ROTATION SYMMETRY: A plane figure has rotation symmetry point O if there are a number of positions the shape can take, when rotated through some positive angle less than one complete revolution and still look the same. For example, an equilateral triangle or any regular polygon has rotational symmetry about its centre.

ROTATIONAL DIFFUSION: A process by which the equilibrium statistical distribution of the overall orientation of molecules or particles is maintained or restored. Rotational diffusion may be compared to translational diffusion through which the equilibrium statistical distribution of position in space is maintained or restored.

ROTATIONAL GRAZING (OFF GRAZING): The division of pasturing range into several fields or paddocks and then moving the livestock around in an ordered sequence.

ROTATIONAL INERTIA (ANGULAR MASS; MASS MOMENT OF INERTIA; MOMENT OF INERTIA): A measure of an object's resistance to changes to its rotation, which is the sum of all the

products formed by multiplying the magnitude of each element of mass by the square of its distance from the axes. Its symbol is I . The moment of inertia of an object about a given axis describes how difficult it is to change its angular motion about that axis. If a body consists of particles of mass m , a perpendicular distance r from an axis of rotation, the moment of inertia I of the body about that axis of rotation is given by $I = \sum mr^2$. The basic equation is $T = I\alpha$, where T is the torque causing angular acceleration α about the specified axis. The moment of inertia of a body is an important quantity because it is the analogue for rotational motion of mass for linear motion.

ROTATIONAL MOTION: The rotation of a particle about a fixed point in space or the motion of a rigid body in such a way that all of its particles move in circles around an axis of rotation with a common angular velocity. Examples of rotational motion are: (i) the rotation of a satellite about a planet; (ii) the varying speed of rotation of the flywheel of a sewing machine; (iii) the motion of a pendulum; (iv) the fixed speed of rotation of the Earth around its axis. The laws relating to the rotation of a body about an axis are analogous to those describing linear motion. The angular displacement (θ) of a body is the angle in radians through which a point or line has been rotated in a specified sense about a specified axis. The speed of rotation or angular velocity (ω) is the rate of change of angular displacement with time (t) which remains constant in a uniform circular motion. In such a case, the angular displacement (θ) experienced by the particle or rotating body in a time t is $\theta = \omega t$, where ω is the constant angular velocity.

ROTATIONAL SPECTROSCOPY (MICROWAVE SPECTROSCOPY): The study in the electromagnetic radiation spectrum, especially in the microwave region, of absorption and emission of electromagnetic radiation by molecules associated with a corresponding change in the rotational quantum number of the molecule. Rotational spectroscopy gives information about inter-atomic distances. Electromagnetic radiation interactions with matter involve absorption or emission of electromagnetic radiation energy by means of transitions between quantised or discrete, levels of energy for electrons, bond vibrations, molecular rotations and electron and nuclear spins in atoms and molecules. The transitions between different rotational energy levels in molecules correspond to the microwave and far-infrared regions of the electromagnetic spectrum. It is only possible for there to be transitions between rotational energy levels in pure rotational spectra if the molecule has a permanent dipole moment.

ROTATIONAL SYMMETRY: The property of a plane figure that does not change its identity when rotated, but remains identical to some rotational transform of itself although its position may be altered. Its order of symmetry is the number of its transforms that are identical figures but are distinct in orientation, so that an equilateral triangle has rotational symmetry of order 3, since each of its three sides may be taken as the base. A circle has rotational symmetry and reflection symmetry.

ROTATOR PHASE TRANSITION: A transition from a fully-ordered crystal of globular or quasi-spherical molecules to a crystal that retains translational order but exhibits dynamic orientational disorder and is usually mechanically soft (plastic phase). For example, the transition of CBr_4 to a plastic phase which exists from 319 to 365 K.

ROTATORY POWER: The quantitative measure of optical activity.

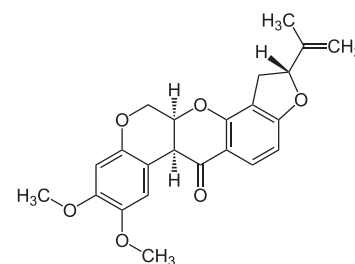
ROTAVIRUS: A wheel shaped double-stranded RNA virus in the family Reoviridae that causes diarrhoea in infants and young children, piglets, calves and foal. The virus is transmitted by the faecal-oral route. It infects and damages the cells that line the small intestine and causes gastroenteritis. There are five species of this virus, known as A, B, C, D and E. Rotavirus A is the most common and causes more than 90% of infections in humans.

ROTAXANE: A compound with two or more independent portions not bonded to each other but linked by a linear portion threaded through a ring and maintained in this position by bulky end groups. Rotaxanes are examples of compounds with mechanical bonding. A number of natural peptide rotaxanes such as cystine knot peptides, cyclotides

or lasso-peptides such as microcin J25 are protein and a variety of peptides with rotaxane substructure have been identified. Synthesis of new rotaxanes is a matter of interest because of their possible use as 'molecular machine' in nanotechnology such as molecular switches or information storage units.

ROTENOIDS: Naturally occurring substances containing a *cis*-fused tetrahydrochromeno[3,4-*b*]chromene nucleus. Many rotenoids contain an additional ring, for example, rotenone.

ROTENONE (TUBATOXIN; PARADERIL): The active ingredient of the insecticide derris, which is an odourless white crystalline compound with the chemical formula $\text{C}_{23}\text{H}_{22}\text{O}_6$, density 1.27 g/cm^3 at 20°C (293 K), melting point $165\text{--}166^\circ\text{C}$ (438-439 K) and boiling point $210\text{--}220^\circ\text{C}$ (483-493 K) at 0.5 mmHg. It is extracted from the roots of several plants such as derris and cubé and used as an insecticide and pesticide.



Chemical Structure of Rotenone

ROTH'S THEOREM (THUE-SIEGEL-ROTH THEOREM): The culminating theorem of rational approximation that for any algebraic irrational number α and for any $\kappa > 2$, there are only finitely many rational numbers p/q solving $|\alpha - p/q| < Cq^{-\kappa}$.

ROTIFERA: A phylum of microscopic pseudocoelomate, mainly free-living aquatic animals having characteristics such as a crown of ciliate at the anterior end and a mastax, which is a gizzard-like structure derived from the modified pharynx.

ROTOMETER: A device, based on Stokes law, for measuring the rate of fluid flow. It is a tapered vertical tube having a circular cross-section in which a float moves in a vertical path to a height dependent upon the rate of fluid flow upward through the tube.

ROTOR: 1. The rotating part of an electric generator, motor, turbine, etc, which rotates because the wires and magnetic field of the motor are arranged so that a torque is developed about the rotor's axis. The stationary part of an electric motor is called the **stator**. 2. A rotary part of a machine or vehicle, especially a hub with a number of radiating airfoils that is rotated in an approximately horizontal plane to provide the lift for a rotary-wing aircraft.

ROTUND: 1. Round or spherical. 2. Ripe, as for example, a ripe fruit. 3. Plump or fat.

ROUCHÉ'S THEOREM: In mathematics, it states that if the complex-valued functions f and g are holomorphic in and on some closed contour, with $|g|$ dominating $|f|$ on contour, then f and $f + g$ have the same number of zeros inside the contour, where each zero is counted as many times as its multiplicity.

This theorem assumes that the contour K is simple, that is, without self-intersections. It is named after Eugène Rouché (1832-1910), French mathematician.

ROUEN: A breed of common domestic ducks descended from and resembling the mallard. The drake or male duck has a green head and neck, rich claret-coloured breast and grey-black body.

ROUGH ENDOPLASMIC RETICULUM (GRANULAR ENDOPLASMIC RETICULUM; ROUGH E.R.): A type of the endoplasmic reticulum studded with ribosomes its surface giving it a rough appearance under the microscope.

ROUGH FELL: A hardy moorland breed of horned sheep originating in the UK and closely related to the Swaledale. It has a dark coloured face with irregular patterns.



Rotund leaf

ROUGH GRAZING: Underdeveloped grazing, common in mountain, heath and moorland areas.

ROUGH STALKED MEADOW GRASS (ROUGH BLUEGRASS): A type of perennial grass, *Poa trivialis*, of the Poa family common in lowland pasture on rich moist soils. It is highly palatable and useful for grazing on heavy and damp soil. When found in cereal crops it is treated as a weed.

ROUGHAGE (DIETARY FIBRE): A material of plant origin that cannot be digested or be broken down by enzymes in the small intestine of the digestive system. It is an important component of human balanced diet consisting mainly of cellulose and plant cell walls. Although indigestible by man it adds bulk to food and helps the alimentary canal to retain water and keep it moving by peristalsis and prevents constipation, minimises intestinal disorders and may serve as an aid in dieting. Foods high in fibre include legumes, green, leafy vegetables, whole fruits and unrefined foods such as bran and sprouted seeds.

ROUGING GLOVE: A glove impregnated with herbicides and used to destroy wild oats.

ROULETTE: 1. The curve traced by a fixed point on a closed convex curve as that curve rolls without slipping along a second curve. 2. A gambling game in which the players bet on which slot of a rotating disk a small ball will come to rest in.

ROUND: 1. Shaped like a circle, a cylinder or a sphere. In topology, the terms circle and sphere refer to topological objects and not geometric ones, so that the surface of an egg shape is a sphere. A round sphere is, topologically speaking, not a tautology, but a sphere with constant curvature (a sphere in the sense of geometry). 2. Also known as **rounding**, it is expressing by an integer or whole number, with no fraction or approximating a number to its nearest place value. It is used to make a number simpler but keeping its value close to its real value by either increasing or decreasing it to the next digit.

Round down is to approximate a number to a certain number of significant digits, a whole number, number of tens, hundreds, etc., by replacing the remaining digits by zeros. For example, 156 can be rounded down to 150 or 100. **Round up** is to approximate a number to a certain number of significant digits, a whole number, number of tens, hundreds, etc., by increasing the relevant digit by one and replacing the remainder by zeros. For example, 575 can be rounded up by a digit to 576, by 10 to 585 or by 24 to 600.

ROUND AND ROUND PLOUGHING: A ploughing method in which fields are ploughed from the centre to the outside or from the edge to the centre.

ROUND ANGLE (PERIGON): An angle of 360° .

ROUND BALER: A tractor drawn machine in which grass is rolled up inside the baler using rubber belts, fixed rollers or a combination of the two. The crop is passed over a system of belts from a round bale. When the bale is complete, twine is wrapped round it to hold its shape and thrown out of the machine.

ROUND ROBIN: In time division multiplex system, it is a method of allocating processor time to each program by letting the programs take turns in a rotational order.

ROUND WINDOW (FENESTRA ROTUNDA; FENESTRA OF COCHLEA): An opening covered by the secondary tympanic membrane between the middle ear and the cochlea, situated below the oval window. It allows fluid in the cochlea to move, which in turn ensures that hair cells of the basilar membrane will be stimulated resulting in hearing.

ROUND-OFF ERROR (ROUNDING ERROR): In mathematics, it is the difference between the precise value of some quantity and its exact mathematical value due to rounding up or down. **Round-off error** is the initial errors accumulated during a calculation as a combined effect of inherent round-off due to the fact that one works

on a fixed precision device and the propagated round-off due to the level of precision used and cancellation or other errors.

ROUNDWORM: A common name for any of a phylum of parasitic, unsegmented terrestrial, freshwater or marine worms (nematodes) including *Ancylostoma duodenale*, *Ascaris lumbricoides*, *Enterobius vermicularis* and *Strongyloides stercoralis* that infest the intestines of humans and some animals. Roundworms are cylindrical, tapering animals with simple bodies consisting of an interior gut and a muscular outer wall, separated by a fluid-filled cavity called a **pseudocoel**.

ROUTER: A hardware device that directs data from a local area network (LAN) to another network connection. It takes incoming packets, analysing the packets and then directing them to the appropriate locations, moving the packets to another network, converting the packets to be moved across a different network interface, dropping the packets or performing any other number of other actions.

ROUTING INFORMATION PROTOCOL (RIP): A protocol used by computers and other network devices such as routers to manage router information within a self-contained network such as a local area network (LAN) or an interconnected group of LANs. The algorithm used in RIP was first developed in 1969 and used to route information in ARPANET (Advanced Research Projects Agency Network, the network that became the basis for the Internet). If compatible, many networks have upgraded from RIP to OSPF or another more current routing protocol.

ROW: In mathematics, it is a horizontal linear array of numbers or terms, especially in a matrix. For example, in the 3×3 matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$
 shown in the diagram, the rows are [1 2 3], [4 5 6] and [7, 8, 9]; the columns are [1, 4, 7], [2, 5, 8] and [3, 6, 9].

ROW CROP: A crop planted in rows wide enough to allow cultivators between the rows.

ROW CROP ROTATION: A lightweight tractor used mostly by market gardeners and farmers who grow row crops. It has a narrow turning circle and adjustable wheel track widths.

ROW INTERCROPPING: The practice of cultivating different crops simultaneously in adjacent rows on the same field. It is one type of intercropping.

ROW EQUIVALENCE: In mathematics, two matrices are row equivalent if one can be obtained from the other by a finite sequence of elementary row operations, which include (i) row addition; (ii) row multiplication (multiplying all entries of a row by a non-zero constant); and (iii) row switching (interchanging two rows of a matrix).

ROW RANK: In mathematics, it is the maximum number of linearly independent row space in the matrix. Column rank is the maximum number of linearly independent column space in the matrix. The row rank, column rank and the rank of the matrix coincide.

ROW SPACE: In a matrix, it is the vector space spanned by the rows of a matrix. This space has dimension equal to the row rank of the matrix.

ROW SPACING: In agriculture, it is the lateral spacing between the centre of one seed tube outlet to the centre of the next outlet for crops that are planted in a row or drill.

ROW VECTOR: A $1 \times n$ matrix having only one row.

ROW-REDUCED ECHELON FORM (REDUCED ECHELON FORM): A matrix that is in a row echelon form (has undergone Gaussian elimination), with the following properties: (i) all zero rows are at the bottom of the matrix; (ii) the first entry of each non-zero row after

the first occurs to the right of the leading entry of the previous row; and (iii) the first non-zero element in each non-zero row is 1 and is the only non-zero element in its column.

ROYAL JELLY: A highly nutritious food substance secreted from the pharyngeal and mandibular glands of worker bees that serves as food for all young larvae and as the only food for larvae that will develop into queen bees.

ROYAL PALM (CUBAN ROYAL PALM; FLORIDA ROYAL PALM; *Roystonea regia*): A large and attractive tree planted throughout the tropics and subtropics as an ornamental tree, belonging to the family, Arecaceae. It is found in southern Florida, Mexico and some other parts of Central America and the Caribbean. It towers about 20-30 metres tall with a smooth trunk and rows of circular leaf scars. The stem diameter is about 47 centimetres. Its distinctive crownshaft and canopy of pinnately compound leaves are clearly visible. It produces white unisexual flowers with pinkish anthers. The seeds of royal palm are used to feed livestock. The leaves are used for thatching and the wood for construction. The roots are used for the treatment of diabetes.

ROYAL POINCIANA (FLAMBOYANT; *Delonix regia*): A many-branched, broad, spreading, flat-crowned deciduous tree in the family Fabaceae, subfamily Caesalpinioideae valued as an ornamental tree. It is native to the western forests of Madagascar, but has been introduced into tropical and sub-tropical regions worldwide. The tree is often broader than tall and grows to about 9-15 metres tall with a dense foliage serving as shade. The leaves are feather-like, bi-pinnately compound, 30.5-50.8 centimetres long with 11-25 pairs of primary leaflets, each divided into 10-25 pairs of secondary leaflets. It produces large, red-orange flowers at the ends of branches in corymbs, have four petals and a fifth petal called the standard petal. The flowers develop into leguminous fruits which are green and flaccid when unripe but turn dark-brown and woody when mature.

RRKM THEORY (RICE-RAMSPERGER-KASSEL-MARCUS THEORY): A theory of unimolecular chemical reactions that enables the computation of simple estimates of the unimolecular reaction rates from a few characteristics of the potential energy surface. The theory improved the Rice-Ramsperger-Kassel theory by the US chemists, O. K. Rice and R. A. Marcus in 1905. The RRK theory was generalised into the RRKM theory in 1952 by R. A. Marcus who took the transition state theory of the individual vibrational frequencies, rotations and zero-point energies of the energised species and activated complexes developed by Eyring in 1935 into account. The RRKM theory is very successful in explaining the results of experiments for unimolecular reactions for a variety of reactions.

RS-232 INTERFACE (RECOMMENDED STANDARD-232): A standard type of computer interface approved by the Electronic Industries Alliance (EIA) used to connect computers to serial devices. It connects to modems, mice, screens and serial printers through an EIA-232 port. The device that connects to the interface is called **Data Communications Equipment** (DCE) and the device to which it connects such as a computer is called **Data Terminal Equipment** (DTE).

RUBBER: 1. Natural rubber is natural polymer or hydrocarbon, which is a strong elastic material made by drying the sap obtained from the milky secretion (latex) of various plants, especially the tree *Hevea brasiliensis*. It is an elastic and good insulator. Its many uses include, as tyres and gloves to water proofing. 2. **Synthetic rubber** is a synthetic elastomer compound with a structure resembling that of natural rubber in essential chemical and physical properties consisting of long-chain polymer molecules. It is built of monomers such as

isoprene, which are compounds of relatively low molecular weight that form the building units of huge molecules called polymers. The monomers are mixed in various proportions to be co-polymerised for a wide range of physical, mechanical and chemical properties. Examples of chemicals used in the manufacture of synthetic rubber are 1,3-butadiene, chloroprene, isobutylene. An example of synthetic rubber is butyl rubber.

RUBELLA (GERMAN MEASLES): A mild viral disease, usually contracted in childhood causing skin rash, lymph node enlargement (swollen glands) and, especially in adults it causes joint pain. It is caused by the rubella virus (Rubivirus). It is spread through contact with fluid droplets expelled from the nose or throat of an infected person.

RUBESCENT (RUFESCENT): To become red or reddish.

RUBIDIUM: A soft reactive silvery-white metal in Group 1 (IA) of the periodic table (the alkali metals), that ignites spontaneously in air and reacts violently with water. Its chemical symbol is Rb, melting point 39 °C (312 K), boiling point 688 °C (961 K), relative density 1.53, atomic number 37 and atomic weight 85.468. It is the most active of the alkali metals after caesium. It is derived mostly from lepidolite and carnallite. It is also found in small quantities in tea, coffee, tobacco and other plants. Trace quantities of rubidium may be required by living organisms. Rubidium is used in making certain catalysts and in photoelectric cells.

RUBIDIUM-STRONTIUM DATING: A method of dating geological specimens such as rocks, minerals and meteorites, based on the decay of the unstable radioisotope rubidium-87 that was present in the rock at the time of its formation into the stable isotope strontium-87. The method is also frequently used to check potassium-argon dates, because the strontium daughter element is not diffused by mild heating, as is argon. Natural rubidium contains 27.85% of rubidium-87, which has a half life of 4.7×10^{11} years.

RUBIGINOSE: Rust coloured or reddish brown.

RUBUS: A genus of flowering plants in the rose family, Rosaceae, subfamily Rosoideae, which includes cane fruits such as raspberries and blackberries. Most plants in this genus are characterised by woody stems with prickles like roses, spines, bristles and gland-tipped hairs.

RUBY: Natural pink to red impure form of the mineral corundum (aluminium oxide, Al_2O_3) with Mohs scale hardness of 9, relative density 4 and a melting point of 2044 °C (2373 K). The red colour with the chemical formula of $\text{Al}_2\text{O}_3:\text{Cr}$ (aluminium oxide with chromium) is caused mainly by the presence of the element chromium. Varieties of gem-quality corundum are called **sapphires**. Imperfect rubies are used as bearings in watches, etc. Synthetic rubies are used as gems or in industry.

RUDACEOUS: A term used to describe a coarse-grained sedimentary rock. Examples of rudaceous rocks include breccia, conglomerate and gravel, as well as types of stony boulder clay. The particles in the rock range in size from large grains of sand to fragments of gravel.

RUDDLE: A red colouring material on a harness temporarily worn by rams so that ewes which have been mated will be marked and identified.

RUDERAL: A plant species first to colonise a land with a high level of disturbance. The disturbance may be natural such as wildfires or avalanches; or due to human influence such as road construction, building construction or mining; or agricultural such as abandoned farming fields or abandoned irrigation ditches. They reproduce rapidly but are poor competitors against more robust types.

RUDIMENTARY: 1. Elementary or basic. 2. Incompletely or imperfectly developed; vestigial.

RUFFINI'S CAPSULE (RUFFINI'S CORPUSCLE): A flattened capsule containing specialised nerve endings, found in skin and in joint and ligament. It responds to pressure and stretching, providing information about certain types of touch stimuli and movements of joints and limbs. They are also thought to be heat receptors sensitive to temperature increases from 25-45 °C (298-318 K).

RUFIOUS (RUFUS): Reddish-brown.

RUGA (RUGAE): A fold or wrinkle, as in the lining of the stomach.

RUGULOSE: Slightly wrinkled; having many small wrinkles.

RULE: 1. In mathematics, a widely accepted procedure for solving a problem, for example, the transformation rules. 2. Another name for ruler, a straight-edge, used for drawing lines or measuring linear distance. 3. Any linear scale, such as used on maps to show the relationship between a particular distance on a map and the actual distance on the Earth.

RULE OF DETACHMENT (MODUS PONENDO PONENS; MODUS PONENS): In logic, the principle or rule of inference that whenever a conditional statement and its antecedent are both given to be true, then its consequent may be validly inferred. In other words, a single conditional statement is made and a hypothesis (P) is stated. The conclusion (Q) is then deduced from the statement and the hypothesis. The most basic form is listed below:

$P \rightarrow Q$ (conditional statement)

P (hypothesis stated)

Q (conclusion deduced)

For example, "If an angle satisfies $90^\circ < A < 180^\circ$, then A is an obtuse angle." A = 120° . A is an obtuse angle.

RULE OF FALSE POSITION (FALSE POSITION; REGULA FALSI): A method of solving equations by using a trial and error values for the variables and then adjusting the values accordingly. For example, to find a solution of $x^2 + y^2 = 50$, test values like $x = 5$, $y = 6$ might be attempted, which gives $x^2 + y^2 = 25 + 36 = 61$, which is more than 50; therefore smaller values for the variables would have to be guessed until the equation is solved. There are two types of false position methods: (i) **simple false position** for solving problems involving direct proportion that can be written algebraically in the form $ax = b$, for the value of x to be found if the values of a and b are known. (ii) **Double false position** is used for solving more difficult problems that can be written algebraically in the form: $f(x) = b$ for the value of x to be determined, if it is known that $f(x_1) = b_1$ and $f(x_2) = b_2$.

RULE OF INFERENCE: In logic, a function which takes premises, analyses their syntax and returns a conclusion. In other words, it is drawing a correct conclusion from one or more premises. For example, the rule of inference called modus ponens takes two premises, "If $P \Rightarrow Q$ " and "P" are true and the conclusion "Q" is true.

RULE OF THREE: In mathematics, the rule that the product of the extremes of a proportion is equal to the product of its means, from which an unknown quantity to be found. Alternatively, it is a form of cross-multiplication that the value of one unknown quantity in a proportion can be found by multiplying the denominator of each ratio by the numerator of the other. For example, if the values of three numbers a , b and c , are known and the value of a third number, d is to be found such that $\frac{a}{b} = \frac{c}{d}$, written proportionally as $a:b::c:d$, then

$d = \frac{cb}{a}$. Algebraically, the equation (proportion) can be multiplied by bd , which gives $ad = bc$.

RULED SURFACE: A surface that is built up from an infinite number of perfectly straight lines. A cylinder, for example, is a ruled surface of parallel straight lines. A cone is a ruled surface of straight lines that meet at the apex of the cone.

RULER (RULE): A long narrow flat piece of plastic, metal or wood with straight edges marked in centimetres or inches or a combination of the two with their fractions such as millimetre. It is used for measuring things and for drawing straight lines.

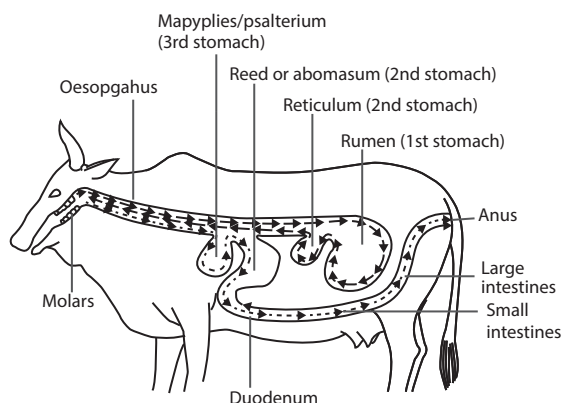
RULES OF GOOD HUSBANDRY (AGRICULTURE ACT 1947): A set of rules in agriculture passed in the United Kingdom, which if they are deemed to have been broken by a tenant, can give a landlord the excuse to evict him.

RUMEN (PAUNCH): The larger part of the reticulorumen, which is the first chamber in the alimentary canal of ruminant animals, where microorganisms break down plant cellulose before the food is returned to the mouth as cud for additional chewing.

RUMEN LIQUOR: Concentrated liquid found in the rumen of an animal. It is used to test the digestibility of feed or the nutrient balance of an animal's diet.

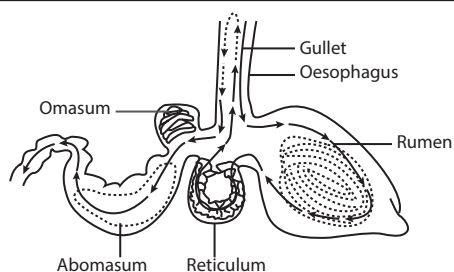
RUMFORD, COUNT BENJAMIN THOMPSON (1753 -1814): American-born British physicist, who acted as a British spy during the American Revolution, forced to flee in 1775, first to England and then to Munich, where he was made a count of the Holy Roman Empire. Whilst there, he observed the boring of cannon barrels, which led him to his best known proposition that friction produces heat and proposed that heat should be considered a form of motion. Rumford applied his understanding of heat to improving fireplaces so that they better reflected heat and carried away smoke. He invented the kitchen stove, drip coffeepot, pressure cooker and thermal underwear.

RUMINANT: Any of various cud-chewing hoofed mammals of the suborder Ruminantia of the order Artiodactyla, with even number of toes and a stomach with multiple chambers, which include rumen, also known as paunch (the first stomach), reticulum (commonly known as tripe), omasum and abomasum (the true stomach). The rumen and reticulum are known as the **reticulorumen**. **Rumination** is the process of digesting food, practised by ruminants. They chew roughly and swallow the food, which is stored in the first compartment, the rumen (the first stomach). During resting time the semi-digested food known as cud or digesta is regurgitated by reverse contraction of the oesophagus into the mouth to be re-chewed. This process continues until it is ready to be sent to the true stomach (the abomasum). From the rumen, it goes to the reticulum, from there it goes to the abomasum (the true stomach), where digesta is digested here and is finally moved into the small intestine, where the digestion and absorption of nutrients occurs, thus extracting the maximum possible food value. Examples of rumens are cattle, antelopes, goats, deer, etc. The digestive system of ruminants and the digestive tract of cattle are shown in the following diagrams:



The feed (grass) is ingested into the rumen. The partly digested food (cud) returns to the mouth from the rumen during rest period to be chewed again and again until it is ready to be sent to the true stomach (the abomasum). From the rumen it goes to the reticulum, then to the abomasum, and to the small intestine.

The digestive system of ruminants



The digestive tract of cattle (ruminant)

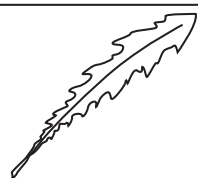
RUMINATE: 1. To chew food that is regurgitated from the first stomach to the mouth of ruminants such as cattle, antelopes, goats and deer and chewed again. Such a food is known as **digesta** or **cud**. 2. A term used to describe endosperm that is irregularly grooved or ridged and so appears chewed as, for example, seen in many members of the Myristicaceae.

RUN: 1. In mathematics, it is how far a line goes along (for a given distance going up). In mathematics, the vertical distance between an upper point and lower point on a graph or the difference between the values of the abscissas of a pair of points. The ratio of rise to run (rise/run) for a given pair of points gives the slope of the line segment joining the points. 2. Of a sequence of data from a distribution, to be divisible into successive strings of observations with a common property. For example, the sequence 12, 16, 8, 14, 17, 11, 3, 5, 9, 18, 10, 8 runs with respect to parity.

RUNCINATE: Having tooth-like projections or sharp pinnatifids such that the segments point away from the apex downward. Runcinate leaves, for example, are seen in the dandelion.

RUNGE-KUTTA METHOD: A numerical approximation technique for solving ordinary differential equations, used in computer analysis. The technique is named after the German mathematicians, Carl Runge (1856-1927) and Martin Kutta (1867-1944).

RUNNER: 1. Horizontal connections between organisms that may be part of the organism or of its skeleton. 2. An aerial stem that grows



Runcinate leaf

horizontally along the soil surface or below ground and gives rise to new plants either from axillary or terminal buds. It is a means of vegetative reproduction or propagation.

RUNNER BEAN (SCARLET RUNNER BEAN): A perennial vine, *Phaseolus coccineus*, belonging to the subfamily Papilionoideae of the family Fabaceae with tuberous roots and edible bean. The vine is characterised by cotyledons that stay in the ground during germination.

RUNOFF: 1. Rainwater or melted snow from the surface of land that is not absorbed by the soil and flows into streams and rivers and finally into oceans or landlocked bodies of water. 2. The flow of excess fertiliser, garden soil or pesticide from farmland into rivers or other waterways.

RUNOFF RATE: 1. The amount of excess fertiliser or pesticide from farmland that flow into rivers in a specific period. 2. Also called **runoff intensity**, it is the excess of rainfall intensity over infiltration capacity, usually expressed in centimetres of rainfall per hour.

RUNT: An individual animal born in multiple births that is smaller and weaker than average for its kind. A runt in a litter faces obvious disadvantages, including difficulties in competing with its siblings for survival and possible rejection from its mother.

RUNTIME: In computing, it refers to the period of time when a program is running. It begins when a program is opened or executed and ends with the program is quit or closed.

RUNTIME ERROR: A program error that occurs while the program is running. The term is often used in contrast to other types of program errors, such as syntax errors and compile time errors.

RUNTIME VERSION: In computer science, it is a copy of a program that is provided with another application, so that the latter can be run, although it does not provide the full functionality of the program. An example is the provision of run-time versions of Microsoft Windows with Windows applications for those users who do not have the full version of Windows.

RUPTILE: Dehiscing or splitting irregularly.

RUSH: A common weed growing near water, in moors and marshes.

RUSHLIKE: Plants that resemble grasses and possess inconspicuous flowers.

RUSSELL-SAUNDERS COUPLING (L-S COUPLING): A type of coupling in systems involving many fermions (electrons, protons and neutrons), in which the energies associated with electrostatic repulsion are much greater than the energies associated with spin-orbit coupling. Russell-Saunders coupling is useful mainly for the lighter atoms with atomic numbers less than 57. For heavier atoms, j-j coupling (a coupling in which the energies of electrostatic repulsion and spin-orbit coupling are similar in size) gives a better description.

RUSSET: A type of dessert apple with a rough brown skin.

RUSSETING: The formation of brown patches on the skin of an apple, in response to various chemicals. Russetting of leaves, tubers and other plant organs may also occur in response to damage or infection.

RUST: 1. The reddish brown powdery hydrated iron oxides ($\text{Fe}_2\text{O}_3 \cdot \text{XH}_2\text{O}$), which are products of corrosion of iron or steel which has been exposed to water or moisture in the air and oxygen or other strong oxidants or acids. In the presence of salt or salt water, for example, in ocean going vessels, iron or steel tends to rust more quickly, as a result of the electro-chemical reactions. 2. A disease of tropical fish in aquariums caused by the protozoa *Oodinium limneticum*, characterised by loss of lustre of the skin surface. 3. Rusts is a group of parasitic fungi of the order Uredinales, commonly attacking the leaves and

stems of cereal crops, forming reddish patches of spores on the host plant. It is easily detected by the reddish-brown spots they produce on the plants.

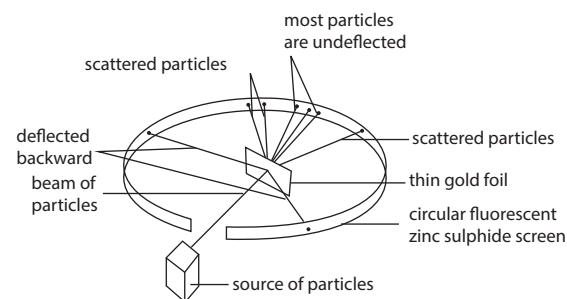
RUT: A period of intensive sexual activity that occurs in the males of several ruminant animals such as cattle, antelope, goats, deer and sheep.

RUTHENIUM: A hard silvery-white metallic element in Group 8 (VIII B) of the Periodic Table (the transition element) with the chemical symbol Ru, melting point 2,310°C (2583 K), boiling point 3,900°C (4173 K), relative density 12.41, atomic number 44 and atomic weight 101.07. It is found in platinum ores and is used to harden platinum and palladium for jewellery and in alloys for nonmagnetic wear-resistant instrument pivots and electrical contacts.

RUTHENIUM RED: A temporary stain that turns mucilage and certain gums pink. It is used as a test for pectin in the middle lamella of plant cells.

RUTHERFORD AND ROYDS' APPARATUS: An apparatus designed by Ernest Rutherford (1871-1937), British physicist and Thomas Royds (1884 - 1955) in 1909 to confirm that an alpha particle was a helium-4 nucleus. Rutherford and Royds's apparatus was designed to allow alpha particles, which were products of a radon-gas (the radioactive decay of radium), to pass a thin (1/100 mm thick) glass capillary tube filled with radium into an evacuated larger glass capillary tube in which the thin glass capillary tube was enclosed. Alpha particles trapped in the outer larger glass capillary tube were then able to acquire electrons either by ionisation of some traces of air or by collisions with glass. The apparatus was left for about a week before the gas trapped in outer tube was analysed. Electrical discharge through the gas produced the characteristic spectrum of helium.

RUTHERFORD SCATTERING (RUTHERFORD ALPHA PARTICLE SCATTERING): The elastic scattering of charged particles by the Coulomb interaction exerted by the atomic nuclei of the sample. It was explained by Ernest Rutherford in 1911 and led to the development of the planetary Rutherford model of the atom and eventually the Bohr model. The first experiment was carried out by Ernest Rutherford's assistants Hans Geiger and Ernest Marsden in 1909 when they bombarded a thin gold foil of approximately 8.6×10^{-6} cm thick having a circular fluorescent zinc sulfide screen around it with a beam of high energy alpha particles from a radioactive source. In the experiment, the following were observed: (i) most of the fast, highly charged alpha particles passed straight through the foil; (ii) some of the alpha particles are deflected through small angles; and (iii) a very small number of the alpha particles rebound off the gold foil. This experiment led to the discovery of the proper structure of an atom. The classical Rutherford scattering process of alpha particles against gold nuclei is an example of "elastic scattering" because neither the alpha particles nor the gold nuclei are internally excited. Rutherford scattering is also sometimes referred to as Coulomb scattering because it relies only upon static electric particles (Coulomb) forces and the minimal distance between particles is set only by this potential. **Rutherford backscattering** is any technique using high energy particles directed toward a sample, in which the bombarding particles are detected and recorded as function of energy and/or angle. The technique is mostly used for determining depth distributions of elements based on the energy of the backscattered particle. In general, He^+ or H^+ particles are used at energies in the order of 100 keV to some MeV.



Rutherford Scattering Experiment

RUTHERFORD, ERNEST (1871-1937): A New Zealand-born British physicist, who became known as the father of nuclear physics. After studies at Canterbury College, he moved to Britain to attend Cambridge University, where he worked with J.J. Thomson at the Cavendish Laboratory, where he became the head in 1919. He researched into the transmutation of elements and demonstrated that radioactivity was the spontaneous disintegration of atoms and discovered and named two types of radioactivity: alpha decay and beta decay. He later identified the alpha particle as a helium atom and used it in postulating the existence of the atomic nucleus and pioneered the Rutherford model of the atom, through his discovery of the Rutherford scattering with his gold foil experiment. He was awarded the Nobel Prize in Chemistry in 1908. Element 104, rutherfordium, is named in his honour.

RUTHERFORDIUM (UNNILQUADIUM): An artificially produced radioactive metal with the symbol Rf and atomic number 104. It is made by bombarding an actinide with carbon, oxygen or neon atoms. Its most long-lived isotopes have mass numbers of 253, 255, 257 and 259 with half-lives of 1.8, 1.6, 4.7 and 3.4 seconds, respectively. It was named after Ernest Rutherford (1871-1937), New Zealand-born British physicist, who became known as the father of nuclear physics.

RUTILE: A natural form of titanium(IV) oxide, which is a dark reddish brown titanium dioxide with the chemical formula TiO_2 , relative density 4.2, melting point 1825°C (2098 K) and hardness of 6.5 on Mohs scale. It forms needle-shaped crystals and obtained from igneous and metamorphic rocks and the main source of titanium. It is used as a gem and in porcelain and glass manufacture as a colouring agent and in making some steels and copper alloys.

RYDBERG CONSTANT: The most accurately measured of the fundamental constants, relating to the wave number of atomic spectrum lines, named after Swedish physicist, Johannes Rydberg (1854-1919). Its symbol is R_∞ with a value of $1.097373157 \times 10^7 \text{ m}^{-1}$ (1.097373157×10^7 per metre) is based on the premise that the nucleus of the atom emitting light is exceedingly massive compared with a single orbiting electron (hence the infinity symbol ∞). It serves as a universal scaling factor for any spectroscopic transition and as an important cornerstone in the determination of other constants.

RYDBERG FORMULA: A formula named after Swedish physicist, Johannes Rydberg (1854-1919). It is used in atomic physics to describe the wavelengths of spectral lines of many chemical elements. The value of its constant, named the Rydberg constant with the symbol R_∞ is $1.097373157 \times 10^7 \text{ m}^{-1}$ (1.097373157×10^7 per metre). His findings were combined with Bohr's model of the atom to give the formula: $1/\lambda = RZ^2(1/n_1^2 - 1/n_2^2)$, where λ is the wavelength of the photon (wavenumber = $1/\text{wavelength}$), R is the Rydberg's constant, Z is the atomic number of the atom, n_1 and n_2 are integers where $n_2 > n_1$. This formula works very well for transitions between

energy levels of a hydrogen atom with only one electron. For atoms with multiple electrons, this formula begins to break down and give results that are incorrect.

RYDBERG ORBITAL: For an atom, an orbital with principal quantum number greater than that of any occupied orbital of the ground state. For a molecular entity, a molecular orbital which correlates with a Rydberg atomic orbital in an atomic fragment produced by dissociation. Typically, the extension of the Rydberg orbital is large compared to the size of the atom or molecular entity.

RYDBERG SPECTRUM: The ultraviolet absorption spectrum of a gas in the ultraviolet region, which is made up of a series of lines that become closer towards shorter wavelengths and eventually merges into a continuous absorption region produced by transitions of atoms of a given element from the ground state to states in which a single electron occupies an orbital farther from the nucleus.

RYDBERG STATE: An electronic state that arises by the excitation of a valence electron to a diffuse atomic-like orbital. For example, the

lowest lying Rydberg state ($n, 3S$) of NH_3 can be described as resulting from the excitation of a non-bonding orbital to a $3S$ Rydberg orbital.

RYDBERG TRANSITION: An electronic transition described approximately as promotion of an electron from a 'bonding' orbital to a Rydberg orbital.

RYE: A tall hardy annual cereal crop, *Secale cereale* and its edible grain, with bluish green leaves grown in temperate areas. It is high in carbohydrates and provides small quantities of protein, potassium and B vitamins. It is used to make flour, whisky and as fodder.

RYE GRASS: A short-lived perennial, leafy, tufted plant, native to Europe and belonging to the family Gramineae (grass family) of the species *Lolium perenne* and *Lolium multiflorum* and classified in the division Magnoliophyta, class Liliatae order Cyperales and cultivated as forage, as cover crop and for lawns.

RYELAND: A rare breed of sheep and one of the oldest of British sheep breeds, raised primarily for meat. It is a medium-sized animal, white faced and without horns with very symmetrical shape and a thick growth of wool.

ADVERT SPACE

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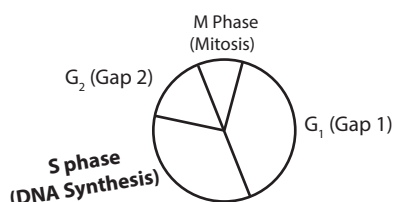
+233 (0)24 891 9104



S: 1. Symbol for second (abbreviated s or sec), the base unit of time in the International System of Units (SI). 2. The symbol for siemens, the SI unit of electrical conductance, which is equal to a conductance of 1 ohm^{-1} . 3. Abbreviation for South.

S PHASE (SYNTHESIS PHASE): The period in the cell cycle during interphase, between G₁ phase and G₂ phase when the chromosome (DNA) is replicated.

During S phase, the centrosome is also duplicated. These two events are unconnected, but require many of the same factors to progress. The end result is the existence of duplicated genetic material in the cell, which will eventually be divided into two.



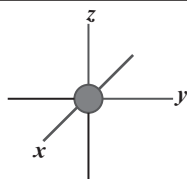
Cell cycle showing S phase

S-BLOCK ELEMENTS: Metallic elements that form Groups 1 (IA) and 2 (IIA) of the Periodic Table. They include the alkali and alkaline earth metals. All the s-block elements are metals except hydrogen and helium and characterised by high reactivity. They are powerful reducing agents. They are called s-block elements because in the atomic ground state the highest energy electron or valence electrons are in the s-orbitals. Except in hydrogen and helium, the s-electrons are very easily lost to form positive ions.

S-LCD (SUPER LIQUID CRYSTAL DISPLAY): A type of LCD display technology for mobile device displays, which unlike regular LCD does not have an air gap between the outer glass and the display element. This produces less glare giving it a closer and better display even when outdoors. It has lower power consumption.

S-MATRIX THEORY (SCATTERING-MATRIX THEORY): A theory initiated to describe strong interactions in terms of their scattering properties. S-matrix uses general properties such as causality, unitarity and analyticity quantum mechanics and the special theory of relativity. The theory was initiated by Werner Heisenberg, a German physicist in 1943.

s-ORBITAL: The orbital shape, used to describe electron configurations of atomic orbital. It is a spherically symmetrical atomic orbital with angular momentum quantum number $l = 0$. The "s" stands for "sharp", which describes the characteristics of the spectroscopic lines. Others are p-orbital (p_rincipal), d-orbital (d_iffuse) and f-orbital (f_undamental) with angular momentum quantum number $l = 1, 2$ and 3 respectively.



S-WAVE (SECONDARY WAVE; SHEAR WAVE): A seismic body wave that travels through the interior of the earth and is usually the second conspicuous wave to reach a seismograph. It shakes the ground back and forth perpendicular to the direction the wave is moving. S-waves move slower than the corresponding P-wave and cannot move through liquids at all. P-wave is a longitudinal earthquake wave that travels through the interior of the earth and is usually the first conspicuous wave to be recorded by a seismograph.

S.I. BASE UNITS: The seven units of measure defined by the International System of Units (SI) as a basic set from which all other SI units are derived. These are length, time, mass, electric current, thermodynamic temperature, amount of substance and luminous intensity. For length, its S.I. unit is the metre with the symbol m; for time, the S.I. unit is second with the symbol s; for mass the S.I. unit is kilogram with the symbol kg; for amount of substance, the S.I. unit is mole with the symbol mol; for luminous intensity, the S.I. unit is candela with the symbol cd; for electric current, the S.I. unit is ampere with the symbol A; for thermodynamic temperature, the S.I. unit is kelvin with the symbol K.

S.I. PREFIX: A prefix used to form decimal multiple and submultiples of the Syst me International d'Units (SI) conventions for scientific measurement in order to avoid using very large or very small numeric values. Every prefix has a unique symbol that is attached to the unit symbol. For example, the prefix kilo-, may be added to gram to indicate multiplication by one thousand; one kilogram is equal to one thousand grams.

SI Prefix	Symbol	Scientific Notation	SI Prefix	Symbol	Scientific Notation
yotta	Y	10^{24}	giga	G	10^9
zetta	Z	10^{21}	mega	M	10^6
exa	E	10^{18}	kilo	k	10^3
peta	P	10^{15}	hecto	h	10^2
tera	T	10^{12}	deka	da	10

SI Prefix	Symbol	Scientific Notation	SI Prefix	Symbol	Scientific Notation
deci	d	10^{-1}	pico	p	10^{-12}
centi	c	10^{-2}	femto	f	10^{-15}
milli	m	10^{-3}	atto	a	10^{-18}
micro	u	10^{-6}	zepto	z	10^{-21}
nano	n	10^{-9}	yocto	z	10^{-24}

Table of SI Prefixes

S.I. RULES AND CONVENTIONS: A list of the rules and conventions for the use of SI to ensure that there is no confusion introduced by peculiar usage, as dictated by National Institute of Standards and Technology (NIST) Special Publication. Some of the rules and conventions for writing unit names and symbols include the following: (i) Only units of the SI and those units recognised for use with the SI should be used to express the values of quantities. (ii) All unit names are written in small letters (newton or kilogram) except Celsius. (iii) The unit symbol is in lower case unless the name of the unit is derived from a proper name, in which case the first letter of the symbol is in upper case. (iv) Unit symbols are unaltered in the plural. (v) Unit symbols and unit names should not be mixed. (vi) Abbreviations such as sec for either s or second or mps for either m/s or metre per second are not allowed. (vii) For unit values more than 1 or less